MediFind PH: OTC Medicine Product Leaflets

Volume 2: Cold, Cough, Allergy and Sore Throat

33 product leaflets compiled in registration-file order. Each leaflet is included unchanged after this contents page and has its own bookmark.

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13	Mucosolvan (Ambroxol Hydrochloride) 30 mg Tablet	DR-XY41618	29
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15	Neozep Non-Drowsy (Phenylephrine Hydrochloride + Paracetamol) 10 mg / 500 mg Tablet	DR-XY30216	31
16	Neozep Forte (Phenylephrine hydrochloride + Chlorphenamine maleate + Paracetamol) 10 mg/ 2 mg/ 500 mg Tablet	DR-XY29559	37
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18	Robitussin (Guafenesin) 200 mg Softgel Capsule	DR-XY28913	40
19	Robitussin DM (Dextromethorphan Hydrobromide + Guaifenesin) 15 mg/100 mg per 5 mL Syrup	DR-8402-L	41
20	Sinecod Forte (Butamirate Citrate) 50 mg Sustained-Release Tablet	DRP-8129	42
21	Sinecod Forte (Butamirate Citrate) 7.5 mg/5 mL Syrup	DR-XY25225	43
22	Sinutab Cold Day & Night (Paracetamol + Phenylephrine Hydrochloride (Day Tablet) and Paracetamol + Phenylephrine Hydrochloride + Chlorpheniramine Maleate (Night Tablet)) 500mg/5mg Film-coated Tablet (Day Tablet); 500mg/5mg / 2mg Film-coated Tablet (Night Tablet) Film-Coated Tablet	DR-XY48369	44
23	Solmux (Carbocisteine) 500 mg Capsule	DRHR-1965	45
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25	Strepsils (Dichlorobenzyl Alcohol + Amylmetacresol) 1.2 mg/600 mcg Lozenge	DRHR-1455	50
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No.	Product	Reg. No.	Page
27	Tuseran (Dextromethorphan hydrobromide + Phenylephrine hydrochloride + Paracetamol) 15 mg/10 mg/325 mg Capsule	DR-XY41123	54
28	Tuseran Forte (Dextromethorphan hydrobromide + Phenylephrine hydrochloride + Paracetamol) 15 mg/10 mg/325 mg Capsule	DR-XY29720	55
29	Vicks VapoDrops (Menthol + Glycyrrhiza glabra (Liquorice) Root + Zingiber officinale (Ginger) Rhizome + Phyllantus emblica (Indian Gooseberry) Fruit) 7 mg/ 15mg/ 10 mg/ 10 mg Lozenge (Orange Menthol Flavor)	DR-XY47938	56
30	Vicks Inhaler (Menthol + Eucalyptus Oil) 108.8 mg/108.8 mg per 0.5 mL Inhaler (Lavender Scent)	DRHR-2545	79
31	Vicks VapoRub (Camphor + Menthol + Eucalyptus Oil) 5.26 g/2.82 g/1.33 g (5.26 %/2.82 %/1.33 % w/w) per 100 g Ointment (Lavender)	DRHR-2535	81
32	Virlix (Cetirizine Dihydrochloride) 10 mg Tablet	DR-XY13333	83
33	Virlix (Cetirizine Dihydrochloride) 10 mg/mL Solution (Oral Drops)	DR-XY16259	84

Allerkid (Cetirizine dihydrochloride)

2.5 mg/ mL Syrup (Oral Drops) | Over-the-Counter (OTC) Drug | FDA Reg. No. DRP-13730

Allerkid: Official FDA Philippines registration record

Brand name	Allerkid
Generic name	Cetirizine dihydrochloride
Dosage strength and form	2.5 mg/ mL Syrup (Oral Drops)
Classification	Over-the-Counter (OTC) Drug
Pharmacologic category	Antihistamine (Piperazine Derivative)
Registration number	DRP-13730
Manufacturer	Amherst Laboratories, Inc.

Allerkid: OTC forms registered under the Allerkid brand (capsule availability)

OTC capsule form registered under the Allerkid brand: NO. 1 OTC / Household Remedy product(s) registered for this brand on the FDA Verification Portal:

Brand name	Dosage strength and form	Classification	Reg. No.
Allerkid	2.5 mg/ mL Syrup (Oral Drops)	Over-the-Counter (OTC) Drug	DRP-13730

Allerkid: Official leaflet (Package Insert / Patient Information Leaflet)

An official leaflet for this registration is published on the FDA Verification Portal: PIL_01 (https://drive.google.com/file/d/1ZS_apF63hbLBGUBZTUN8qf9McbYXzIZ1/view?usp=drivesdk). Its text (OCR) and the original leaflet pages follow.

Source

FDA Philippines Verification Portal, https://verification.fda.gov.ph (search: "Allerkid"), retrieved 2026-10-22. Fields above are reproduced from the official registration record without modification.

Compiled for an academic RAG case study. Not medical advice. The portal states its information is for verification purposes. Verify current status on the portal before relying on it.

Allerkid: Official leaflet text (OCR transcription)

Machine transcription of the official FDA leaflet for Reg. No. DRP-13730. OCR may contain errors; the original leaflet pages are attached after this section.

Allerkid leaflet PIL_01

FORMULATIONS
Each mL Syrup (Oral Drops) contains:
Cetirizine dihydrochloride .....2.5 mg
Each 5 mL (1 teaspoonful) Syrup contains:
Cetirizine dihydrochloride .....5 mg
PRODUCT DESCRIPTIONS
Cetirizine dihydrochloride (Allerkid®) 2.5 mg/mL syrup (oral drops): colorless to slightly yellow-colored, clear syrup with fruity grape odor, and sweet grape taste using the TasteRite®Technology.
Cetirizine dihydrochloride (Allerkid®) 5 mg/5 mL syrup: colorless to slightly yellow-colored, clear syrup with fruity grape odor, and sweet grape taste using the TasteRite®Technology.
TasteRite® technology is a unique tastemasking system developed specifically for liquid dosage forms. This technology significantly reduces the bitterness of medicine so that children taste the flavor and not the medicine.
WHAT IS IN THE MEDICINE?
This medicine contains cetirizine, an anti-allergy, which blocks the action of histamine, a substance that causes allergic symptoms. Cetirizine is less likely to cause drowsiness and dizziness than the older class of anti-allergy medicines.
STRENGTH OF THE MEDICINE
See formulation
WHAT IS THIS MEDICINE USED FOR?
This medicine is used for the relief of symptoms associated with:
• allergic rhinitis such as sneezing; runny, itchy nose; and itchy, watery eyes
• allergic conjunctivitis
• skin symptoms of allergy such as itch and rash
HOW MUCH AND HOW OFTEN SHOULD YOU USE THIS MEDICINE?
This medicine should be taken orally (by mouth).
Age Group
Oral Dose of Cetirizine Dihydrochloride
2.5 mg per mL Syrup
(Oral Drops) 5 mg per 5 mL Syrup
Infants 6 months –
less than 12 months
2.5 mg (1 mL)
once a day -
Children:
12 months –
less than 2 years
once or twice a day (every 12 hours) -
once a day or
2.5 mg (1 mL)
twice a day
(every 12 hours)
2.5 mL (1/2 teaspoonful)
Or, as recommended by a doctor.
Note: Dose adjustment is needed in those with kidney and liver disease
WHEN SHOULD YOU NOT TAKE THIS MEDICINE?
• If you are allergic to cetirizine or other ingredients in the product
• If you have serious kidney problems
UNDESIRABLE EFFECTS
The most common undesirable effects of cetirizine include:
For children ages 6 months to 12 years: abdominal pain, bronchospasm (sudden, uncontrolled narrowing of airways in lungs), cough, diarrhea, epistaxis (nose bleeding), excessive tiredness, headache, nausea, pharyngitis (sore throat), rhinitis, somnolence (drowsiness/sleepiness), vomiting.
The following undesirable effects have also been reported with cetirizine use:
Infections and infestations: Bronchitis (inflammation of the bronchi), conjunctivitis (irritation and redness of the thin membrane covering the eye), pneumonia (inflammation of the lungs)
Neoplasms benign, malignant and unspecified (including cysts and polyps): Nasal polyp
Blood and lymphatic system disorders: Hemolytic anemia (anemia resulting from destruction of red blood cells), lymphadenopathy (swelling of the lymph nodes), thrombocytopenia (decreased number of platelets)

Immune system disorders: Anaphylaxis (rare, life-threatening allergic reactions), angioedema (rapid swelling of the skin), facial edema, glomerulonephritis (inflammation of the kidney), hypersensitivity, periorbital edema (accumulation of fluid surrounding the eye socket), urticaria (hives)
Metabolism and nutrition disorders: Anorexia (decreased appetite), edema (swelling), increased appetite, generalized edema, leg edema, peripheral edema
Psychiatric disorders: Aggression, anxiety, confusion, depersonalization, depression, dysphonia (disorder of the voice), emotional lability, enuresis (involuntary urination), euphoria (exaggerated feeling of well-being), hallucination, hyperventilation (rapid or deep breathing), insomnia, nervousness, suicidal ideation
Nervous system disorders: Abnormal coordination, abnormal thinking, agitation, amnesia, blurred vision, convulsions, dysgeusia (changed taste), hyperesthesia (increased sensitivity to any stimulation), hyperkinesia (hyperactivity), hypoesthesia (abnormally decreased sensitivity), impaired concentration, involuntary movements (dyskinesia, dystonia, tics), memory impairment, migraine, myelitis (inflammation of the spinal cord), oculogyration (uncontrolled circular movement of the eyeballs), orofacial dyskinesia (involuntary movement of the mouth and face), paradoxical central nervous system stimulation, paralysis, paresthesia (abnormal sensation such as burning, prickling or tingling), paroniria (nightmare), ptosis (drooping of eyelids), seizures, sleep disorders, tremor, vertigo, visual field defect
Eye disorders: Blindness, eye accommodation (focusing) disorders, eye bleeding, eye pain, glaucoma, xerophthalmia (dry eye)
Ear and labyrinth disorders: Deafness, otalgia (ear pain), tinnitus (ringing in the ears)
Cardiac disorders: Cardiac failure, chest pain, dyspnea (shortness of breath), heart failure, palpitation, tachycardia (rapid heart rate)
Vascular disorders: Anaphylactic shock (severe allergic reaction), hemorrhoids, hot flushes, hypertension (increased blood pressure), pallor (pale skin), rectal bleeding, syncope (fainting/temporary loss of consciousness)
Respiratory, thoracic and mediastinal disorders: Increased sputum, parosmia (distorted sense of smell), respiratory disorder, rhinitis (inflammation of the nose), sinusitis, upper respiratory tract infection
Gastrointestinal disorders: Changed taste, constipation, diarrhea, dry mouth, dyspepsia (indigestion, feeling of fullness), enlarged abdomen, eructation (burping), flatulence (gassiness), gastritis (inflammation of the stomach), increased salivation, melena (black stools), stomatitis/ulcerative stomatitis (mouth sores), swelling of the tongue, taste loss, tongue discoloration, tooth caries
Hepatobiliary disorders: Abnormal liver function (elevated liver enzymes and bilirubin), cholestasis (inflammation of the gall bladder), hepatitis
Skin and subcutaneous tissue disorders: Acne, alopecia (baldness), bullous eruption (large blisters), dermatitis (skin inflammation), dry skin, eczema, fixed drug eruption, furunculosis (boils), hyperkeratosis (thickened skin), hypertrichosis (excessive hair growth), photosensitivity, pruritus (itchiness), purpura (small purple-red skin marks), rash, seborrhea (oily skin), skin disorder, skin nodule
Musculoskeletal and connective tissue disorders: Back pain, hypertonia (muscle stiffness), rigors (shaking), twitching
Renal and urinary disorders: Cystitis (inflammation of the urinary bladder), difficulty in urinating, dysuria (painful urination), hematuria (blood in the urine), polyuria (excessive urination), urinary incontinence, urinary retention, urinary tract infection, urination frequency
Reproductive system and breast disorders: Breast pain in women, decreased libido, dysmenorrhea, intermenstrual bleeding, leukorrhea (whitish vaginal discharge), menorrhagia (heavy menstrual bleeding), vaginitis (inflammation of the vagina)
General disorders and administration site conditions: Asthenia (weakness), ataxia (incoordination), fever, malaise (feeling of bodily discomfort), pain
Investigations: Increased weight
Injury, poisoning and procedural complications: Accidental injury, ototoxicity (ear poisoning)

WHAT OTHER MEDICINE OR FOOD SHOULD BE AVOIDED WHILE TAKING THIS MEDICINE?

Do not use this product together with alcohol, medicines for anxiety or sleep disorder (e.g., Alprazolam, Diazepam, Chlordiazepoxide, Temazepam, Triazolam, etc.), medicines for depression (e.g., Amitriptyline, Doxepin, Nortriptyline, Fluoxetine, Sertraline, Paroxetine, etc.), other cold or allergy medicines, or any other medicines that cause drowsiness, sleepiness and relaxation because of the possibility for increased toxicity.
The extent of absorption of cetirizine is not reduced with food, although the rate of absorption is decreased by 1 hour.
Tell your doctor if you are taking other medicines, including vitamins, supplements, and herbal medicines.

WHAT SHOULD YOU DO IF YOU MISS A DOSE?

If you miss a dose, just take the next dose if still needed for the condition being treated and the subsequent doses at the recommended time or schedule.
Do not double the dose.

HOW SHOULD YOU KEEP THIS MEDICINE?
• Store at temperatures not exceeding 30°C
• Keep the product out of sight and reach of children
SIGNS AND SYMPTOMS OF OVERDOSE
Adverse events reported after intake of at least 5 times the recommended daily dose are: confusion, diarrhea, dizziness, excessive tiredness, headache, feeling of bodily discomfort, mydriasis (pupil dilatation), itchiness, restlessness, sedation, somnolence (drowsiness/sleepiness), stupor (decreased mental alertness), tachycardia (rapid heart rate), tremor, and urinary retention.
WHAT TO DO WHEN YOU HAVE TAKEN MORE THAN THE RECOMMENDED DOSAGE?
If you have taken more than the recommended dosage, consult a doctor.
CARE THAT SHOULD BE TAKEN WHEN TAKING THIS MEDICINE
• May cause drowsiness. Exercise caution when engaging in activities requiring mental alertness.
• Avoid excessive alcohol consumption when taking this medicine.
• Do not take more than the recommended dose.
• Do not use after the expiry date on the label.
WHEN SHOULD YOU CONSULT YOUR DOCTOR?
• Ask a doctor before use if:
o You have kidney or liver disease
o You are epileptic or at risk of convulsions
o You have difficulty passing urine
o You are pregnant or breastfeeding
o Are due for an allergy testing
• Stop use and ask a doctor if:
o Symptoms worsen or do not improve
o New symptoms occur
o Any undesirable effect occurs
ADVERSE DRUG REACTION REPORTING STATEMENT
For suspected adverse drug reaction, seek medical attention immediately and report to the FDA at www.fda.gov.ph AND Unilab at +632-8-UNILAB-1 (+632-8-864522-1) for Metro Manila or toll-free +1-800-10-UNILAB-1 for provinces, or e-mail productsafety@unilab.com.ph. By reporting undesirable effects, you can help provide more information on the safety of this medicine.
AVAILABILITY
Cetirizine dihydrochloride (Allerkid®) 2.5 mg/mL syrup (oral drops):
Amber Glass Bottle x 10 mL with Dropper (Box of 1’s)
Cetirizine dihydrochloride (Allerkid®) 5 mg/5 mL syrup:
Amber Glass Bottle x 30 mL and 60 mL (Box of 1's)
DATE OF FIRST AUTHORIZATION FOR ALLERKID® 2.5 MG/ML SYRUP (ORAL DROPS), DRP-13730: April 2010
DATE OF FIRST AUTHORIZATION FOR ALLERKID® 5 MG/5 ML SYRUP, DRP-13729: July 2010
DATE OF REVISION OF PATIENT INFORMATION LEAFLET: January 2023
Manufactured by AMHERST LABORATORIES, INC.
UNILAB Pharma Campus
Barangay Mamplasan, Biñan, Laguna, Philippines
for UNILAB, Inc.
No. 66 United Street, Mandaluyong City
Metro Manila, Philippines
Antihistamine
(Piperazine Derivative)
ARTWORK INFORMATION: P300000040060 IN ALLERKID DRP/SYR LINE SS PH
NEW Reg. No. DRP-13730 Pantone/s:
Old FG/SKU Code PMS No.
151994-03, 152011-03, 152012-04,
151995-04, 153202-04
Basis for
Revision:
New FG/SKU Code KG/PLG No. N/A
Existing Barcode Dimension CCR No.:
New Barcode Existing PM Code P300000024177
Method of Proofing Performed (Division) Global Vision Inspection (*Provide GV Report to QA) Manual Proofing
Reference Checklist (Division):
CPR/CPN/Approval of country PMS Knifeguide or PL Guide Global Vision Report
Superseded Artwork FDA Approved artwork PM code and FG code CCDS
Evaluation Checklist (QA): PM code and FG code Global Vision Report (if applicable)
CPR/Approval of country Dimension (PMS & KG/PLG) Barcode (SAP) Color and Pantone
Photo-eye(tubes/Sachet) Approved CCR Space for LEM CCDS
Unilab logo and seal(CBG Guideline) Code bars (ISFG) Manufacturing Clause FDA Approved AW/Notification

Marketing Division: Name, Signature, Date QA-SMS-PM: Disposition:
APPROVED
DISAPPROVED
Remarks:
Noted By:
U-F-SQM-044/01
Kathyrene Visco
spread
EXISTING rev no.
151994, 152011,
152012, 151995
N/A
Black
PCPR Conversion
Argee Guilas
Ma. Lourdes Genuino
Div. Pharmacist
In-house
Artist/Agency Med. Director Evaluated
Brand By:
Personnel
Div. Head/ Prod.
Dir.

CARE THAT SHOULD BE TAKEN WHEN TAKING THIS MEDICINE
- May cause drowsiness. Exercise caution when engaging in activities requiring mental alertness.
- Avoid excessive alcohol consumption when taking this medicine.
- Do not take more than the recommended dose.
- Do not use after the expiry date on the label.

WHEN SHOULD YOU CONSULT YOUR DOCTOR?
- Ask a doctor before use if:

o You have kidney or liver disease
o You are epileptic or at risk of convulsions
o You have difficulty passing urine
o You are pregnant or breastfeeding
o Are due for an allergy testing

- Stop use and ask a doctor if:

o Symptoms worsen or do not improve
o New symptoms occur
o Any undesirable effect occurs

ADVERSE DRUG REACTION REPORTING STATEMENT
For suspected adverse drug reaction, seek medical attention immediately and report to the FDA at www.fda.gov.ph AND Unilab at +632-8-**UNILAB**-1 (+632-8-**864522**-1) for Metro Manila or toll-free +1-800-10-UNILAB-1 for provinces, or e-mail productsafety@unilab.com.ph. By reporting undesirable effects, you can help provide more information on the safety of this medicine.

AVAILABILITY
Cetirizine dihydrochloride (Allerkid®) 2.5 mg/mL syrup (oral drops):
Amber Glass Bottle x 10 mL with Dropper (Box of 1's)
Cetirizine dihydrochloride (Allerkid®) 5 mg/5 mL syrup:
Amber Glass Bottle x 30 mL and 60 mL (Box of 1's)

DATE OF FIRST AUTHORIZATION FOR ALLERKID® 2.5 MG/ML SYRUP (ORAL DROPS), DRP-13730: April 2010
DATE OF FIRST AUTHORIZATION FOR ALLERKID® 5 MG/5 ML SYRUP, DRP-13729: July 2010
DATE OF REVISION OF PATIENT INFORMATION LEAFLET: January 2023
Manufactured by AMHERST LABORATORIES, INC.
UNILAB Pharma Campus
Barangay Mamplasan, Biñan, Laguna, Philippines
for **UNILAB, Inc.**
No. 66 United Street, Mandaluyong City
Metro Manila, Philippines

UNILAB Consumer Care
8-UNILAB-1
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Provincial Toll Free: 1-800-10-8645221
Email: info@unilab.com.ph
Website: www.unilab.com.ph

UNILAB
Trusted Quality Healthcare

P300000040060 Reg. IPOPHIL

APPROVED
By Kathyrene Visco at 11:24 am, Sep 07, 2023

UNILAB

CETIRIZINE dihydrochloride
ALLERKID®

2.5 mg/mL Syrup (Oral Drops)
5 mg/ 5 mL Syrup
Antihistamine (Piperazine Derivative)

FORMULATIONS
Each mL Syrup (Oral Drops) contains:
Cetirizine dihydrochloride .....2.5 mg

Each 5 mL (1 teaspoonful) Syrup contains:
Cetirizine dihydrochloride .....5 mg

PRODUCT DESCRIPTIONS
Cetirizine dihydrochloride (Allerkid®) 2.5 mg/mL syrup (oral drops): colorless to slightly yellow-colored, clear syrup with fruity grape odor, and sweet grape taste using the TasteRite®Technology.

Cetirizine dihydrochloride (Allerkid®) 5 mg/5 mL syrup: colorless to slightly yellow-colored, clear syrup with fruity grape odor, and sweet grape taste using the TasteRite®Technology.

TasteRite® technology is a unique tastemasking system developed specifically for liquid dosage forms. This technology significantly reduces the bitterness of medicine so that children taste the flavor and not the medicine.

WHAT IS IN THE MEDICINE?
This medicine contains cetirizine, an anti-allergy, which blocks the action of histamine, a substance that causes allergic symptoms. Cetirizine is less likely to cause drowsiness and dizziness than the older class of anti-allergy medicines.

STRENGTH OF THE MEDICINE
See formulation

WHAT IS THIS MEDICINE USED FOR?
This medicine is used for the relief of symptoms associated with:
- allergic rhinitis such as sneezing; runny, itchy nose; and itchy, watery eyes
- allergic conjunctivitis
- skin symptoms of allergy such as itch and rash

HOW MUCH AND HOW OFTEN SHOULD YOU USE THIS MEDICINE?
This medicine should be taken orally (by mouth).

Age Group	Oral Dose of Cetirizine Dihydrochloride	
	2.5 mg per mL Syrup (Oral Drops)	5 mg per 5 mL Syrup
Infants 6 months – less than 12 months	2.5 mg (1 mL) once a day	—
Children:		
12 months – less than 2 years	2.5 mg (1 mL) once or twice a day (every 12 hours)	—
2 – 5 years	5 mg (2 mL) once a day or 2.5 mg (1 mL) twice a day (every 12 hours)	5 mL (1 teaspoonful) once a day or 2.5 mL (1/2 teaspoonful) twice a day (every 12 hours)
6 – 12 years	—	10 mL (2 teaspoonsful) once a day or 5 mL (1 teaspoonful) twice a day (every 12 hours)
Or, as recommended by a doctor.		

APPROVED
By Maria Lourdes G. Genuino,MD at 6:39 pm, Sep 07, 2023

Note: Dose adjustment is needed in those with kidney and liver disease

WHEN SHOULD YOU NOT TAKE THIS MEDICINE?
- If you are allergic to cetirizine or other ingredients in the product
- If you have serious kidney problems

UNDESIRABLE EFFECTS
The most common undesirable effects of cetirizine include:

For children ages 6 months to 12 years: abdominal pain, bronchospasm (sudden, uncontrolled narrowing of airways in lungs), cough, diarrhea, epistaxis (nose bleeding), excessive tiredness, headache, nausea, pharyngitis (sore throat), rhinitis, somnolence (drowsiness/sleepiness), vomiting.

The following undesirable effects have also been reported with cetirizine use:

Infections and infestations: Bronchitis (inflammation of the bronchi), conjunctivitis (irritation and redness of the thin membrane covering the eye), pneumonia (inflammation of the lungs)
Neoplasms benign, malignant and unspecified (including cysts and polyps): Nasal polyp
Blood and lymphatic system disorders: Hemolytic anemia (anemia resulting from destruction of red blood cells), lymphadenopathy (swelling of the lymph nodes), thrombocytopenia (decreased number of platelets)
Immune system disorders: Anaphylaxis (rare, life-threatening allergic reactions), angioedema (rapid swelling of the skin), facial edema, glomerulonephritis (inflammation of the kidney), hypersensitivity, periorbital edema (accumulation of fluid surrounding the eye socket), urticaria (hives)
Metabolism and nutrition disorders: Anorexia (decreased appetite), edema (swelling), increased appetite, generalized edema, leg edema, peripheral edema
Psychiatric disorders: Aggression, anxiety, confusion, depersonalization, depression, dysphonia (disorder of the voice), emotional lability, enuresis (involuntary urination), euphoria (exaggerated feeling of well-being), hallucination, hyperventilation (rapid or deep breathing), insomnia, nervousness, suicidal ideation
Nervous system disorders: Abnormal coordination, abnormal thinking, agitation, amnesia, blurred vision, convulsions, dysgeusia (changed taste), hyperesthesia (increased sensitivity to any stimulation), hyperkinesia (hyperactivity), hypoesthesia (abnormally decreased sensitivity), impaired concentration, involuntary movements (dyskinesia, dystonia, tics), memory impairment, migraine, myelitis (inflammation of the spinal cord), oculogyration (uncontrolled circular movement of the eyeballs), orofacial dyskinesia (involuntary movement of the mouth and face), paradoxical central nervous system stimulation, paralysis, paresthesia (abnormal sensation such as burning, prickling or tingling), paroniria (nightmare), ptosis (drooping of eyelids), seizures, sleep disorders, tremor, vertigo, visual field defect
Eye disorders: Blindness, eye accommodation (focusing) disorders, eye bleeding, eye pain, glaucoma, xerophthalmia (dry eye)
Ear and labyrinth disorders: Deafness, otalgia (ear pain), tinnitus (ringing in the ears)
Cardiac disorders: Cardiac failure, chest pain, dyspnea (shortness of breath), heart failure, palpitation, tachycardia (rapid heart rate)
Vascular disorders: Anaphylactic shock (severe allergic reaction), hemorrhoids, hot flushes, hypertension (increased blood pressure), pallor (pale skin), rectal bleeding, syncope (fainting/temporary loss of consciousness)
Respiratory, thoracic and mediastinal disorders: Increased sputum, parosmia (distorted sense of smell), respiratory disorder, rhinitis (inflammation of the nose), sinusitis, upper respiratory tract infection
Gastrointestinal disorders: Changed taste, constipation, diarrhea, dry mouth, dyspepsia (indigestion, feeling of fullness), enlarged abdomen, eructation (burping), flatulence (gassiness), gastritis (inflammation of the stomach), increased salivation, melena (black stools), stomatitis/ulcerative stomatitis (mouth sores), swelling of the tongue, taste loss, tongue discoloration, tooth caries
Hepatobiliary disorders: Abnormal liver function (elevated liver enzymes and bilirubin), cholestasis (inflammation of the gall bladder), hepatitis
Skin and subcutaneous tissue disorders: Acne, alopecia (baldness), bullous eruption (large blisters), dermatitis (skin inflammation), dry skin, eczema, fixed drug eruption, furunculosis (boils), hyperkeratosis (thickened skin), hypertrichosis (excessive hair growth), photosensitivity, pruritus (itchiness), purpura (small purple-red skin marks), rash, seborrhea (oily skin), skin disorder, skin nodule
Musculoskeletal and connective tissue disorders: Back pain, hypertonia (muscle stiffness), rigors (shaking), twitching
Renal and urinary disorders: Cystitis (inflammation of the urinary bladder), difficulty in urinating, dysuria (painful urination), hematuria (blood in the urine), polyuria (excessive urination), urinary incontinence, urinary retention, urinary tract infection, urination frequency
Reproductive system and breast disorders: Breast pain in women, decreased libido, dysmenorrhea, intermenstrual bleeding, leukorrhea (whitish vaginal discharge), menorrhagia (heavy menstrual bleeding), vaginitis (inflammation of the vagina)
General disorders and administration site conditions: Asthenia (weakness), ataxia (incoordination), fever, malaise (feeling of bodily discomfort), pain
Investigations: Increased weight
Injury, poisoning and procedural complications: Accidental injury, ototoxicity (ear poisoning)

WHAT OTHER MEDICINE OR FOOD SHOULD BE AVOIDED WHILE TAKING THIS MEDICINE?
Do not use this product together with alcohol, medicines for anxiety or sleep disorder (e.g., Alprazolam, Diazepam, Chlordiazepoxide, Temazepam, Triazolam, etc.), medicines for depression (e.g., Amitriptyline, Doxepin, Nortriptyline, Fluoxetine, Sertraline, Paroxetine, etc.), other cold or allergy medicines, or any other medicines that cause drowsiness, sleepiness and relaxation because of the possibility for increased toxicity.
The extent of absorption of cetirizine is not reduced with food, although the rate of absorption is decreased by 1 hour.

Tell your doctor if you are taking other medicines, including vitamins, supplements, and herbal medicines.

WHAT SHOULD YOU DO IF YOU MISS A DOSE?
If you miss a dose, just take the next dose if still needed for the condition being treated and the subsequent doses at the recommended time or schedule.

Do not double the dose.

HOW SHOULD YOU KEEP THIS MEDICINE?
- Store at temperatures not exceeding 30°C
- Keep the product out of sight and reach of children

SIGNS AND SYMPTOMS OF OVERDOSE
Adverse events reported after intake of at least 5 times the recommended daily dose are: confusion, diarrhea, dizziness, excessive tiredness, headache, feeling of bodily discomfort, mydriasis (pupil dilatation), itchiness, restlessness, sedation, somnolence (drowsiness/sleepiness), stupor (decreased mental alertness), tachycardia (rapid heart rate), tremor, and urinary retention.

WHAT TO DO WHEN YOU HAVE TAKEN MORE THAN THE RECOMMENDED DOSAGE?
If you have taken more than the recommended dosage, consult a doctor.

ARTWORK INFORMATION: P300000040060 IN ALLERKID DRP/SYR LINE SS PH

■ **NEW** ☐ **EXISTING** rev no.

		Reg. No.	DRP-13730 / DRP-13729	Pantone/s:	Black
Old FG/SKU Code	151994, 152011, 152012, 151995	PMS No.	151994-03, 152011-03, 152012-04, 151995-04, 153202-04	Basis for Revision:	PCPR Conversion
New FG/SKU Code	N/A	KG/PLG No.	N/A		
		Dimension	140 x 196 mm (± 1 mm) – spread 140 x 98 mm - folded		
Existing Barcode	N/A			CCR No.:	TMP-CCR-0159
New Barcode	N/A	Existing PM Code	P300000024177		

Method of Proofing Performed (Division) ☐ Global Vision Inspection (*Provide GV Report to QA) ■ Manual Proofing

Reference Checklist (Division):
■ CPR/CPN/Approval of country ■ PMS ☐ Knifeguide or PL Guide ☐ Global Vision Report
☐ Superseded Artwork ☐ FDA Approved artwork ☐ PM code and FG code ☐ CCDS

Evaluation Checklist (QA):
◉ CPR/Approval of country ◉ PM code and FG code ☐ Global Vision Report (if applicable)
☐ Photo-eye(tubes/Sachet) ◉ Dimension (PMS & KG/PLG) ☐ Barcode (SAP) ◉ Color and Pantone
◉ Unilab logo and seal(CBG Guideline) ☐ Approved CCR ☐ Space for LEM ◉ CCDS
◉ Code bars (ISFG) ◉ Manufacturing Clause ◉ FDA Approved AW/Notification

Marketing Division: *Name, Signature, Date*
In-house Artist/Agency: N/A
Brand Personnel: Argee Guilas
Med. Director: Ma. Lourdes Genuino
Div. Head/ Prod. Dir.: N/A
Div. Pharmacist: Kathyrene Visco

QA-SMS-PM:
Evaluated By:
Noted By:

Disposition:
◉ APPROVED
☐ DISAPPROVED
Remarks: GV used during evaluation

U-F-SQM-044/01

APPROVED
UL-CQA&C

APPROVED
By Aila Marie P. Avila at 9:37 am, Sep 29, 2023

APPROVED
By Amalia B. Magadia at 5:22 pm, Sep 29, 2023

Allerta (Loratadine)

10 mg Tablet | Over-the-Counter (OTC) Drug | FDA Reg. No. DR-XY29925

Allerta: Official FDA Philippines registration record

Brand name	Allerta
Generic name	Loratadine
Dosage strength and form	10 mg Tablet
Classification	Over-the-Counter (OTC) Drug
Pharmacologic category	Antihistamine
Registration number	DR-XY29925
Manufacturer	Amherst Laboratories, Inc.

Allerta: OTC forms registered under the Allerta brand (capsule availability)

OTC capsule form registered under the Allerta brand: NO. 1 OTC / Household Remedy product(s) registered for this brand on the FDA Verification Portal:

Brand name	Dosage strength and form	Classification	Reg. No.
Allerta	10 mg Tablet	Over-the-Counter (OTC) Drug	DR-XY29925

Allerta: Official leaflet (Package Insert / Patient Information Leaflet)

No PI/PIL is published on the FDA Verification Portal for registration DR-XY29925 as of 2026-10-22. Indication and dosage are not covered by this record.

Source

FDA Philippines Verification Portal, https://verification.fda.gov.ph (search: "Allerta"), retrieved 2026-10-22. Fields above are reproduced from the official registration record without modification.

Compiled for an academic RAG case study. Not medical advice. The portal states its information is for verification purposes. Verify current status on the portal before relying on it.

Benadryl AH (Diphenhydramine Hydrochloride)

25 mg Capsule | Over-the-Counter (OTC) Drug | FDA Reg. No. DR-4620

Benadryl AH: Official FDA Philippines registration record

Brand name	Benadryl AH
Generic name	Diphenhydramine Hydrochloride
Dosage strength and form	25 mg Capsule
Classification	Over-the-Counter (OTC) Drug
Pharmacologic category	Antihistamine
Registration number	DR-4620
Manufacturer	Interphil Laboratories, Inc.

Benadryl AH: OTC forms registered under the Benadryl brand (capsule availability)

OTC capsule form registered under the Benadryl brand: YES. 7 OTC / Household Remedy product(s) registered for this brand on the FDA Verification Portal:

Brand name	Dosage strength and form	Classification	Reg. No.
Benadryl One	10 mg Film-Coated Tablet	Over-the-Counter (OTC) Drug	DR-XY49097
Benadryl One	10 mg Film-Coated Tablet	Over-the-Counter (OTC) Drug	DR-XY49098
Benadryl One	10 mg film-coated tablet	Over-the-Counter (OTC) Drug	DRP-1469-02
Benadryl EXP	100 mg/5 mL Syrup	Over-the-Counter (OTC) Drug	DR-XY7343
Benadryl	12.5 mg/5 mL Syrup	Over-the-Counter (OTC) Drug	DR-4619
Benadryl AH	25 mg Capsule	Over-the-Counter (OTC) Drug	DR-4620
Benadryl AH	25 mg Capsule	Over-the-Counter (OTC) Drug	DR-XY45645

Benadryl AH: Official leaflet (Package Insert / Patient Information Leaflet)

No PI/PIL is published on the FDA Verification Portal for registration DR-4620 as of 2026-10-22. Indication and dosage are not covered by this record.

Source

FDA Philippines Verification Portal, https://verification.fda.gov.ph (search: "Benadryl"), retrieved 2026-10-22. Fields above are reproduced from the official registration record without modification.

Compiled for an academic RAG case study. Not medical advice. The portal states its information is for verification purposes. Verify current status on the portal before relying on it.

Benadryl One (Cetirizine hydrochloride)

10 mg film-coated tablet | Over-the-Counter (OTC) Drug | FDA Reg. No. DRP-1469-02

Benadryl One: Official FDA Philippines registration record

Brand name	Benadryl One
Generic name	Cetirizine hydrochloride
Dosage strength and form	10 mg film-coated tablet
Classification	Over-the-Counter (OTC) Drug
Pharmacologic category	Antihistamine (Piperazine Derivative)
Registration number	DRP-1469-02
Manufacturer	Interphil Laboratories, Inc.

Benadryl One: OTC forms registered under the Benadryl brand (capsule availability)

OTC capsule form registered under the Benadryl brand: YES. 7 OTC / Household Remedy product(s) registered for this brand on the FDA Verification Portal:

Brand name	Dosage strength and form	Classification	Reg. No.
Benadryl One	10 mg Film-Coated Tablet	Over-the-Counter (OTC) Drug	DR-XY49097
Benadryl One	10 mg Film-Coated Tablet	Over-the-Counter (OTC) Drug	DR-XY49098
Benadryl One	10 mg film-coated tablet	Over-the-Counter (OTC) Drug	DRP-1469-02
Benadryl EXP	100 mg/5 mL Syrup	Over-the-Counter (OTC) Drug	DR-XY7343
Benadryl	12.5 mg/5 mL Syrup	Over-the-Counter (OTC) Drug	DR-4619
Benadryl AH	25 mg Capsule	Over-the-Counter (OTC) Drug	DR-4620
Benadryl AH	25 mg Capsule	Over-the-Counter (OTC) Drug	DR-XY45645

Benadryl One: Official leaflet (Package Insert / Patient Information Leaflet)

No PI/PIL is published on the FDA Verification Portal for registration DRP-1469-02 as of 2026-10-22. Indication and dosage are not covered by this record.

Source

FDA Philippines Verification Portal, https://verification.fda.gov.ph (search: "Benadryl"), retrieved 2026-10-22. Fields above are reproduced from the official registration record without modification.

Compiled for an academic RAG case study. Not medical advice. The portal states its information is for verification purposes. Verify current status on the portal before relying on it.

Bioflu (Phenylephrine Hydrochloride + Chlorphenamine Maleate + Paracetamol)

2.5 mg/ 500 mcg/ 125 mg per 5 mL Suspension | Over-the-Counter (OTC) Drug | FDA Reg. No. DRP-13914

Bioflu: Official FDA Philippines registration record

Brand name	Bioflu
Generic name	Phenylephrine Hydrochloride + Chlorphenamine Maleate + Paracetamol
Dosage strength and form	2.5 mg/ 500 mcg/ 125 mg per 5 mL Suspension
Classification	Over-the-Counter (OTC) Drug
Pharmacologic category	Nasal Decongestant/ Antihistamine/ Analgesic/ Antipyretic
Registration number	DRP-13914
Manufacturer	Amherst Laboratories, Inc.

Bioflu: OTC forms registered under the Bioflu brand (capsule availability)

OTC capsule form registered under the Bioflu brand: YES. 4 OTC / Household Remedy product(s) registered for this brand on the FDA Verification Portal:

Brand name	Dosage strength and form	Classification	Reg. No.
Bioflu	10 mg/2 mg/500 mg Film-Coated Tablet	Over-the-Counter (OTC) Drug	DR-XY34482
Bioflu	2.5 mg/ 500 mcg/ 125 mg per 5 mL Suspension	Over-the-Counter (OTC) Drug	DRP-13914
Bioflu	2.5 mg/500 mcg/125 mg per 5 mL Suspension	Over-the-Counter (OTC) Drug	DR-XY37402
Bioflu Complete	500 mg/10 mg/4 mg Capsule	Over-the-Counter (OTC) Drug	DR-XY46707

Bioflu: Official leaflet (Package Insert / Patient Information Leaflet)

An official leaflet for this registration is published on the FDA Verification Portal: PIL_01 (https://drive.google.com/file/d/163BpED_8Q64k7Q-JnMdRFzea23_SPYTg/view?usp=drivesdk). Its text (OCR) and the original leaflet pages follow.

Source

FDA Philippines Verification Portal, https://verification.fda.gov.ph (search: "Bioflu"), retrieved 2026-10-22. Fields above are reproduced from the official registration record without modification.

Compiled for an academic RAG case study. Not medical advice. The portal states its information is for verification purposes. Verify current status on the portal before relying on it.

Bioflu: Official leaflet text (OCR transcription)

Machine transcription of the official FDA leaflet for Reg. No. DRP-13914. OCR may contain errors; the original leaflet pages are attached after this section.

Bioflu leaflet PIL_01

UNILAB
Skin and subcutaneous tissue disorders: Rare cases of serious skin reactions (i.e., Stevens-Johnson syndrome, toxic epidermal necrolysis and
PHENYLEPHRINE HCl
WHAT OTHER MEDICINE OR FOOD SHOULD BE AVOIDED WHILE TAKING THIS MEDICINE?
CHLORPHENAMINE MALEATE
possibility for increased toxicity.
Concurrent administration with medicines for depression such as monoamine oxidase (MAO) inhibitors (e.g., selegiline, moclobemide) and
PARACETAMOL
tricyclic antidepressants (e.g., amitriptyline, imipramine) may result in hypertensive crisis (sudden, severe increase in blood pressure that
can lead to stroke).
Taking an alpha-adrenergic blocking agent (e.g., phenothiazine drugs, phentolamine) before the administration of this product may decrease the
BIOFLU®
blood pressure.
Medicines that cause sedation such as anxiolytics (e.g., alprazolam, diazepam) and hypnotics (e.g., zolpidem) may potentiate drowsiness.
2.5 mg / 500 mcg / 125 mg per 5 mL Suspension
Paracetamol, when used together with warfarin (a blood-thinning medicine), may cause an increase in the International Nomalized Ratio (INR),
Nasal Decongestant/Antihistamine/Analgesic-Antipyretic
phenobarbital, phenytoin) may increase susceptibility to the harmful effects to the liver.
Eachl t ins:
FORMULATION
Tell your doctor about other medicines you are taking (i.e., other medicines for cough, cold, allergy, pain, or fever, especially other paracetamol-containing products).
Phenylephrine HCl
Chlorphenamine Maleate ....
500 mcg
.2.5 mg
WHAT SHOULD YOU DO IF YOU MISS A DOSE?
Paracetamol .
...125 mg
time or schedule (i.e., every 6 hours).
PRODUCT DESCRIPTION
Do not double the dose.
HOW SHOULD YOU KEEP THIS MEDICINE?
Chlorphenamine maleate, an anti-allergy, relieves symptoms of allergy such as runny nose, sneezing, and itchy, watery eyes.
This medicine contains phenylephrine HCl, chlorphenamine maleate and paracetamol.
WHAT IS IN THE MEDICINE?
also reduces postnasal drip.
Paracetamol is an effective fever reducer and pain reliever.
Phenylephrine HCl
• Increased blood pressure
• Palpitation
Store at temoeratures not exceeding 30°C.
Vomiting
SIGNS AND SYMPTOMS OF OVERDOSAGE
• Paresthesia (sensation of tingling, pricking, or numbness of the skin)
• Headache
• Seizures
WHAT IS THIS MEDICINE USED FOR?
STRENGTH OF THE MEDICINE
See Formulation
This medicine is used for the relief of clogged nose, runny nose, postnasal drip, itchy and watery eyes, sneezing, headache, body aches, and
Paracetamol
Chlorphenamine maleate

Overdosage of Paracetamol usually involves 4 phases with the following signs and symptoms:
• Extreme sleepiness or prolonged drowsiness
Convulsion
Agitation
• Irritability
• Weakness
•Tachycardia
• Hallucination
Cooma
See Formulation
•Convulsion
• Agitation
• Irritability
• Tachycardia
sinus openings and passages.
WHAT IS THIS MEDICINE USED FOR?
This medicine is used for the relief of clogged nose, runny nose, postnasal drip, itchy and watery eyes, sneezing, headache, body aches, and
fever associated with flu, the common cold, allergic rhinitis, sinusitis, and other minor respiratory tract infections. It also helps decongest
Overdosage of Paracetamol usually involves 4 phases with the following signs and symptoms:
Paracetamol
II.
1.
Right upper abdominal pain or tenderness, liver enlargement which may be characterized by abdominal discomfort of "feeling full, elevated bilirubin and liver enzyme concentrations, prolongation of prothrombin time, and occasionally decreased urine output
Eating disorder, nausea, vomiting, malaise, and excessive sweating
This medicine should be taken orally (by mouth).
III.
IV.
Recovery or progression to fatal complete liver failure
Age Group
Chlorphenamine Maleate + Paracetamol
Dose of Phenylephrine HCl +
every 6 hours
Suspension
If you have taken more than the recommended dosage, consult a doctor or contact a poison control center right away, even if you seem
you do not notice any signs or symptoms.
WHAT TO DO WHEN YOU HAVE TAKEN MORE THAN THE RECOMMENDED DOSAGE?
Children 7 to 12 years old
Adults and Children 12 years and older
CARE THAT SHOULD BE TAKEN WHEN TAKING THIS MEDICINE
Children 2 to 6 years old
Or, as recommended by a doctor.
- An adult takes more than 4 g of Paracetamol in 24 hours, which is the maximum daily amount;
• If you have high blood pressure or severe heart disease unless recommended by a doctor
WHEN SHOULD YOU NOT TAKE THIS MEDICINE?
• If you are pregnant or breastfeeding
Do not use with any other medicine containing phenylephrine, chlorphenamine or paracetamol (prescription or nonprescription). If you are
Skin reddening
- An adult has 3 or more alcoholic drinks everyday while using this product.
- Blisters
-Rash
UNDESIRABLE EFFECTS
Phenylephrine HCl
Cardiac disorders: Arrhythmia (irregular heart beat), palpitation
Nervous system disorders: Dizziness, restlessness, tremor (muscle shaking)
• Do not take more than the recommended dose.
• Do not use after the expiry date on the label.
Ask a doctor before use if you have liver or kidney disease.
Respiratory, thoracic and mediastinal disorders: Respiratory distress (noisy, congested breathing)
General disorders and administration site conditions: Weakness
Vascular disorders: Increased blood pressure, pallor
Before taking this medication, tell your doctor if you have:
WHEN SHOULD YOU CONSULT YOUR DOCTOR?

• Glaucoma
• Thyroid problems
Chlorphenamine maleate
• Diabetes
• Liver or kidney disease
Chlorphenamine maleate may cause sleepiness and drowsiness.
Stop use and ask a doctor if:
Psychiatric disorders: Insomnia/sleeplessness, nervousness
• Fever gets worse or lasts more than 3 days
• New symptoms occur
Nervous system disorders: Headache, tremor (muscle shaking)
Eye disorders: Blurred vision, visual disturbance
Cardiac disorders: Chest tightness
Gast itng
Musculoskeletal and connective tissue disorders: Muscle weakness
Ear and labyrinth disorders: Tinnitus (ringing in the ears)
Vascular disorders: Increased or decreased blood pressure
For suspected adverse drug reaction, seek medical attention immediately and report to the FDA at www.fda.gov.ph AND Unilab at
By reporting undesirable effects, you can help provide more information on the safety of this medicine.
AVAILABILITY
Bioflu Suspension (DRP-13914) - in Amber Glass Bottle x 60 mL (Box of 1's)
ADVERSE DRUG REACTION REPORTING STATEMENT
+632-8-UNILAB-1(+632-8-864522-1) for Metro Manila or toll-free +1-800-10-UNILAB-1for provinces, or e-mail
productsafety@unilab.com.ph.
UNILAB Consumer Care
General disorders and administration site conditions: Dryness of the mouth, nose, and throat; irritability
Paracetamol, when taken within the recommended dose and duration of treatment, has low incidence of side effects.
Renal and urinary disorders: Difficulty urinating
Paracetamol
UNILAB Pharma Campus, Barangay Mamplasan
DATE OF FIRST AUTHORIZATION: October 2012
DATE OF REVISION OF PATIENT INFORMATION LEAFLET (PIL): JUne 2023 (CCDS 17-020)
Manufactured by AMHERST LABORATORIES, INC.
UNILAB
8-UNILAB-1
8-864522-1
number of granulocytes), leukopenia (decrease in the number of leukocytes), neutropenia (decrease in the number of neutrophils), pancytopenia
(deficiency of all cellular elements in the blood), thrombocytopenia (decrease in the number of platelets)
Blood and lymphatic system disorders: Changes in the number of white blood cells and platelets such as agranulocytosis (decrease in the
for UNILAB, Inc.
Biñan, Laguna, Philippines
No. 66 United Street, Mandaluyong City
Trusted Quality Healthcare.
Provincial Toll Free: 1-800-10-8645221
Website:w.uniab.com.ph
Email infoounllab.com.ph
Gbast M i nces
Metro Manila, Philippines
Reg. IPOPHIL

UNILAB

PHENYLEPHRINE HCl
CHLORPHENAMINE MALEATE
PARACETAMOL

BIOFLU®

2.5 mg / 500 mcg / 125 mg per 5 mL Suspension
Nasal Decongestant/Antihistamine/Analgesic-Antipyretic

FORMULATION
Each 5 mL (1 teaspoonful) suspension contains:
Phenylephrine HCl ..... 2.5 mg
Chlorphenamine Maleate ..... 500 mcg
Paracetamol ..... 125 mg

PRODUCT DESCRIPTION
Pink colored suspension with strawberry odor and strawberry taste

WHAT IS IN THE MEDICINE?
This medicine contains phenylephrine HCl, chlorphenamine maleate and paracetamol.

Phenylephrine HCl, a nasal decongestant, clears obstructed air passages and nasal sinuses due to congestion making breathing easier. It also reduces postnasal drip.

Chlorphenamine maleate, an anti-allergy, relieves symptoms of allergy such as runny nose, sneezing, and itchy, watery eyes.

Paracetamol is an effective fever reducer and pain reliever.

STRENGTH OF THE MEDICINE
See Formulation

WHAT IS THIS MEDICINE USED FOR?
This medicine is used for the relief of clogged nose, runny nose, postnasal drip, itchy and watery eyes, sneezing, headache, body aches, and fever associated with flu, the common cold, allergic rhinitis, sinusitis, and other minor respiratory tract infections. It also helps decongest sinus openings and passages.

HOW MUCH AND HOW OFTEN SHOULD YOU USE THIS MEDICINE?
This medicine should be taken orally (by mouth).

Age Group	Dose of Phenylephrine HCl + Chlorphenamine Maleate + Paracetamol every 6 hours
	Suspension
Adults and Children 12 years and older	15 mL (3 teaspoonsful)
Children 7 to 12 years old	10 mL (2 teaspoonsful)
Children 2 to 6 years old	5 mL (1 teaspoonful)
	Or, as recommended by a doctor.

WHEN SHOULD YOU NOT TAKE THIS MEDICINE?
- If you are allergic to any ingredient of the product
- If you have high blood pressure or severe heart disease unless recommended by a doctor
- If you have anemia, kidney or liver disease unless recommended by a doctor
- If you are pregnant or breastfeeding

UNDESIRABLE EFFECTS
Phenylephrine HCl
Psychiatric disorders: Anxiety (feeling of uneasiness), insomnia/sleeplessness, nervousness
Nervous system disorders: Dizziness, restlessness, tremor (muscle shaking)
Cardiac disorders: Arrhythmia (irregular heart beat), palpitation
Vascular disorders: Increased blood pressure, pallor
Respiratory, thoracic and mediastinal disorders: Respiratory distress (noisy, congested breathing)
General disorders and administration site conditions: Weakness

Chlorphenamine maleate
Chlorphenamine maleate may cause sleepiness and drowsiness.
Psychiatric disorders: Insomnia/sleeplessness, nervousness
Nervous system disorders: Headache, tremor (muscle shaking)
Eye disorders: Blurred vision, visual disturbance
Ear and labyrinth disorders: Tinnitus (ringing in the ears)
Cardiac disorders: Chest tightness
Vascular disorders: Increased or decreased blood pressure
Gastrointestinal disorders: Constipation, diarrhea, gastrointestinal discomfort, nausea, vomiting
Musculoskeletal and connective tissue disorders: Muscle weakness
Renal and urinary disorders: Difficulty urinating
General disorders and administration site conditions: Dryness of the mouth, nose, and throat; irritability

Paracetamol
Paracetamol, when taken within the recommended dose and duration of treatment, has low incidence of side effects.
Blood and lymphatic system disorders: Changes in the number of white blood cells and platelets such as agranulocytosis (decrease in the number of granulocytes), leukopenia (decrease in the number of leukocytes), neutropenia (decrease in the number of neutrophils), pancytopenia (deficiency of all cellular elements in the blood), thrombocytopenia (decrease in the number of platelets)
Immune system disorders: Allergic reactions which may cause difficulty in breathing, skin rash, angioedema (swelling of the face or throat)
Gastrointestinal disorders: Minor stomach and intestinal disturbances
Skin and subcutaneous tissue disorders: Rare cases of serious skin reactions (i.e., Stevens-Johnson syndrome, toxic epidermal necrolysis and acute generalized exanthematous pustulosis) which may include symptoms such as reddening of the skin, rash, blisters and detachment of the upper surface of the skin have been reported with paracetamol.

WHAT OTHER MEDICINE OR FOOD SHOULD BE AVOIDED WHILE TAKING THIS MEDICINE?
Do not use this product together with sympathomimetic agents (e.g., epinephrine) and general anesthetics (e.g., halothane) because of the possibility for increased toxicity.

Concurrent administration with medicines for depression such as monoamine oxidase (MAO) inhibitors (e.g., selegiline, moclobemide) and tricyclic antidepressants (e.g., amitriptyline, imipramine) may result in hypertensive crisis (sudden, severe increase in blood pressure that can lead to stroke).

Taking an alpha-adrenergic blocking agent (e.g., phenothiazine drugs, phentolamine) before the administration of this product may decrease the effect of phenylephrine. Concomitant use with beta-adrenergic blocking agents (e.g., propranolol) and methyldopa may cause an increase in blood pressure.

Medicines that cause sedation such as anxiolytics (e.g., alprazolam, diazepam) and hypnotics (e.g., zolpidem) may potentiate drowsiness.

Paracetamol, when used together with warfarin (a blood-thinning medicine), may cause an increase in the International Normalized Ratio (INR), which may serve as a sign of increased risk for bleeding. Paracetamol increases the anticoagulation effect of warfarin.

Medicines which stimulate the enzymes responsible for the metabolic activation of paracetamol such as medicines for convulsion (e.g., phenobarbital, phenytoin) may increase susceptibility to the harmful effects to the liver.

The absorption of paracetamol may be accelerated by metoclopramide or domperidone and reduced by cholestyramine.

Tell your doctor about other medicines you are taking (i.e., other medicines for cough, cold, allergy, pain, or fever, especially other paracetamol-containing products).

WHAT SHOULD YOU DO IF YOU MISS A DOSE?
If you miss a dose, just take the next dose if still needed for the condition being treated, and the subsequent doses at the recommended time or schedule (i.e., every 6 hours).

Do not double the dose.

HOW SHOULD YOU KEEP THIS MEDICINE?
Keep the product out of reach and sight of children.
Store at temeperatures not exceeding 30°C.

SIGNS AND SYMPTOMS OF OVERDOSAGE
Phenylephrine HCl
- Increased blood pressure
- Headache
- Seizures
- Palpitation
- Paresthesia (sensation of tingling, pricking, or numbness of the skin)
- Vomiting
- Cerebral hemorrhage (bleeding from a ruptured blood vessel in the brain)

Chlorphenamine maleate
- Extreme sleepiness or prolonged drowsiness
- Weakness
- Hallucination
- Convulsion
- Agitation
- Irritability
- Tachycardia
- Coma

Paracetamol
Overdosage of Paracetamol usually involves 4 phases with the following signs and symptoms:
I. Eating disorder, nausea, vomiting, malaise, and excessive sweating
II. Right upper abdominal pain or tenderness, liver enlargement which may be characterized by abdominal discomfort of "feeling full", elevated bilirubin and liver enzyme concentrations, prolongation of prothrombin time, and occasionally decreased urine output
III. Eating disorder, nausea, vomiting, and malaise recur and signs of liver failure (e.g., jaundice) and possibly kidney failure and cardiomyopathy (disorder of the heart muscle) may develop
IV. Recovery or progression to fatal complete liver failure

WHAT TO DO WHEN YOU HAVE TAKEN MORE THAN THE RECOMMENDED DOSAGE?
If you have taken more than the recommended dosage, consult a doctor or contact a poison control center right away, even if you seem well, because of the risk of delayed, serious liver damage. Quick medical attention is important for adults as well as for children even if you do not notice any signs or symptoms.

CARE THAT SHOULD BE TAKEN WHEN TAKING THIS MEDICINE
- **Liver Warning:** This product contains paracetamol. Severe liver damage may occur if:
  - An adult takes more than 4 g of Paracetamol in 24 hours, which is the maximum daily amount;
  - A child takes more than 5 doses in 24 hours, which is the maximum daily amount of Paracetamol;
  - Taken with other medicines containing Paracetamol (or Acetaminophen);
  - An adult has 3 or more alcoholic drinks everyday while using this product.
- **Allergy Alert:** Paracetamol may cause severe skin reactions. Symptoms may include:
  - Skin reddening
  - Blisters
  - Rash
- If a skin reaction occurs, stop use and seek medical help right away.
- Do not use with any other medicine containing phenylephrine, chlorphenamine or paracetamol (prescription or nonprescription). If you are not sure whether a medicine contains these three active ingredients, ask a doctor.
- Ask a doctor before use if you have liver or kidney disease.
- Ask a doctor before use if you are taking warfarin, a blood thinning medicine.
- May cause drowsiness; patients should observe caution while driving or performing other tasks requiring alertness.
- Do not take more than the recommended dose.
- Do not use after the expiry date on the label.

WHEN SHOULD YOU CONSULT YOUR DOCTOR?
Before taking this medication, tell your doctor if you have:
- High blood pressure or any type of heart problems
- Glaucoma
- Thyroid problems
- Diabetes
- Liver or kidney disease
- An enlarged prostate, bladder problems or difficulty urinating

Stop use and ask a doctor if:
- Fever gets worse or lasts more than 3 days
- New symptoms occur

ADVERSE DRUG REACTION REPORTING STATEMENT
For suspected adverse drug reaction, seek medical attention immediately and report to the FDA at www.fda.gov.ph AND Unilab at +632-8-UNILAB-1 (+632-8-864522-1) for Metro Manila or toll-free +1-800-10-UNILAB-1 for provinces, or e-mail productsafety@unilab.com.ph. By reporting undesirable effects, you can help provide more information on the safety of this medicine.

AVAILABILITY
Bioflu Suspension (DRP-13914) - in Amber Glass Bottle x 60 mL (Box of 1's)

DATE OF FIRST AUTHORIZATION: October 2012
DATE OF REVISION OF PATIENT INFORMATION LEAFLET (PIL): June 2023 (CCDS 17-020)

Manufactured by AMHERST LABORATORIES, INC.
UNILAB Pharma Campus, Barangay Mamplasan
Biñan, Laguna, Philippines
for UNILAB, Inc.
No. 66 United Street, Mandaluyong City
Metro Manila, Philippines

Trusted Quality Healthcare

UNILAB Consumer Care
8-UNILAB-1
8-864522-1
Provincial Toll Free: 1-800-10-8645221
Email: info@unilab.com.ph
Website: www.unilab.com.ph

Reg. IPOPHIL P300000040096

140 mm

98 mm

Bioflu (Phenylephrine Hydrochloride + Chlorphenamine Maleate + Paracetamol)

10 mg/2 mg/500 mg Film-Coated Tablet | Over-the-Counter (OTC) Drug | FDA Reg. No. DR-XY34482

Bioflu: Official FDA Philippines registration record

Brand name	Bioflu
Generic name	Phenylephrine Hydrochloride + Chlorphenamine Maleate + Paracetamol
Dosage strength and form	10 mg/2 mg/500 mg Film-Coated Tablet
Classification	Over-the-Counter (OTC) Drug
Pharmacologic category	Nasal Decongestant/Antihistamine/Analgesic/Antipyretic
Registration number	DR-XY34482
Manufacturer	UNILAB, Inc.

Bioflu: OTC forms registered under the Bioflu brand (capsule availability)

OTC capsule form registered under the Bioflu brand: YES. 4 OTC / Household Remedy product(s) registered for this brand on the FDA Verification Portal:

Brand name	Dosage strength and form	Classification	Reg. No.
Bioflu	10 mg/2 mg/500 mg Film-Coated Tablet	Over-the-Counter (OTC) Drug	DR-XY34482
Bioflu	2.5 mg/ 500 mcg/ 125 mg per 5 mL Suspension	Over-the-Counter (OTC) Drug	DRP-13914
Bioflu	2.5 mg/500 mcg/125 mg per 5 mL Suspension	Over-the-Counter (OTC) Drug	DR-XY37402
Bioflu Complete	500 mg/10 mg/4 mg Capsule	Over-the-Counter (OTC) Drug	DR-XY46707

Bioflu: Official leaflet (Package Insert / Patient Information Leaflet)

No PI/PIL is published on the FDA Verification Portal for registration DR-XY34482 as of 2026-10-22. Indication and dosage are not covered by this record.

Source

FDA Philippines Verification Portal, https://verification.fda.gov.ph (search: "Bioflu"), retrieved 2026-10-22. Fields above are reproduced from the official registration record without modification.

Compiled for an academic RAG case study. Not medical advice. The portal states its information is for verification purposes. Verify current status on the portal before relying on it.

Bioflu Complete (Paracetamol + Phenylephrine Hydrochloride + Chlorphenamine Maleate)

500 mg/10 mg/4 mg Capsule | Over-the-Counter (OTC) Drug | FDA Reg. No. DR-XY46707

Bioflu Complete: Official FDA Philippines registration record

Brand name	Bioflu Complete
Generic name	Paracetamol + Phenylephrine Hydrochloride + Chlorphenamine Maleate
Dosage strength and form	500 mg/10 mg/4 mg Capsule
Classification	Over-the-Counter (OTC) Drug
Pharmacologic category	Nasal Decongestant/Antihistamine/Analgesic/Antipyretic
Registration number	DR-XY46707
Manufacturer	PT Medifarma Laboratories

Bioflu Complete: OTC forms registered under the Bioflu brand (capsule availability)

OTC capsule form registered under the Bioflu brand: YES. 4 OTC / Household Remedy product(s) registered for this brand on the FDA Verification Portal:

Brand name	Dosage strength and form	Classification	Reg. No.
Bioflu	10 mg/2 mg/500 mg Film-Coated Tablet	Over-the-Counter (OTC) Drug	DR-XY34482
Bioflu	2.5 mg/ 500 mcg/ 125 mg per 5 mL Suspension	Over-the-Counter (OTC) Drug	DRP-13914
Bioflu	2.5 mg/500 mcg/125 mg per 5 mL Suspension	Over-the-Counter (OTC) Drug	DR-XY37402
Bioflu Complete	500 mg/10 mg/4 mg Capsule	Over-the-Counter (OTC) Drug	DR-XY46707

Bioflu Complete: Official leaflet (Package Insert / Patient Information Leaflet)

No PI/PIL is published on the FDA Verification Portal for registration DR-XY46707 as of 2026-10-22. Indication and dosage are not covered by this record.

Source

FDA Philippines Verification Portal, https://verification.fda.gov.ph (search: "Bioflu"), retrieved 2026-10-22. Fields above are reproduced from the official registration record without modification.

Compiled for an academic RAG case study. Not medical advice. The portal states its information is for verification purposes. Verify current status on the portal before relying on it.

Claritin (Loratadine)

10 mg Tablet | Over-the-Counter (OTC) Drug | FDA Reg. No. DR-XY46593

Claritin: Official FDA Philippines registration record

Brand name	Claritin
Generic name	Loratadine
Dosage strength and form	10 mg Tablet
Classification	Over-the-Counter (OTC) Drug
Pharmacologic category	Antihistamine
Registration number	DR-XY46593
Manufacturer	PT. Bayer Indonesia

Claritin: OTC forms registered under the Claritin brand (capsule availability)

OTC capsule form registered under the Claritin brand: NO. 1 OTC / Household Remedy product(s) registered for this brand on the FDA Verification Portal:

Brand name	Dosage strength and form	Classification	Reg. No.
Claritin	10 mg Tablet	Over-the-Counter (OTC) Drug	DR-XY46593

Claritin: Official leaflet (Package Insert / Patient Information Leaflet)

An official leaflet for this registration is published on the FDA Verification Portal: PIL_01 (https://drive.google.com/file/d/1Gk7rVWMsF411j8O_uHooCB0y8cp_tZJM/view?usp=drivesdk). Its text (OCR) and the original leaflet pages follow.

Source

FDA Philippines Verification Portal, https://verification.fda.gov.ph (search: "Claritin"), retrieved 2026-10-22. Fields above are reproduced from the official registration record without modification.

Compiled for an academic RAG case study. Not medical advice. The portal states its information is for verification purposes. Verify current status on the portal before relying on it.

Claritin: Official leaflet text (OCR transcription)

Machine transcription of the official FDA leaflet for Reg. No. DR-XY46593. OCR may contain errors; the original leaflet pages are attached after this section.

Claritin leaflet PIL_01

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reactions have been reported very rarely during the post-marketing period are listed in the following table in order of decreasing seriousness of loratadine:

System Organ Class	Adverse Experience Term
Immune system disorders	Anaphylactic reaction including allergic edema and angioedema
Nervous system disorders	Dizziness, Seizures
Cardiac disorders	Tachycardia, Palpitation
Hepatobiliary disorders	Abnormal hepatic function
Skin and subcutaneous tissue disorders	Alopecia

10. What other medicine or food should be avoided while taking this Medicine?
Drug Interactions
Increase in plasma concentrations of this drug has been reported with concomitant use with ketoconazole, erythromycin or cimetidine in controlled clinical trials, but without clinically significant changes (including electrocardiographic).
Other drugs known to inhibit hepatic metabolism should be coadministered with caution.
Laboratory Interactions
The administration of loratadine should be discontinued at least 48 hours before skin tests since antihistamines may prevent or reduce otherwise positive reactions to dermal reactivity index.
Pediatric population
Interaction studies have only been performed in adults.
11. What should you do if you miss a dose?
Take the missed dose as soon as remembered but once daily only.
12. Signs and Symptoms of Overdose?
Intake of loratadine above the labeled dose increases the risk of anticholinergic symptoms. Somnolence, tachycardia, and headache have been reported with overdoses.
13. What to do when you have taken more than the recommended dosage?
In the event of overdosage, general symptomatic and supportive measures should be instituted and maintained for as long as necessary. Loratadine is not eliminated by hemodialysis. It is not known if loratadine is eliminated by peritoneal dialysis.
There is no information to indicate that abuse or dependence occurs with loratadine.
14. How should you keep this Medicine?
Store at temperatures not exceeding 30°C and protect from excessive moisture.
15. When should you consult your doctor?
If undesirable effects listed in number 9 are experienced.
Manufacturer: PT. Bayer Indonesia – Jalan Raya Jakarta Bogor KM 32, Cisalak, Kec. Sukmajaya, Kota Depok, Jawa Barat, Indonesia
Marketing Authorization Holder: Bayer Philippines, Inc. – 8th Floor, Science Hub Tower 1, Campus Avenue Corner Turin St., McKinley Hill Cyberpark, Pinagsama, Taguig City
If you want to report a product complaint or side effect, please contact your health care professional or the Philippine FDA at adr@fda.gov.ph.
Inquiries can also be directed to:
Bayer Philippines, Inc.
Taguig City, Philippines
E-mail: MedInfoCH-PH@bayer.com
drugsafety.philippines@bayer.com
Registration Number: DR-XY46593
Date of First Authorization: 04 September 2019
Date of Revision of PIL: March 2023 (Based on CCDS version 1 dated 20 February 2018)
Antihistamine
Long-Acting, Non-sedating Antihistamine
1. Name of the Product
Loratadine (Claritin®)
2. Description of the Product
Loratadine (Claritin®) tablet contains loratadine, a tri-cyclic antihistamine with selective peripheral H1-receptor antagonistic activity, which is currently used for the relief of symptoms associated with

allergic rhinitis, and urticaria and other dermatologic disorders.
3. What is in the Medicine?
Each tablet contains:
Loratadine (micronized)…….. 10 mg
List of excipients: corn starch, lactose and magnesium stearate.
4. Strength of the Medicine?
Tablet: 10 mg
5. What is this Medicine used for?
Loratadine (Claritin®) products are indicated for the relief of symptoms associated with allergic rhinitis, such as sneezing, nasal discharge (rhinorrhea) and itching, as well as ocular itching and burning, urticaria and other dermatologic disorders. Nasal and ocular signs and symptoms are relieved rapidly after oral administration.
6. How much and how often should you use this Medicine?
Method of Administration: Oral use.
Dosage Regimen:
Adults and Children 6 years of age and over with a body weight greater than 30 kg: One tablet [10 mg] once daily.
Additional information on special populations
Patients with severe hepatic impairment
Patients with severe liver impairment should be administered a lower initial dose because they may have reduced clearance of loratadine. An initial dose of 10 mg every other day is recommended for adults and children weighing more than 30 kg, and for children weighing 30 kg or less, 5 ml (5 mg) is recommended
7. When should you not take this Medicine?
Hypersensitivity to the active substance, metabolites or to any of the excipients.
8. Care that should be taken when taking this Medicine?
Warnings/Precautions
The product should be administered with caution in patients with liver impairment.
Effects on ability to drive or use machines
In clinical trials, loratadine had no or negligible influence on the ability to drive and use machines.
Fertility, Pregnancy and Lactation
Pregnancy
A large amount of data on pregnant women (more than 1000 exposed outcomes) indicate no malformative nor feto/ neonatal toxicity of loratadine. Animal studies do not indicate direct or indirect harmful effects with respect to reproductive toxicity. As a precautionary measure, it is preferable to avoid the use of loratadine during pregnancy.
Lactation
Loratadine is excreted in breast milk. The product is not recommended for use in breast-feeding women.
Fertility
There are no data available on male and female fertility.
9. Undesirable Effects of this Medicine
Summary of the safety profile
Loratadine has no clinically significant sedative properties at the recommended dosage.
In clinical trials, in a range of indications including Allergic Rhinitis (AR) and Chronic Idiopatic Urticaria (CIU), the most frequently reported adverse effects were headache, somnolence, fatigue and dry mouth and gastrointestinal disorders such as nausea, gastritis, and also allergic symptoms like rash. At the recommended dose, loratadine had no clinically significant sedative properties. In addition, other adverse
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148 mm

Loratadine
Claritin® 10 mg Tablet
Antihistamine

Long-Acting, Non-sedating Antihistamine

1. **Name of the Product**
   Loratadine (Claritin®)

2. **Description of the Product**
   Loratadine (Claritin®) tablet contains loratadine, a tri-cyclic antihistamine with selective peripheral H_1-receptor antagonistic activity, which is currently used for the relief of symptoms associated with allergic rhinitis, and urticaria and other dermatologic disorders.

3. **What is in the Medicine?**
   Each tablet contains:
   Loratadine (micronized)…….. 10 mg
   List of excipients: corn starch, lactose and magnesium stearate.

4. **Strength of the Medicine?**
   Tablet: 10 mg

5. **What is this Medicine used for?**
   Loratadine (Claritin®) products are indicated for the relief of symptoms associated with allergic rhinitis, such as sneezing, nasal discharge (rhinorrhea) and itching, as well as ocular itching and burning, urticaria and other dermatologic disorders. Nasal and ocular signs and symptoms are relieved rapidly after oral administration.

6. **How much and how often should you use this Medicine?**
   Method of Administration: Oral use.
   Dosage Regimen:
   Adults and Children 6 years of age and over with a body weight greater than 30 kg: One tablet [10 mg] once daily.
   Additional information on special populations
   Patients with severe hepatic impairment
   Patients with severe liver impairment should be administered a lower initial dose because they may have reduced clearance of loratadine. An initial dose of 10 mg every other day is recommended for adults and children weighing more than 30 kg, and for children weighing 30 kg or less, 5 ml (5 mg) is recommended

7. **When should you not take this Medicine?**
   Hypersensitivity to the active substance, metabolites or to any of the excipients.

8. **Care that should be taken when taking this Medicine?**
   Warnings/Precautions
   The product should be administered with caution in patients with liver impairment.

   Effects on ability to drive or use machines

   In clinical trials, loratadine had no or negligible influence on the ability to drive and use machines.

   Fertility, Pregnancy and Lactation

   Pregnancy

   A large amount of data on pregnant women (more than 1000 exposed outcomes) indicate no malformative nor feto/ neonatal toxicity of loratadine. Animal studies do not indicate direct or indirect harmful effects with respect to reproductive toxicity. As a precautionary measure, it is preferable to avoid the use of loratadine during pregnancy.

   Lactation

   Loratadine is excreted in breast milk. The product is not recommended for use in breast-feeding women.

   Fertility

   There are no data available on male and female fertility.

9. **Undesirable Effects of this Medicine**
   Summary of the safety profile
   Loratadine has no clinically significant sedative properties at the recommended dosage.
   In clinical trials, in a range of indications including Allergic Rhinitis (AR) and Chronic Idiopatic Urticaria (CIU), the most frequently reported adverse effects were headache, somnolence, fatigue and dry mouth and gastrointestinal disorders such as nausea, gastritis, and also allergic symptoms like rash. At the recommended dose, loratadine had no clinically significant sedative properties. In addition, other adverse

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reactions have been reported very rarely during the post-marketing period are listed in the following table in order of decreasing seriousness of loratadine:

System Organ Class	Adverse Experience Term
Immune system disorders	Anaphylactic reaction including allergic edema and angioedema
Nervous system disorders	Dizziness, Seizures
Cardiac disorders	Tachycardia, Palpitation
Hepatobiliary disorders	Abnormal hepatic function
Skin and subcutaneous tissue disorders	Alopecia

10. **What other medicine or food should be avoided while taking this Medicine?**
    Drug Interactions
    Increase in plasma concentrations of this drug has been reported with concomitant use with ketoconazole, erythromycin or cimetidine in controlled clinical trials, but without clinically significant changes (including electrocardiographic).
    Other drugs known to inhibit hepatic metabolism should be coadministered with caution.
    Laboratory Interactions
    The administration of loratadine should be discontinued at least 48 hours before skin tests since antihistamines may prevent or reduce otherwise positive reactions to dermal reactivity index.
    Pediatric population
    Interaction studies have only been performed in adults.

11. **What should you do if you miss a dose?**
    Take the missed dose as soon as remembered but once daily only.

12. **Signs and Symptoms of Overdose?**
    Intake of loratadine above the labeled dose increases the risk of anticholinergic symptoms. Somnolence, tachycardia, and headache have been reported with overdoses.

13. **What to do when you have taken more than the recommended dosage?**
    In the event of overdosage, general symptomatic and supportive measures should be instituted and maintained for as long as necessary. Loratadine is not eliminated by hemodialysis. It is not known if loratadine is eliminated by peritoneal dialysis.
    There is no information to indicate that abuse or dependence occurs with loratadine.

14. **How should you keep this Medicine?**
    Store at temperatures not exceeding 30°C and protect from excessive moisture.

15. **When should you consult your doctor?**
    If undesirable effects listed in number 9 are experienced.

Manufacturer: PT. Bayer Indonesia – Jalan Raya Jakarta Bogor KM 32, Cisalak, Kec. Sukmajaya, Kota Depok, Jawa Barat, Indonesia

Marketing Authorization Holder: Bayer Philippines, Inc. – 8th Floor, Science Hub Tower 1, Campus Avenue Corner Turin St., McKinley Hill Cyberpark, Pinagsama, Taguig City

If you want to report a product complaint or side effect, please contact your health care professional or the Philippine FDA at adr@fda.gov.ph.

Inquiries can also be directed to:
Bayer Philippines, Inc.
Taguig City, Philippines
E-mail: MedInfoCH-PH@bayer.com
drugsafety.philippines@bayer.com

Registration Number: DR-XY46593
Date of First Authorization: 04 September 2019
Date of Revision of PIL: March 2023 (Based on CCDS version 1 dated 20 February 2018)

Bayer

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Decolgen (Phenylephrine Hydrochloride + Chlorphenamine Maleate + Paracetamol)

10 mg/2 mg/500 mg Tablet | Over-the-Counter (OTC) Drug | FDA Reg. No. DR-XY29386

Decolgen: Official FDA Philippines registration record

Brand name	Decolgen
Generic name	Phenylephrine Hydrochloride + Chlorphenamine Maleate + Paracetamol
Dosage strength and form	10 mg/2 mg/500 mg Tablet
Classification	Over-the-Counter (OTC) Drug
Pharmacologic category	Nasal Decongestant/Antihistamine/Analgesic/Antipyretic
Registration number	DR-XY29386
Manufacturer	UNILAB, Inc.

Decolgen: OTC forms registered under the Decolgen brand (capsule availability)

OTC capsule form registered under the Decolgen brand: YES. 5 OTC / Household Remedy product(s) registered for this brand on the FDA Verification Portal:

Brand name	Dosage strength and form	Classification	Reg. No.
Decolgen	10 mg/2 mg/500 mg Tablet	Over-the-Counter (OTC) Drug	DR-XY29386
Decolgen	2 mg/10 mg/500 mg Tablet	Over-the-Counter (OTC) Drug	DR-XY41124
Decolgen Sinus Relief	200 mg / 25 mg / 2 mg Capsule	Over-the-Counter (OTC) Drug	DRP-16859-01
DecolgenR Max	200mg / 25 mg / 2mg Capsule	Over-the-Counter (OTC) Drug	DR-XY47703
Decolgen Forte	25 mg/2 mg/500 mg Tablet	Over-the-Counter (OTC) Drug	DR-XY45716

Decolgen: Official leaflet (Package Insert / Patient Information Leaflet)

No PI/PIL is published on the FDA Verification Portal for registration DR-XY29386 as of 2026-10-22. Indication and dosage are not covered by this record.

Source

FDA Philippines Verification Portal, https://verification.fda.gov.ph (search: "Decolgen"), retrieved 2026-10-22. Fields above are reproduced from the official registration record without modification.

Compiled for an academic RAG case study. Not medical advice. The portal states its information is for verification purposes. Verify current status on the portal before relying on it.

Decolgen (Chlorphenamine Maleate + Phenylephrine Hydrochloride + Paracetamol)

2 mg/10 mg/500 mg Tablet | Over-the-Counter (OTC) Drug | FDA Reg. No. DR-XY41124

Decolgen: Official FDA Philippines registration record

Brand name	Decolgen
Generic name	Chlorphenamine Maleate + Phenylephrine Hydrochloride + Paracetamol
Dosage strength and form	2 mg/10 mg/500 mg Tablet
Classification	Over-the-Counter (OTC) Drug
Pharmacologic category	H1-Receptor Antagonist (Antihistamine)/Sympathomimetic/Analgesic/Antipyretic
Registration number	DR-XY41124
Manufacturer	PT. Medifarma Laboratories

Decolgen: OTC forms registered under the Decolgen brand (capsule availability)

OTC capsule form registered under the Decolgen brand: YES. 5 OTC / Household Remedy product(s) registered for this brand on the FDA Verification Portal:

Brand name	Dosage strength and form	Classification	Reg. No.
Decolgen	10 mg/2 mg/500 mg Tablet	Over-the-Counter (OTC) Drug	DR-XY29386
Decolgen	2 mg/10 mg/500 mg Tablet	Over-the-Counter (OTC) Drug	DR-XY41124
Decolgen Sinus Relief	200 mg / 25 mg / 2 mg Capsule	Over-the-Counter (OTC) Drug	DRP-16859-01
DecolgenR Max	200mg / 25 mg / 2mg Capsule	Over-the-Counter (OTC) Drug	DR-XY47703
Decolgen Forte	25 mg/2 mg/500 mg Tablet	Over-the-Counter (OTC) Drug	DR-XY45716

Decolgen: Official leaflet (Package Insert / Patient Information Leaflet)

No PI/PIL is published on the FDA Verification Portal for registration DR-XY41124 as of 2026-10-22. Indication and dosage are not covered by this record.

Source

FDA Philippines Verification Portal, https://verification.fda.gov.ph (search: "Decolgen"), retrieved 2026-10-22. Fields above are reproduced from the official registration record without modification.

Compiled for an academic RAG case study. Not medical advice. The portal states its information is for verification purposes. Verify current status on the portal before relying on it.

Difflam (Benzydamine Hydrochloride + Cetylpyridinium Chloride)

3mg/1.33mg Lozenge (Honey and Lemon Flavor) | Over-the-Counter (OTC) Drug | FDA Reg. No. DR-XY46608

Difflam: Official FDA Philippines registration record

Brand name	Difflam
Generic name	Benzydamine Hydrochloride + Cetylpyridinium Chloride
Dosage strength and form	3mg/1.33mg Lozenge (Honey and Lemon Flavor)
Classification	Over-the-Counter (OTC) Drug
Pharmacologic category	Non-Steroidal Anti-Inflammatory Drug (NSAID) / Antiseptic
Registration number	DR-XY46608
Manufacturer	Unique Pharmaceuticals Laboratories (Division of J.B. Chemicals & Pharmaceuticals Ltd.)

Difflam: OTC forms registered under the Difflam brand (capsule availability)

OTC capsule form registered under the Difflam brand: NO. 11 OTC / Household Remedy product(s) registered for this brand on the FDA Verification Portal:

Brand name	Dosage strength and form	Classification	Reg. No.
Difflam AB	1.2 mg/600 mcg Lozenge (Orange Flavor)	Over-the-Counter (OTC) Drug	DR-XY49758
Difflam for Children	1.5 mg/mL (0.15% w/v) Oromucosal Spray (Mint Flavor)	Over-the-Counter (OTC) Drug	DR-XY49570
Difflam Gargle	10 mg/mL (1% w/v) Oral Solution (Fresh Mint Flavor)	Household Remedy (HR)	DRHR-2172
Difflam H-extraTM	2.4mg Lozenge (Purple Grape Flavor)	Over-the-Counter (OTC) Drug	DR-XY47789
Difflam-C	22.5 mg/18 mg per 15 mL Solution	Over-the-Counter (OTC) Drug	DR-XY41264
Difflam Forte	3 mg/mL Throat Spray	Over-the-Counter (OTC) Drug	DR-XY35579
Difflam Orange	3mg / 1.33mg Lozenge	Over-the-Counter (OTC) Drug	DR-XY40920
Difflam Eucalyptus & Menthol	3mg / 1.33mg Lozenge	Over-the-Counter (OTC) Drug	DR-XY40921
Difflam Lemon	3mg Lozenges	Over-the-Counter (OTC) Drug	DRHR-2315
Difflam	3mg/1.33mg Lozenge (Honey and Lemon Flavor)	Over-the-Counter (OTC) Drug	DR-XY46608
Difflam	3mg/1.33mg Lozenge (Raspberry Flavor)	Over-the-Counter (OTC) Drug	DR-XY46609

Difflam: Official leaflet (Package Insert / Patient Information Leaflet)

No PI/PIL is published on the FDA Verification Portal for registration DR-XY46608 as of 2026-10-22. Indication and dosage are not covered by this record.

Source

FDA Philippines Verification Portal, https://verification.fda.gov.ph (search: "Difflam"), retrieved 2026-10-22. Fields above are reproduced from the official registration record without modification.

Compiled for an academic RAG case study. Not medical advice. The portal states its information is for verification purposes. Verify current status on the portal before relying on it.

Difflam-C (Benzydamine hydrochloride + Chlorhexidine gluconate.)

22.5 mg/18 mg per 15 mL Solution | Over-the-Counter (OTC) Drug | FDA Reg. No. DR-XY41264

Difflam-C: Official FDA Philippines registration record

Brand name	Difflam-C
Generic name	Benzydamine hydrochloride + Chlorhexidine gluconate.
Dosage strength and form	22.5 mg/18 mg per 15 mL Solution
Classification	Over-the-Counter (OTC) Drug
Pharmacologic category	Non-Steroidal Anti-Inflammatory Drug (NSAID) / Antiseptic
Registration number	DR-XY41264
Manufacturer	Ensign Laboratories Pty. Ltd.

Difflam-C: OTC forms registered under the Difflam brand (capsule availability)

OTC capsule form registered under the Difflam brand: NO. 11 OTC / Household Remedy product(s) registered for this brand on the FDA Verification Portal:

Brand name	Dosage strength and form	Classification	Reg. No.
Difflam AB	1.2 mg/600 mcg Lozenge (Orange Flavor)	Over-the-Counter (OTC) Drug	DR-XY49758
Difflam for Children	1.5 mg/mL (0.15% w/v) Oromucosal Spray (Mint Flavor)	Over-the-Counter (OTC) Drug	DR-XY49570
Difflam Gargle	10 mg/mL (1% w/v) Oral Solution (Fresh Mint Flavor)	Household Remedy (HR)	DRHR-2172
Difflam H-extraTM	2.4mg Lozenge (Purple Grape Flavor)	Over-the-Counter (OTC) Drug	DR-XY47789
Difflam-C	22.5 mg/18 mg per 15 mL Solution	Over-the-Counter (OTC) Drug	DR-XY41264
Difflam Forte	3 mg/mL Throat Spray	Over-the-Counter (OTC) Drug	DR-XY35579
Difflam Orange	3mg / 1.33mg Lozenge	Over-the-Counter (OTC) Drug	DR-XY40920
Difflam Eucalyptus & Menthol	3mg / 1.33mg Lozenge	Over-the-Counter (OTC) Drug	DR-XY40921
Difflam Lemon	3mg Lozenges	Over-the-Counter (OTC) Drug	DRHR-2315
Difflam	3mg/1.33mg Lozenge (Honey and Lemon Flavor)	Over-the-Counter (OTC) Drug	DR-XY46608
Difflam	3mg/1.33mg Lozenge (Raspberry Flavor)	Over-the-Counter (OTC) Drug	DR-XY46609

Difflam-C: Official leaflet (Package Insert / Patient Information Leaflet)

No PI/PIL is published on the FDA Verification Portal for registration DR-XY41264 as of 2026-10-22. Indication and dosage are not covered by this record.

Source

FDA Philippines Verification Portal, https://verification.fda.gov.ph (search: "Difflam"), retrieved 2026-10-22. Fields above are reproduced from the official registration record without modification.

Compiled for an academic RAG case study. Not medical advice. The portal states its information is for verification purposes. Verify current status on the portal before relying on it.

Mucosolvan (Ambroxol Hydrochloride)

30 mg Tablet | Over-the-Counter (OTC) Drug | FDA Reg. No. DR-XY41618

Mucosolvan: Official FDA Philippines registration record

Brand name	Mucosolvan
Generic name	Ambroxol Hydrochloride
Dosage strength and form	30 mg Tablet
Classification	Over-the-Counter (OTC) Drug
Pharmacologic category	Mucolytic
Registration number	DR-XY41618
Manufacturer	Sanofi Vietnam Shareholding Company

Mucosolvan: OTC forms registered under the Mucosolvan brand (capsule availability)

OTC capsule form registered under the Mucosolvan brand: YES. 4 OTC / Household Remedy product(s) registered for this brand on the FDA Verification Portal:

Brand name	Dosage strength and form	Classification	Reg. No.
Mucosolvan	15 mg/5 mL Syrup	Over-the-Counter (OTC) Drug	DR-XY42649
Mucosolvan	30 mg Tablet	Over-the-Counter (OTC) Drug	DR-XY41618
Mucosolvan	30 mg/ 5 mL Syrup	Over-the-Counter (OTC) Drug	DR-XY42650
Mucosolvan 24 Hrs	75 mg Sustained Release Capsule	Over-the-Counter (OTC) Drug	DR-XY41726

Mucosolvan: Official leaflet (Package Insert / Patient Information Leaflet)

No PI/PIL is published on the FDA Verification Portal for registration DR-XY41618 as of 2026-10-22. Indication and dosage are not covered by this record.

Source

FDA Philippines Verification Portal, https://verification.fda.gov.ph (search: "Mucosolvan"), retrieved 2026-10-22. Fields above are reproduced from the official registration record without modification.

Compiled for an academic RAG case study. Not medical advice. The portal states its information is for verification purposes. Verify current status on the portal before relying on it.

Mucosolvan 24 Hrs (Ambroxol Hydrochloride)

75 mg Sustained Release Capsule | Over-the-Counter (OTC) Drug | FDA Reg. No. DR-XY41726

Mucosolvan 24 Hrs: Official FDA Philippines registration record

Brand name	Mucosolvan 24 Hrs
Generic name	Ambroxol Hydrochloride
Dosage strength and form	75 mg Sustained Release Capsule
Classification	Over-the-Counter (OTC) Drug
Pharmacologic category	Mucolytic
Registration number	DR-XY41726
Manufacturer	Sanofi Winthrop Industrie

Mucosolvan 24 Hrs: OTC forms registered under the Mucosolvan brand (capsule availability)

OTC capsule form registered under the Mucosolvan brand: YES. 4 OTC / Household Remedy product(s) registered for this brand on the FDA Verification Portal:

Brand name	Dosage strength and form	Classification	Reg. No.
Mucosolvan	15 mg/5 mL Syrup	Over-the-Counter (OTC) Drug	DR-XY42649
Mucosolvan	30 mg Tablet	Over-the-Counter (OTC) Drug	DR-XY41618
Mucosolvan	30 mg/ 5 mL Syrup	Over-the-Counter (OTC) Drug	DR-XY42650
Mucosolvan 24 Hrs	75 mg Sustained Release Capsule	Over-the-Counter (OTC) Drug	DR-XY41726

Mucosolvan 24 Hrs: Official leaflet (Package Insert / Patient Information Leaflet)

No PI/PIL is published on the FDA Verification Portal for registration DR-XY41726 as of 2026-10-22. Indication and dosage are not covered by this record.

Source

FDA Philippines Verification Portal, https://verification.fda.gov.ph (search: "Mucosolvan"), retrieved 2026-10-22. Fields above are reproduced from the official registration record without modification.

Compiled for an academic RAG case study. Not medical advice. The portal states its information is for verification purposes. Verify current status on the portal before relying on it.

Neozep Non-Drowsy (Phenylephrine Hydrochloride + Paracetamol)

10 mg / 500 mg Tablet | Over-the-Counter (OTC) Drug | FDA Reg. No. DR-XY30216

Neozep Non-Drowsy: Official FDA Philippines registration record

Brand name	Neozep Non-Drowsy
Generic name	Phenylephrine Hydrochloride + Paracetamol
Dosage strength and form	10 mg / 500 mg Tablet
Classification	Over-the-Counter (OTC) Drug
Pharmacologic category	Nasal Decongestant / Analgesic / Antipyretic
Registration number	DR-XY30216
Manufacturer	UNILAB, Inc.

Neozep Non-Drowsy: OTC forms registered under the Neozep brand (capsule availability)

OTC capsule form registered under the Neozep brand: NO. 8 OTC / Household Remedy product(s) registered for this brand on the FDA Verification Portal:

Brand name	Dosage strength and form	Classification	Reg. No.
Neozep Z+ Forte	10 mg / 2 mg / 325 mg / 10 mg (equivalent to 27.446 mg Zinc Sulfate Monohydrate) Tablet	Over-the-Counter (OTC) Drug	DR-XY47429
Neozep Z+ Non Drowsy	10 mg / 325 mg / 10 mg (equivalent to 27.446 mg Zinc Sulfate Monohydrate) Tablet	Over-the-Counter (OTC) Drug	DR-XY47428
Neozep Non-Drowsy	10 mg / 500 mg Tablet	Over-the-Counter (OTC) Drug	DR-XY30216
Neozep Forte	10 mg/ 2 mg/ 500 mg Tablet	Over-the-Counter (OTC) Drug	DR-XY29559
Neozep Forte	10 mg/ 2 mg/ 500 mg Tablet	Over-the-Counter (OTC) Drug	DR-XY41125
Neozep Non-Drowsy 325	10 mg/325 mg Tablet	Over-the-Counter (OTC) Drug	DR-XY46287
Neozep Non-Drowsy	10 mg/500 mg Tablet	Over-the-Counter (OTC) Drug	DR-XY42978
Neozep Forte 325	325 mg 10 mg / 2 mg Tablet	Over-the-Counter (OTC) Drug	DRP-14673

Neozep Non-Drowsy: Official leaflet (Package Insert / Patient Information Leaflet)

An official leaflet for this registration is published on the FDA Verification Portal: PIL_01 (https://drive.google.com/file/d/1hS_ZY3XTat_-Jtv7Lx_zbhXtjAMqjMEW/view?usp=drivesdk). Its text (OCR) and the original leaflet pages follow.

Source

FDA Philippines Verification Portal, https://verification.fda.gov.ph (search: "Neozep"), retrieved 2026-10-22. Fields above are reproduced from the official registration record without modification.

Compiled for an academic RAG case study. Not medical advice. The portal states its information is for verification purposes. Verify current status on the portal before relying on it.

Neozep Non-Drowsy: Official leaflet text (OCR transcription)

Machine transcription of the official FDA leaflet for Reg. No. DR-XY30216. OCR may contain errors; the original leaflet pages are attached after this section.

Neozep Non-Drowsy leaflet PIL_01

UNILAB
PHENYLEPHRINE HCl
PARACETAMOL
Neozep Non-Drowsy
Nasal Decongeslant/ Analgeslc / Anlipyrelic
FORMULATION
Eech tablot contains:
Phenylophrine hydrochlorido
10 mg
Paracetamol
500 mg
PRODUCT DESCRIPTION
Phenylephfine HCl + Paracelamol (Neozep Non-Drowsy) Tablet is a green. 0 662* x 0.300* elliptical-shaped, plain tablet, debéssed with "NEOZEP" on one side and plain on the olher side.
WHAT IS IN THE MEDICINE?
This medicine conlains Phenylephrine HCl and Paracetamol.
Phenyluphrine HCi, a nasal decongestant, clears obstrided air passages and nasal sinuses due to congestion making breathing easier. tt also reduces postnasal drip
Paracetamol is an effective fever reducer and pain reliever
breathing easier. It also reduces postnasal drip
STRENGTH OF THE MEDICINE See Formulation
WHAT IS THIS MEDICINE USED FOR?
This medicine is used for the relief of clogged noso, postnasal drip, headache, body aches, and fever associated with the common cold. sinusitis, flu. and olher minor respiratory tract iníections. Il also helps decongest sinus openings and passages.
HOW MUCH AND HOW OFTEN SHOULD YOU USE THIS MEDICINE?
Adults and Children 12 years and older: Orally. 1 tablet every 6 hours, or as reconinended by a doctor.
WHEN SHOULD YOU NOT TAKE THIS MEDICINE?
· If you are allergic lo any ingredient in the product
• If you have high blood pressure or severe hear disease uinless recommended by a doctor
• If you have anernla, kidney or tiver disease unless reconmended by a doctor
· If you are pregnant or breastfeeding
UNDESIRABLE EFFECTS
Phenylephrine HCi
Psychiatric disosdors: Anxicty (focling of uneasincss). insomnia/slecp dislurbance. ncrvousncss
Nervous system disordors: Dizziness. restlossness, Iremor (muscle shaking)
Cardlac disorders: Arrhythmia (irregular heart beal), palpitalion
Vascular disorders: Palior
Respiratory, thoracic and mediastinal disorders: Respiratory dislress (noisy, congesled breathing)
General disorders and administration slte conditions: Waakness
Invest/gations: Increased blood pressure
Paracetamol
Paracelamol, when laken within the recommended dose and duration of treatment, has low incidence of side effects.
Blood and lymphat/c system d/sordors: Changes in the number of white blood cells and platelets such as agranulocytosis (decrease In lhe number of granulocyles), leukopenia (decrease in the number of leukocytes). neutropenia (decrcase In the number of nettrophils), pancylopenia (deficiency of all cellular elements in Ihe blood). thrombocylopenia (docreaso in Ihe number of plalelets)
Immune system disorders: Alergic reactions which may cause difficulty in breathing. skin rash, angioedema (swelling of the lace or throal)
Gastrointestinal disordors: Minor stomach and intestinat disturbances
Skin and subcutaneous tissue disorders: Rare cases of serious skin reactlons (i.e., Slevens-Johnson syndrome. toxic epidermal necrolysls and acute generalized exanthematous pustulosis) whiclhi may include symptoms such as skin reddening, blisters or rash
WHAT OTHER MEDICINE OR FOOD SHOULD BE AVOIDED WHILE TAKING THIS MEDICINE?
halothane) because of lhe possibility for increased toxicty.
Concurrent administration with medicines for depression such as monoamine oxidase (MAO) inhibitors (e.g., selegiline. halothane) becauseof Ihe possibility for increased toxicrty.
severe increase in blood pressure that can lead to stroke).
Taking an alpha-adrenergic blocking agent (e.g., phenothiazine drugs, phentolamine) belore the administraton of this propranolol) and methyldopa may cause an increase in blood

Paracetamol, when used together with warfarin (a blood-thinning medicine). may cause an increase in the Inlemational Normalized Ratio (INR), which may serve as a sign of increased risk for bleeding. Paracetamol increases the anticoagulation effect of warfarin.
Medicines which stimulate the enzymes responsible for the melabolic activation of paracetamot such as medicnes for The absorplion of paracetamol may be accelerated by metoclopramide or domperidone and reduced by cholestyramine.
Tell your doctor aboul other medicines you are taking (ie., other medicines for cough, cold. alergy. pain, or fever, especialty other paracetamol-containing products)
WHAT SHOULD YOU DO IF YOU MISS A DOSE?
If you miss a dose, just take the next dose if still necded for the condition being treated, and the subscqucnt doses at lhe recommended time or schedule (i.e., every 6 houirs).
Do not double the dose.
HOW SHOULD YOU KEEP THIS MEDICINE?
Keep the product out of reach and sight of children. Store at temperatures not exceeding30°C.
SIGNS AND SYMPTOMS OF OVERDOSAGE
Phenylephrine HCl
·Increased blood pressure
• Headache
• Seizures
• Patpitation
• Paresthesia (sensation of tingling. pricking or numbness of the skin)
• Vomiting
• Cerebral hemorrhage (bleeding from a ruptured blood vessel in the brain)
Paracetamol
Overdosage of paracetamol usualy involves 4 phases with the following signs and synptoms.
1. Eating disorder. nausea, vomiting. malaise. and excessivc sweating
il Right upper abdominal pain or tendemess. liver enlargement which may be characterized by abdominal
1. Eating disorder. nausea, vomiling. malaise. and excessivc sweating
il Right upper abdominal pain or tenderness. liver enlargement which may be characterized by abdominal discomfort of "feeling fuil", elevated bilirubin and liver enzyme concentrations, prolongation of prolhronibin time. and occasionatly decreased unne output
III. Ealing disorder. nausea. voniiting. and malaise recur and signs of liver (ailure (e g., jaundice) and possibly kidney failure and cardionyopaty (disorder of tle lieart muscle) nay develop
IV. Recovery or progression lo fatal complete liver failture
WHAT TO DO WHEN YOU HAVE TAKEN MORE THAN THE RECOMMENDED DOSAGE?
tf you have laken more than the recommended dosage. consult a doctor or contacl a polson control center right away, even if you seem well, because of lhe nsk of delayed, serious liver damage. Ouick medical altention is important for adults as well as for children even if you do not nolice any signs or s ymptoms
CARE THAT SHOULD BE TAKEN WHEN TAKING THIS MEDICINE
• Liver Warning: This product contains paracelamol. Severe liver damage may occur if:
• An adult or child 12 years and older lakes more than 4 g of paracetamol in 24 hours, which is the maxirnum daily aniount:
• Taken with other medicines containing paracelamol (or acetaninophen);
• An adull has 3 or more alcoholic drinks everyday while using this producl.
• Do not use wth any other medicine containing paracalamol or phenylephnne (prescriptiont or nonprescnption).
If you are not sure whether a medicine contains these two active ingredients, ask a doctor.
· Ask a doctor before use if the patient has liver or kidney disease.
• Ask a doctor before use if the palient is laking warfann, a blood thinning medicine.
· Do not take more than the recommended dose.
• Do not use after lhe expiry dale on the label.
WHEN SHOULD YOU CONSULT YOUR DOCTOR?
• Before taking this medicalion. tell your doctor if you havo:
• High blood pressuro or any lypo of heart problems • Glaucoma
• Thyroid problems
· Diabetes
· Liver or kidney discase· An enlarged prostato, bladder problems or difficulty urinating
Stop uso and ask a doctor if:
• Fever gets worse or lasts more than 3 days
• New symploms occur
For suspected adverse drug reaction. seek medical attention immediately and report Io the FDA at www fda.gov.ph AND Unitab at (+ 632) 858-1 000 or productsafety@unitab.com.ph. By reporting undesirable effects, you can help provide more Information on the safety of this medicine.
AVAILABILITY: Box of 100 lablets (Aluminum/Clear PVC blister pack x 10's)-DR-XY30216
UNILAB
DATE OF REVISION OF PATIENT INFORMATION LEAFLET (PIL): Oclober 2019
DATE OF FIRST AUTHORIZATION: 18 NoV 2013

Manufactured by UNILAB. Inc.
Trusted Quality Healthcare
DATE OF FIRST AUTHORIZATION: 18 Nov 2013
Manufactured by UNILAB, Inc.
No 66 United Street, Mandaluyong City, Metro Manila, Philippines
Reg. IPOPHIL

UNILAB

PHENYLEPHRINE HCl PARACETAMOL

Neozep® Non-Drowsy

10 mg / 500 mg Tablet

Nasal Decongestant / Analgesic / Antipyretic

FORMULATION
Each tablet contains:
Phenylephrine hydrochloride ..... 10 mg
Paracetamol ..... 500 mg

PRODUCT DESCRIPTION
Phenylephrine HCl + Paracetamol (Neozep® Non-Drowsy) Tablet is a green, 0.662" x 0.300" elliptical-shaped, plain tablet, debossed with "NEOZEP" on one side and plain on the other side.

WHAT IS IN THE MEDICINE?
This medicine contains Phenylephrine HCl and Paracetamol.
Phenylephrine HCl, a nasal decongestant, clears obstructed air passages and nasal sinuses due to congestion making breathing easier. It also reduces postnasal drip.
Paracetamol is an effective fever reducer and pain reliever.

STRENGTH OF THE MEDICINE See Formulation

WHAT IS THIS MEDICINE USED FOR?
This medicine is used for the relief of clogged nose, postnasal drip, headache, body aches, and fever associated with the common cold, sinusitis, flu, and other minor respiratory tract infections. It also helps decongest sinus openings and passages.

HOW MUCH AND HOW OFTEN SHOULD YOU USE THIS MEDICINE?
Adults and Children 12 years and older: Orally, 1 tablet every 6 hours, or as recommended by a doctor.

WHEN SHOULD YOU NOT TAKE THIS MEDICINE?
- If you are allergic to any ingredient in the product
- If you have high blood pressure or severe heart disease unless recommended by a doctor
- If you have anemia, kidney or liver disease unless recommended by a doctor
- If you are pregnant or breastfeeding

UNDESIRABLE EFFECTS
Phenylephrine HCl
Psychiatric disorders: Anxiety (feeling of uneasiness), insomnia/sleep disturbance, nervousness
Nervous system disorders: Dizziness, restlessness, tremor (muscle shaking)
Cardiac disorders: Arrhythmia (irregular heart beat), palpitation
Vascular disorders: Pallor
Respiratory, thoracic and mediastinal disorders: Respiratory distress (noisy, congested breathing)
General disorders and administration site conditions: Weakness
Investigations: Increased blood pressure
Paracetamol
Paracetamol, when taken within the recommended dose and duration of treatment, has low incidence of side effects.
Blood and lymphatic system disorders: Changes in the number of white blood cells and platelets such as agranulocytosis (decrease in the number of granulocytes), leukopenia (decrease in the number of leukocytes), neutropenia (decrease in the number of neutrophils), pancytopenia (deficiency of all cellular elements in the blood), thrombocytopenia (decrease in the number of platelets)
Immune system disorders: Allergic reactions which may cause difficulty in breathing, skin rash, angioedema (swelling of the face or throat)
Gastrointestinal disorders: Minor stomach and intestinal disturbances
Skin and subcutaneous tissue disorders: Rare cases of serious skin reactions (i.e., Stevens-Johnson syndrome, toxic epidermal necrolysis and acute generalized exanthematous pustulosis) which may include symptoms such as skin reddening, blisters or rash

WHAT OTHER MEDICINE OR FOOD SHOULD BE AVOIDED WHILE TAKING THIS MEDICINE?
Do not use this product together with sympathomimetic agents (e.g., epinephrine) and general anesthetics (e.g., halothane) because of the possibility for increased toxicity.
Concurrent administration with medicines for depression such as monoamine oxidase (MAO) inhibitors (e.g., selegiline, moclobemide) and tricyclic antidepressants (e.g., amitriptyline, imipramine) may result in hypertensive crisis (sudden, severe increase in blood pressure that can lead to stroke).

Taking an alpha-adrenergic blocking agent (e.g., phenothiazine drugs, phentolamine) before the administration of this product may decrease the effect of phenylephrine. Concomitant use with beta-adrenergic blocking agents (e.g., propranolol) and methyldopa may cause an increase in blood
Paracetamol, when used together with warfarin (a blood-thinning medicine), may cause an increase in the International Normalized Ratio (INR), which may serve as a sign of increased risk for bleeding. Paracetamol increases the anticoagulation effect of warfarin.
Medicines which stimulate the enzymes responsible for the metabolic activation of paracetamol such as medicines for convulsion (e.g., phenobarbital, phenytoin) may increase susceptibility to the harmful effects to the liver.
The absorption of paracetamol may be accelerated by metoclopramide or domperidone and reduced by cholestyramine.
Tell your doctor about other medicines you are taking (i.e., other medicines for cough, cold, allergy, pain, or fever, especially other paracetamol-containing products).

WHAT SHOULD YOU DO IF YOU MISS A DOSE?
If you miss a dose, just take the next dose if still needed for the condition being treated, and the subsequent doses at the recommended time or schedule (i.e., every 6 hours).
Do not double the dose.

HOW SHOULD YOU KEEP THIS MEDICINE?
Keep the product out of reach and sight of children. Store at temperatures not exceeding 30°C.

SIGNS AND SYMPTOMS OF OVERDOSAGE
Phenylephrine HCl
- Increased blood pressure
- Headache
- Seizures
- Palpitation
- Paresthesia (sensation of tingling, pricking or numbness of the skin)
- Vomiting
- Cerebral hemorrhage (bleeding from a ruptured blood vessel in the brain)

Paracetamol
Overdosage of paracetamol usually involves 4 phases with the following signs and symptoms.
I. Eating disorder, nausea, vomiting, malaise, and excessive sweating
II. Right upper abdominal pain or tenderness, liver enlargement which may be characterized by abdominal discomfort of "feeling full", elevated bilirubin and liver enzyme concentrations, prolongation of prothrombin time, and occasionally decreased urine output
III. Eating disorder, nausea, vomiting, and malaise recur and signs of liver failure (e.g., jaundice) and possibly kidney failure and cardiomyopathy (disorder of the heart muscle) may develop
IV. Recovery or progression to fatal complete liver failure

WHAT TO DO WHEN YOU HAVE TAKEN MORE THAN THE RECOMMENDED DOSAGE?
If you have taken more than the recommended dosage, consult a doctor or contact a poison control center right away, even if you seem well, because of the risk of delayed, serious liver damage. Quick medical attention is important for adults as well as for children even if you do not notice any signs or symptoms.

CARE THAT SHOULD BE TAKEN WHEN TAKING THIS MEDICINE
- **Liver Warning:** This product contains paracetamol. Severe liver damage may occur if:
  - An adult or child 12 years and older takes more than 4 g of paracetamol in 24 hours, which is the maximum daily amount;
  - Taken with other medicines containing paracetamol (or acetaminophen);
  - An adult has 3 or more alcoholic drinks everyday while using this product.
- Do not use with any other medicine containing paracetamol or phenylephrine (prescription or nonprescription). If you are not sure whether a medicine contains these two active ingredients, ask a doctor.
- Ask a doctor before use if the patient has liver or kidney disease.
- Ask a doctor before use if the patient is taking warfarin, a blood thinning medicine.
- Do not take more than the recommended dose.
- Do not use after the expiry date on the label.

WHEN SHOULD YOU CONSULT YOUR DOCTOR?
- Before taking this medication, tell your doctor if you have:
  - High blood pressure or any type of heart problems
  - Glaucoma
  - Thyroid problems
  - Diabetes
  - Liver or kidney disease
  - An enlarged prostate, bladder problems or difficulty urinating

Stop use and ask a doctor if:
- Fever gets worse or lasts more than 3 days
- New symptoms occur

For suspected adverse drug reaction, seek medical attention immediately and report to the FDA at www.fda.gov.ph AND Unilab at (+632) 858-1000 or productsafety@unilab.com.ph. By reporting undesirable effects, you can help provide more information on the safety of this medicine.

AVAILABILITY: Box of 100 tablets (Aluminum/Clear PVC blister pack x 10's)-DR-XY30216

DATE OF REVISION OF PATIENT INFORMATION LEAFLET (PIL): October 2019
DATE OF FIRST AUTHORIZATION: 18 Nov 2013

Manufactured by **UNILAB, Inc.**
No. 66 United Street, Mandaluyong City, Metro Manila, Philippines

UNILAB
Trusted Quality Healthcare

Reg. iPOPHIL

PANTONE 3425 C

Neozep Forte (Phenylephrine hydrochloride + Chlorphenamine maleate + Paracetamol)

10 mg/ 2 mg/ 500 mg Tablet | Over-the-Counter (OTC) Drug | FDA Reg. No. DR-XY29559

Neozep Forte: Official FDA Philippines registration record

Brand name	Neozep Forte
Generic name	Phenylephrine hydrochloride + Chlorphenamine maleate + Paracetamol
Dosage strength and form	10 mg/ 2 mg/ 500 mg Tablet
Classification	Over-the-Counter (OTC) Drug
Pharmacologic category	Nasal Decongestant/ Antihistamine/ Analgesic-Antipyretic
Registration number	DR-XY29559
Manufacturer	UNILAB, Inc.

Neozep Forte: OTC forms registered under the Neozep brand (capsule availability)

OTC capsule form registered under the Neozep brand: NO. 8 OTC / Household Remedy product(s) registered for this brand on the FDA Verification Portal:

Brand name	Dosage strength and form	Classification	Reg. No.
Neozep Z+ Forte	10 mg / 2 mg / 325 mg / 10 mg (equivalent to 27.446 mg Zinc Sulfate Monohydrate) Tablet	Over-the-Counter (OTC) Drug	DR-XY47429
Neozep Z+ Non Drowsy	10 mg / 325 mg / 10 mg (equivalent to 27.446 mg Zinc Sulfate Monohydrate) Tablet	Over-the-Counter (OTC) Drug	DR-XY47428
Neozep Non-Drowsy	10 mg / 500 mg Tablet	Over-the-Counter (OTC) Drug	DR-XY30216
Neozep Forte	10 mg/ 2 mg/ 500 mg Tablet	Over-the-Counter (OTC) Drug	DR-XY29559
Neozep Forte	10 mg/ 2 mg/ 500 mg Tablet	Over-the-Counter (OTC) Drug	DR-XY41125
Neozep Non-Drowsy 325	10 mg/325 mg Tablet	Over-the-Counter (OTC) Drug	DR-XY46287
Neozep Non-Drowsy	10 mg/500 mg Tablet	Over-the-Counter (OTC) Drug	DR-XY42978
Neozep Forte 325	325 mg 10 mg / 2 mg Tablet	Over-the-Counter (OTC) Drug	DRP-14673

Neozep Forte: Official leaflet (Package Insert / Patient Information Leaflet)

No PI/PIL is published on the FDA Verification Portal for registration DR-XY29559 as of 2026-10-22. Indication and dosage are not covered by this record.

Source

FDA Philippines Verification Portal, https://verification.fda.gov.ph (search: "Neozep"), retrieved 2026-10-22. Fields above are reproduced from the official registration record without modification.

Compiled for an academic RAG case study. Not medical advice. The portal states its information is for verification purposes. Verify current status on the portal before relying on it.

Neozep Z+ Forte (Phenylephrine Hydrochloride + Chlorphenamine Maleate + Paracetamol + Zinc)

10 mg / 2 mg / 325 mg / 10 mg (equivalent to 27.446 mg Zinc Sulfate Monohydrate) Tablet | Over-the-Counter (OTC) Drug | FDA Reg. No. DR-XY47429

Neozep Z+ Forte: Official FDA Philippines registration record

Brand name	Neozep Z+ Forte
Generic name	Phenylephrine Hydrochloride + Chlorphenamine Maleate + Paracetamol + Zinc
Dosage strength and form	10 mg / 2 mg / 325 mg / 10 mg (equivalent to 27.446 mg Zinc Sulfate Monohydrate) Tablet
Classification	Over-the-Counter (OTC) Drug
Pharmacologic category	Nasal Decongestant / Antihistamine / Analgesic - Antipyretic / Mineral
Registration number	DR-XY47429
Manufacturer	PT Medifarma Laboratories

Neozep Z+ Forte: OTC forms registered under the Neozep brand (capsule availability)

OTC capsule form registered under the Neozep brand: NO. 8 OTC / Household Remedy product(s) registered for this brand on the FDA Verification Portal:

Brand name	Dosage strength and form	Classification	Reg. No.
Neozep Z+ Forte	10 mg / 2 mg / 325 mg / 10 mg (equivalent to 27.446 mg Zinc Sulfate Monohydrate) Tablet	Over-the-Counter (OTC) Drug	DR-XY47429
Neozep Z+ Non Drowsy	10 mg / 325 mg / 10 mg (equivalent to 27.446 mg Zinc Sulfate Monohydrate) Tablet	Over-the-Counter (OTC) Drug	DR-XY47428
Neozep Non-Drowsy	10 mg / 500 mg Tablet	Over-the-Counter (OTC) Drug	DR-XY30216
Neozep Forte	10 mg/ 2 mg/ 500 mg Tablet	Over-the-Counter (OTC) Drug	DR-XY29559
Neozep Forte	10 mg/ 2 mg/ 500 mg Tablet	Over-the-Counter (OTC) Drug	DR-XY41125
Neozep Non-Drowsy 325	10 mg/325 mg Tablet	Over-the-Counter (OTC) Drug	DR-XY46287
Neozep Non-Drowsy	10 mg/500 mg Tablet	Over-the-Counter (OTC) Drug	DR-XY42978
Neozep Forte 325	325 mg 10 mg / 2 mg Tablet	Over-the-Counter (OTC) Drug	DRP-14673

Neozep Z+ Forte: Official leaflet (Package Insert / Patient Information Leaflet)

No PI/PIL is published on the FDA Verification Portal for registration DR-XY47429 as of 2026-10-22. Indication and dosage are not covered by this record.

Source

FDA Philippines Verification Portal, https://verification.fda.gov.ph (search: "Neozep"), retrieved 2026-10-22. Fields above are reproduced from the official registration record without modification.

Compiled for an academic RAG case study. Not medical advice. The portal states its information is for verification purposes. Verify current status on the portal before relying on it.

Robitussin (Guafenesin)

200 mg Softgel Capsule | Over-the-Counter (OTC) Drug | FDA Reg. No. DR-XY28913

Robitussin: Official FDA Philippines registration record

Brand name	Robitussin
Generic name	Guafenesin
Dosage strength and form	200 mg Softgel Capsule
Classification	Over-the-Counter (OTC) Drug
Pharmacologic category	Expectorant
Registration number	DR-XY28913
Manufacturer	Catalent Australia Pty. Ltd.

Robitussin: OTC forms registered under the Robitussin brand (capsule availability)

OTC capsule form registered under the Robitussin brand: YES. 2 OTC / Household Remedy product(s) registered for this brand on the FDA Verification Portal:

Brand name	Dosage strength and form	Classification	Reg. No.
Robitussin DM	15 mg/100 mg per 5 mL Syrup	Over-the-Counter (OTC) Drug	DR-8402-L
Robitussin	200 mg Softgel Capsule	Over-the-Counter (OTC) Drug	DR-XY28913

Robitussin: Official leaflet (Package Insert / Patient Information Leaflet)

No PI/PIL is published on the FDA Verification Portal for registration DR-XY28913 as of 2026-10-22. Indication and dosage are not covered by this record.

Source

FDA Philippines Verification Portal, https://verification.fda.gov.ph (search: "Robitussin"), retrieved 2026-10-22. Fields above are reproduced from the official registration record without modification.

Compiled for an academic RAG case study. Not medical advice. The portal states its information is for verification purposes. Verify current status on the portal before relying on it.

Robitussin DM (Dextromethorphan Hydrobromide + Guaifenesin)

15 mg/100 mg per 5 mL Syrup | Over-the-Counter (OTC) Drug | FDA Reg. No. DR-8402-L

Robitussin DM: Official FDA Philippines registration record

Brand name	Robitussin DM
Generic name	Dextromethorphan Hydrobromide + Guaifenesin
Dosage strength and form	15 mg/100 mg per 5 mL Syrup
Classification	Over-the-Counter (OTC) Drug
Pharmacologic category	Antitussive/Expectorant
Registration number	DR-8402-L
Manufacturer	Interphil Laboratories, Inc.

Robitussin DM: OTC forms registered under the Robitussin brand (capsule availability)

OTC capsule form registered under the Robitussin brand: YES. 2 OTC / Household Remedy product(s) registered for this brand on the FDA Verification Portal:

Brand name	Dosage strength and form	Classification	Reg. No.
Robitussin DM	15 mg/100 mg per 5 mL Syrup	Over-the-Counter (OTC) Drug	DR-8402-L
Robitussin	200 mg Softgel Capsule	Over-the-Counter (OTC) Drug	DR-XY28913

Robitussin DM: Official leaflet (Package Insert / Patient Information Leaflet)

No PI/PIL is published on the FDA Verification Portal for registration DR-8402-L as of 2026-10-22. Indication and dosage are not covered by this record.

Source

FDA Philippines Verification Portal, https://verification.fda.gov.ph (search: "Robitussin"), retrieved 2026-10-22. Fields above are reproduced from the official registration record without modification.

Compiled for an academic RAG case study. Not medical advice. The portal states its information is for verification purposes. Verify current status on the portal before relying on it.

Sinecod Forte (Butamirate Citrate)

50 mg Sustained-Release Tablet | Over-the-Counter (OTC) Drug | FDA Reg. No. DRP-8129

Sinecod Forte: Official FDA Philippines registration record

Brand name	Sinecod Forte
Generic name	Butamirate Citrate
Dosage strength and form	50 mg Sustained-Release Tablet
Classification	Over-the-Counter (OTC) Drug
Pharmacologic category	Cough Supressant
Registration number	DRP-8129
Manufacturer	LLOYD LABORATORIES, INC.

Sinecod Forte: OTC forms registered under the Sinecod brand (capsule availability)

OTC capsule form registered under the Sinecod brand: NO. 2 OTC / Household Remedy product(s) registered for this brand on the FDA Verification Portal:

Brand name	Dosage strength and form	Classification	Reg. No.
Sinecod Forte	50 mg Sustained-Release Tablet	Over-the-Counter (OTC) Drug	DRP-8129
Sinecod Forte	7.5 mg/5 mL Syrup	Over-the-Counter (OTC) Drug	DR-XY25225

Sinecod Forte: Official leaflet (Package Insert / Patient Information Leaflet)

No PI/PIL is published on the FDA Verification Portal for registration DRP-8129 as of 2026-10-22. Indication and dosage are not covered by this record.

Source

FDA Philippines Verification Portal, https://verification.fda.gov.ph (search: "Sinecod"), retrieved 2026-10-22. Fields above are reproduced from the official registration record without modification.

Compiled for an academic RAG case study. Not medical advice. The portal states its information is for verification purposes. Verify current status on the portal before relying on it.

Sinecod Forte (Butamirate Citrate)

7.5 mg/5 mL Syrup | Over-the-Counter (OTC) Drug | FDA Reg. No. DR-XY25225

Sinecod Forte: Official FDA Philippines registration record

Brand name	Sinecod Forte
Generic name	Butamirate Citrate
Dosage strength and form	7.5 mg/5 mL Syrup
Classification	Over-the-Counter (OTC) Drug
Pharmacologic category	Antitussive
Registration number	DR-XY25225
Manufacturer	INTERPHIL LABORATORIES, INC.

Sinecod Forte: OTC forms registered under the Sinecod brand (capsule availability)

OTC capsule form registered under the Sinecod brand: NO. 2 OTC / Household Remedy product(s) registered for this brand on the FDA Verification Portal:

Brand name	Dosage strength and form	Classification	Reg. No.
Sinecod Forte	50 mg Sustained-Release Tablet	Over-the-Counter (OTC) Drug	DRP-8129
Sinecod Forte	7.5 mg/5 mL Syrup	Over-the-Counter (OTC) Drug	DR-XY25225

Sinecod Forte: Official leaflet (Package Insert / Patient Information Leaflet)

No PI/PIL is published on the FDA Verification Portal for registration DR-XY25225 as of 2026-10-22. Indication and dosage are not covered by this record.

Source

FDA Philippines Verification Portal, https://verification.fda.gov.ph (search: "Sinecod"), retrieved 2026-10-22. Fields above are reproduced from the official registration record without modification.

Compiled for an academic RAG case study. Not medical advice. The portal states its information is for verification purposes. Verify current status on the portal before relying on it.

Sinutab Cold Day & Night (Paracetamol + Phenylephrine Hydrochloride (Day Tablet) and Paracetamol + Phenylephrine Hydrochloride + Chlorpheniramine Maleate (Night Tablet))

500mg/5mg Film-coated Tablet (Day Tablet); 500mg/5mg / 2mg Film-coated Tablet (Night Tablet) Film-Coated Tablet | Over-the-Counter (OTC) Drug | FDA Reg. No. DR-XY48369

Sinutab Cold Day & Night: Official FDA Philippines registration record

Brand name	Sinutab Cold Day & Night
Generic name	Paracetamol + Phenylephrine Hydrochloride (Day Tablet) and Paracetamol + Phenylephrine Hydrochloride + Chlorpheniramine Maleate (Night Tablet)
Dosage strength and form	500mg/5mg Film-coated Tablet (Day Tablet); 500mg/5mg / 2mg Film-coated Tablet (Night Tablet) Film-Coated Tablet
Classification	Over-the-Counter (OTC) Drug
Pharmacologic category	Analgesic/Antipyretic/Nasal Decongestant; Analgesic/Antipyretic/Nasal Decongestant/Antihistamine
Registration number	DR-XY48369
Manufacturer	Probiotec Pharma Pty Limited

Sinutab Cold Day & Night: OTC forms registered under the Sinutab brand (capsule availability)

OTC capsule form registered under the Sinutab brand: NO. 1 OTC / Household Remedy product(s) registered for this brand on the FDA Verification Portal:

Brand name	Dosage strength and form	Classification	Reg. No.
Sinutab Cold Day & Night	500mg/5mg Film-coated Tablet (Day Tablet); 500mg/5mg / 2mg Film-coated Tablet (Night Tablet) Film-Coated Tablet	Over-the-Counter (OTC) Drug	DR-XY48369

Sinutab Cold Day & Night: Official leaflet (Package Insert / Patient Information Leaflet)

No PI/PIL is published on the FDA Verification Portal for registration DR-XY48369 as of 2026-10-22. Indication and dosage are not covered by this record.

Source

FDA Philippines Verification Portal, https://verification.fda.gov.ph (search: "Sinutab"), retrieved 2026-10-22. Fields above are reproduced from the official registration record without modification.

Compiled for an academic RAG case study. Not medical advice. The portal states its information is for verification purposes. Verify current status on the portal before relying on it.

Solmux (Carbocisteine)

500 mg Capsule | Household Remedy (HR) | FDA Reg. No. DRHR-1965

Solmux: Official FDA Philippines registration record

Brand name	Solmux
Generic name	Carbocisteine
Dosage strength and form	500 mg Capsule
Classification	Household Remedy (HR)
Pharmacologic category	Mucolytic
Registration number	DRHR-1965
Manufacturer	UNILAB, Inc.

Solmux: OTC forms registered under the Solmux brand (capsule availability)

OTC capsule form registered under the Solmux brand: YES. 4 OTC / Household Remedy product(s) registered for this brand on the FDA Verification Portal:

Brand name	Dosage strength and form	Classification	Reg. No.
Solmux Throat	1.5mg/1.2 mg per mL Oromucosal Spray	Over-the-Counter (OTC) Drug	DR-XY48921
Solmux TF	100 mg/5 mL Syrup	Household Remedy (HR)	DRHR-1839
Solmux TF	200 mg/5 mL Syrup	Household Remedy (HR)	DRHR-1838
Solmux	500 mg Capsule	Household Remedy (HR)	DRHR-1965

Solmux: Official leaflet (Package Insert / Patient Information Leaflet)

An official leaflet for registration DRHR-2096, another registration of the same brand, strength and form is published on the FDA Verification Portal: PIL_01 (https://drive.google.com/file/d/1HgxSLuw28oXgBXmMOA2s7Q6YI511b0pu/view?usp=drivesdk). Its text (OCR) and the original leaflet pages follow.

Source

FDA Philippines Verification Portal, https://verification.fda.gov.ph (search: "Solmux"), retrieved 2026-10-22. Fields above are reproduced from the official registration record without modification.

Compiled for an academic RAG case study. Not medical advice. The portal states its information is for verification purposes. Verify current status on the portal before relying on it.

Solmux: Official leaflet text (OCR transcription)

Machine transcription of the official FDA leaflet for Reg. No. DRHR-1965. OCR may contain errors; the original leaflet pages are attached after this section.

Solmux leaflet PIL_01

Carbocisteine
Solmux
40 mg/mL Syrup (Oral Drops)
100 mg/5 mL Pediatric Syrup
500 mg/ 5 mL Solmux® Forte Adult Suspension
500 mg Capsule
MUCOLYTIC
FORMULATIONS
Each mL Syrup (Oral Drops) contains:
Carbocisteine .....
40 mg
Each 5 mL (1 teaspoonful) Pediatric Syrup contains:
..100 mg
….. 40 mg
Carbocisteine ....
Each 5 mL (1 teaspoonful) Forte Adult Suspension contains:
Carbocisteine ...
.500 mg
Each capsule contains:
Carbocisteine, PP
PRODUCT DESCRIPTIONS
Carbocisteine (Solmux®) 40 mg/mL Syrup (Oral Drops), 100 mg/5 mL Pediatric Syrup, are red, clear syrups with strawberry odor and sweet strawberry taste using the TasteRite® technology.
TasteRite® technology is a unique tastemasking system developed specifically for liquid dosage forms to improve the taste of the medicine.
odor, and strawberry taste; it uses sorbitol as sweetenér and does not contain sucrose or table sugar.
body.
WHAT ARE IN THE MEDICINES?
These medicines contain carbocisteine, a medicine that causes phlegm to be less thick and sticky and thus easier significantly reduced the ability of bacteria to adhere to human pharyngeal cells.
STRENGTH OF THE MEDICINES
See Formulations
WHAT ARE THESE MEDICINES USED FOR?
Carbocisteine (Solmux) is a mucolytic agent used to relieve cough characterized by excessive or sticky sputum or phlegm to help treat réspiratory tract disorders such as acute bronchitis. Carbocisteine also reduces the number of
WHAT ARE THESE MEDICINES OSED FOR?
exacerbations associated with chronic obstructive pulmonary disease (CoPD) and inhibits the adherence of bacteria and virus to human respiratory cells. Carbocisteine is also used in the treatment of otitis media with effusion (glue ear).
WHEN SHOULD YOU NOT TAKE THESE MEDICINES?
• If you are allergic to any ingredient in the products
• Avoid carbocisteine in patients with active stomach or intestinal ulcer
UNDESIRABLE EFFECTS
Carbocisteine is generally safe.
Nausea, stomach discomfort, diarrhea, and skin rashes are the most common undesirable effects.
Carbocisteine rarely causes dizziness, insomnia, headache, palpitations, mild lowering of blood glucose, dryness of mouth, abnormal heart rhythm (atrial fibrillation), and"gassiness".
HOW MUCH AND HOW OFTEN SHOULD YOU USE THESE MEDICINES?
Oral Dose of Carbocisteine (Solmux)
Every 6 hours
Every 8 hours
Age Group
40 mg/mL
100 mg/5 mL
500 mg/5 mL
500 mg
(Oral Drops)
Syrup
Pediatric
ForteAdult Suspension

Gapsule
Children
1.5 mL
(1 teaspoonful)
5 mL
(2 teaspoonsful)
10 mL
(3 teaspoonsful)
15 mL
Adults and
Children over
(1 teaspoonful)
5mL
1 capsule
12 years
Or, as recommended by a doctor.
WHAT OTHER MEDICINE OR FOOD SHOULD BE AVOIDED WHILE TAKING THESE MEDICINES?
Always tell your doctor if you are taking other medicines, including herbal medicines and food supplements.
WHAT SHOULD YOU DO IF YOU MISS A DOSE?
If you miss a dose, just take the next dose if still needed for the condition being treated, and the subsequent doses at the recommended time or schedule, i.e., every 8 hours (for adults and children 2 years and older) or every 6 hours (for children 1 to 2 years old). Do not double the dose.
HOW SHOULD YOU KEEP THESE MEDICINES?
Keep the product out of reach and sight of children. Store at temperatures not exceeding 30°C.
SIGNS AND SYMPTOMS OF OVERDOSAGE
Carbocisteine overdose is usually associated with stomach or intestinal discomfort.
WHAT TO DO IF YOU HAVE TAKEN MORE THAN THE RECOMMENDED DOSAGE?
If you have taken more than the recommended dosage, consult a doctor or contact a poison control center right away.
CARE THAT SHOULD BE TAKEN WHEN TAKING THESE MEDICINES?
• These medicines should be given with care in patients with a history of stomach or intestinal ulcer and gastrointestinal bleeding.
Do not take more than the recommended dose.
Do not use after the expiry date on the label.
WHEN SHOULD YOU CONSULT YOUR DOCTOR?
• If you are pregnant or breastfeeding, consult your doctor before using this product
• Stop taking the medicine and consult your doctor immediately if you develop any of the following:
• Skin rashes
• Stomach or intestinal bleeding
• If symptoms do not improve or new symptoms appear
00XXXXx
For suspected adverse drug reaction, seek medical attention immediately and report to FDA at www.fda.gov.ph AND Unilab at (+632) 858-1000 or productsafety@unilab.com.ph. By reporting undesirable effects, you can help provide informetion en the sefety of this medici
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more information on the safety of this medicine.
AVAILABILITY, REGISTRATIÓN NO., DATE OF FIRST AUTHORIZATION
Carbocisteine (Solmux®) 40 mg/mL Syrup (Oral Drops) available in Amber Glass Bottle x 15 mL (Box of 1`s), DRHR-793, 07 February 2011
Carbocisteine (Solmux®) 100 mg/5 mL Pediatric Syrup, available in Amber Glass Bottle x 60 mL (Box of 1`s), Reg. IPOPHIL
DRHR-792.07 Februarv2011
Carbocisteine (Solmux® Forte) 500 mg / 5 mL Adult Suspension, available in 60 mL Amber Boston Bottle with Aluminum Pilfer Proof Cap (Box of 1 s), DRHR-720, 07 February 2011
Carbocisteine (Solmux®) 500 mg Capsule, available in Box x 4's and 100s (Flex foil strip x 4's), DRHR-2096, 18 November 2021
*Carbocisteine (Solmux®) 500 mg Capsule, available in Box x 4`s and 100s (Foil strip x 4s), DRHR-717, 27 June 2005
Date of Revision of Patient Information Leaflet: August 2018
Manufactured by AMHERST LABORATORIES, INC.
UNILAB
UNILAB Pharma Campus, Barangay Mamplasan, Biñan, Laguna, Philippines
for UNILAB, Inc., No. 66 United Street, Mandaluyong City, Metro Manila, Philippines
*Manufactured by UNILAB, Inc., No. 66 United Street, Mandaluyong City, Metro Manila, Philippines
Trusted Quality Healthcare

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Carbocisteine Solmux®

40 mg/mL Syrup (Oral Drops)
100 mg/5 mL Pediatric Syrup
500 mg / 5 mL Solmux® Forte Adult Suspension
500 mg Capsule

MUCOLYTIC

FORMULATIONS

Each mL Syrup (Oral Drops) contains:
Carbocisteine ..... 40 mg
Each 5 mL (1 teaspoonful) Pediatric Syrup contains:
Carbocisteine ..... 100 mg
Each 5 mL (1 teaspoonful) Forte Adult Suspension contains:
Carbocisteine ..... 500 mg
Each capsule contains:
Carbocisteine, PP ..... 500 mg

PRODUCT DESCRIPTIONS

Carbocisteine (Solmux®) 40 mg/mL Syrup (Oral Drops), 100 mg/5 mL Pediatric Syrup, are red, clear syrups with strawberry odor and sweet strawberry taste using the TasteRite® technology.
TasteRite® technology is a unique tastemasking system developed specifically for liquid dosage forms to improve the taste of the medicine.

Carbocisteine (Solmux® Forte) 500 mg/5 mL Adult Suspension is a pink, thick suspension with fruity strawberry odor, and strawberry taste; it uses sorbitol as sweetener and does not contain sucrose or table sugar.

Carbocisteine (Solmux®) 500 mg Capsule is a hard gelatin capsule with ivory opaque cap and rich yellow opaque body.

WHAT ARE IN THE MEDICINES?

These medicines contain carbocisteine, a medicine that causes phlegm to be less thick and sticky and thus easier to cough up and expel. Carbocisteine, in an *in vitro* study (outside the living organism), was shown to have significantly reduced the ability of bacteria to adhere to human pharyngeal cells.

STRENGTH OF THE MEDICINES

See Formulations

WHAT ARE THESE MEDICINES USED FOR?

Carbocisteine (Solmux) is a mucolytic agent used to relieve cough characterized by excessive or sticky sputum or phlegm to help treat respiratory tract disorders such as acute bronchitis. Carbocisteine also reduces the number of exacerbations associated with chronic obstructive pulmonary disease (COPD) and inhibits the adherence of bacteria and virus to human respiratory cells. Carbocisteine is also used in the treatment of otitis media with effusion (glue ear).

WHEN SHOULD YOU NOT TAKE THESE MEDICINES?

- If you are allergic to any ingredient in the products
- Avoid carbocisteine in patients with active stomach or intestinal ulcer

UNDESIRABLE EFFECTS

Carbocisteine is generally safe.

Nausea, stomach discomfort, diarrhea, and skin rashes are the most common undesirable effects.

Carbocisteine rarely causes dizziness, insomnia, headache, palpitations, mild lowering of blood glucose, dryness of mouth, abnormal heart rhythm (atrial fibrillation), and "gassiness".

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HOW MUCH AND HOW OFTEN SHOULD YOU USE THESE MEDICINES?

Age Group	Oral Dose of Carbocisteine (Solmux)			
	Every 6 hours	Every 8 hours		
	40 mg/mL Syrup (Oral Drops)	100 mg/5 mL Pediatric Syrup	500 mg/5 mL Forte Adult Suspension	500 mg Capsule
Children 1 to 2 years	1.5 mL	-	-	-
2 to 3 years	-	5 mL (1 teaspoonful)	-	-
4 to 7 years	-	10 mL (2 teaspoonsful)	-	-
8 to 12 years	-	15 mL (3 teaspoonsful)	-	-
Adults and Children over 12 years	-	-	5 mL (1 teaspoonful)	1 capsule
	Or, as recommended by a doctor.			

WHAT OTHER MEDICINE OR FOOD SHOULD BE AVOIDED WHILE TAKING THESE MEDICINES?

Always tell your doctor if you are taking other medicines, including herbal medicines and food supplements.

WHAT SHOULD YOU DO IF YOU MISS A DOSE?

If you miss a dose, just take the next dose if still needed for the condition being treated, and the subsequent doses at the recommended time or schedule, i.e., every 8 hours (for adults and children 2 years and older) or every 6 hours (for children 1 to 2 years old). Do not double the dose.

HOW SHOULD YOU KEEP THESE MEDICINES?

Keep the product out of reach and sight of children. Store at temperatures not exceeding 30ºC.

SIGNS AND SYMPTOMS OF OVERDOSAGE

Carbocisteine overdose is usually associated with stomach or intestinal discomfort.

WHAT TO DO IF YOU HAVE TAKEN MORE THAN THE RECOMMENDED DOSAGE?

If you have taken more than the recommended dosage, consult a doctor or contact a poison control center right away.

CARE THAT SHOULD BE TAKEN WHEN TAKING THESE MEDICINES?

- These medicines should be given with care in patients with a history of stomach or intestinal ulcer and gastrointestinal bleeding.
- Do not take more than the recommended dose.
- Do not use after the expiry date on the label.

WHEN SHOULD YOU CONSULT YOUR DOCTOR?

- If you are pregnant or breastfeeding, consult your doctor before using this product
- Stop taking the medicine and consult your doctor immediately if you develop any of the following:
  - Skin rashes
  - Stomach or intestinal bleeding
- If symptoms do not improve or new symptoms appear

For suspected adverse drug reaction, seek medical attention immediately and report to FDA at www.fda.gov.ph AND Unilab at (+632) 858-1000 or productsafety@unilab.com.ph. By reporting undesirable effects, you can help provide more information on the safety of this medicine.

AVAILABILITY, REGISTRATION NO., DATE OF FIRST AUTHORIZATION

Carbocisteine (Solmux®) 40 mg/mL Syrup (Oral Drops) available in Amber Glass Bottle x 15 mL (Box of 1`s), DRHR-793, 07 February 2011
Carbocisteine (Solmux®) 100 mg/5 mL Pediatric Syrup, available in Amber Glass Bottle x 60 mL (Box of 1`s), DRHR-792. 07 February 2011
Carbocisteine (Solmux® Forte) 500 mg / 5 mL Adult Suspension, available in 60 mL Amber Boston Bottle with Aluminum Pilfer Proof Cap (Box of 1`s), DRHR-720, 07 February 2011
Carbocisteine (Solmux®) 500 mg Capsule, available in Box x 4`s and 100`s (Flex foil strip x 4`s), DRHR-2096, 18 November 2021
*Carbocisteine (Solmux®) 500 mg Capsule, available in Box x 4`s and 100`s (Foil strip x 4`s), DRHR-717, 27 June 2005

Date of Revision of Patient Information Leaflet: August 2018

Manufactured by AMHERST LABORATORIES, INC.
UNILAB Pharma Campus, Barangay Mamplasan, Biñan, Laguna, Philippines
for **UNILAB, Inc.**, No. 66 United Street, Mandaluyong City, Metro Manila, Philippines

*Manufactured by **UNILAB, Inc.**, No. 66 United Street, Mandaluyong City, Metro Manila, Philippines

UNILAB
Trusted Quality Healthcare

Reg. (PO)PHL P3000000XXXXX

Solmux TF (Carbocisteine)

200 mg/5 mL Syrup | Household Remedy (HR) | FDA Reg. No. DRHR-1838

Solmux TF: Official FDA Philippines registration record

Brand name	Solmux TF
Generic name	Carbocisteine
Dosage strength and form	200 mg/5 mL Syrup
Classification	Household Remedy (HR)
Pharmacologic category	Mucolytic
Registration number	DRHR-1838
Manufacturer	Amherst Laboratories, Inc.

Solmux TF: OTC forms registered under the Solmux brand (capsule availability)

OTC capsule form registered under the Solmux brand: YES. 4 OTC / Household Remedy product(s) registered for this brand on the FDA Verification Portal:

Brand name	Dosage strength and form	Classification	Reg. No.
Solmux Throat	1.5mg/1.2 mg per mL Oromucosal Spray	Over-the-Counter (OTC) Drug	DR-XY48921
Solmux TF	100 mg/5 mL Syrup	Household Remedy (HR)	DRHR-1839
Solmux TF	200 mg/5 mL Syrup	Household Remedy (HR)	DRHR-1838
Solmux	500 mg Capsule	Household Remedy (HR)	DRHR-1965

Solmux TF: Official leaflet (Package Insert / Patient Information Leaflet)

No PI/PIL is published on the FDA Verification Portal for registration DRHR-1838 as of 2026-10-22. Indication and dosage are not covered by this record.

Source

FDA Philippines Verification Portal, https://verification.fda.gov.ph (search: "Solmux"), retrieved 2026-10-22. Fields above are reproduced from the official registration record without modification.

Compiled for an academic RAG case study. Not medical advice. The portal states its information is for verification purposes. Verify current status on the portal before relying on it.

Strepsils (Dichlorobenzyl Alcohol + Amylmetacresol)

1.2 mg/600 mcg Lozenge | Household Remedy | FDA Reg. No. DRHR-1455

Strepsils: Official FDA Philippines registration record

Brand name	Strepsils
Generic name	Dichlorobenzyl Alcohol + Amylmetacresol
Dosage strength and form	1.2 mg/600 mcg Lozenge
Classification	Household Remedy
Pharmacologic category	Antiseptic
Registration number	DRHR-1455
Manufacturer	Reckitt Benckiser Healthcare Manufcaturing (Thailand) Ltd.

Strepsils: OTC forms registered under the Strepsils brand (capsule availability)

OTC capsule form registered under the Strepsils brand: NO. 17 OTC / Household Remedy product(s) registered for this brand on the FDA Verification Portal:

Brand name	Dosage strength and form	Classification	Reg. No.
Strepsils Orange with Vitamin C	1.2 mg / 600 mcg / 100 mg Lozenge	Over-the-Counter (OTC) Drug	DRHR-1497
Strepsils	1.2 mg / 600 mcg Lozenge	Household Remedy (HR)	DRHR-1702
Strepsils Orange with Vitamin C	1.2 mg/ 600 mcg / 100 mg Lozenge	Over-the-Counter (OTC) Drug	DRHR-1558
Strepsils	1.2 mg/ 600 mcg Lozenge	Household Remedy	DRHR-1456
Strepsils	1.2 mg/ 600 mcg Lozenge	Household Remedy	DRHR-1527
Strepsils	1.2 mg/600 mcg Lozenge	Household Remedy	DRHR-1455
Strepsils	1.2 mg/600 mcg Lozenge	Household Remedy (HR)	DRHR-1643
Strepsils	1.2 mg/600 mcg Lozenge	Household Remedy (HR)	DRHR-1644
Strepsils	1.2 mg/600 mcg Lozenge	Over-the-Counter (OTC) Drug	DRHR-1557
Strepsils	1.2 mg/600 mcg Lozenge	Over-the-Counter (OTC) Drug	DRHR-1767
Strepsils Chesty Cough	15 mg lozenge	Over-the-Counter (OTC) Drug	DR-XY42567
Strepsils Max	2.4 mg Lozenge	Over-the-Counter (OTC) Drug	DR-XY40212
Strepsils Dry Cough	5 mg Lozenge	Household Remedy (HR)	DRHR-1636
Strepsils MaxPro	8.75 mg Lozenge (honey and lemon flavor)	Over-the-Counter (OTC) Drug	DR-XY45838
Strepsils MaxPro	8.75 mg Lozenge (honey and lemon flavor)	Over-the-Counter (OTC) Drug	DR-XY45839

Brand name	Dosage strength and form	Classification	Reg. No.
Strepsils MaxPro Direct	8.75 mg/0.5 mL Oromucosal Spray (Honey and Lemon Flavor)	Over-the-Counter (OTC) Drug	DR-XY49658
Strepsils MaxPro Direct	8.75 mg/0.5 mL Oromucosal Spray (Mint and Cherry Flavor)	Over-the-Counter (OTC) Drug	DR-XY49229

Strepsils: Official leaflet (Package Insert / Patient Information Leaflet)

No PI/PIL is published on the FDA Verification Portal for registration DRHR-1455 as of 2026-10-22. Indication and dosage are not covered by this record.

Source

FDA Philippines Verification Portal, https://verification.fda.gov.ph (search: "Strepsils"), retrieved 2026-10-22. Fields above are reproduced from the official registration record without modification.

Compiled for an academic RAG case study. Not medical advice. The portal states its information is for verification purposes. Verify current status on the portal before relying on it.

Strepsils Max (Hexylresorcinol)

2.4 mg Lozenge | Over-the-Counter (OTC) Drug | FDA Reg. No. DR-XY40212

Strepsils Max: Official FDA Philippines registration record

Brand name	Strepsils Max
Generic name	Hexylresorcinol
Dosage strength and form	2.4 mg Lozenge
Classification	Over-the-Counter (OTC) Drug
Pharmacologic category	Antiseptic
Registration number	DR-XY40212
Manufacturer	Reckitt Benckiser Healthcare International Ltd.

Strepsils Max: OTC forms registered under the Strepsils brand (capsule availability)

OTC capsule form registered under the Strepsils brand: NO. 17 OTC / Household Remedy product(s) registered for this brand on the FDA Verification Portal:

Brand name	Dosage strength and form	Classification	Reg. No.
Strepsils Orange with Vitamin C	1.2 mg / 600 mcg / 100 mg Lozenge	Over-the-Counter (OTC) Drug	DRHR-1497
Strepsils	1.2 mg / 600 mcg Lozenge	Household Remedy (HR)	DRHR-1702
Strepsils Orange with Vitamin C	1.2 mg/ 600 mcg / 100 mg Lozenge	Over-the-Counter (OTC) Drug	DRHR-1558
Strepsils	1.2 mg/ 600 mcg Lozenge	Household Remedy	DRHR-1456
Strepsils	1.2 mg/ 600 mcg Lozenge	Household Remedy	DRHR-1527
Strepsils	1.2 mg/600 mcg Lozenge	Household Remedy	DRHR-1455
Strepsils	1.2 mg/600 mcg Lozenge	Household Remedy (HR)	DRHR-1643
Strepsils	1.2 mg/600 mcg Lozenge	Household Remedy (HR)	DRHR-1644
Strepsils	1.2 mg/600 mcg Lozenge	Over-the-Counter (OTC) Drug	DRHR-1557
Strepsils	1.2 mg/600 mcg Lozenge	Over-the-Counter (OTC) Drug	DRHR-1767
Strepsils Chesty Cough	15 mg lozenge	Over-the-Counter (OTC) Drug	DR-XY42567
Strepsils Max	2.4 mg Lozenge	Over-the-Counter (OTC) Drug	DR-XY40212
Strepsils Dry Cough	5 mg Lozenge	Household Remedy (HR)	DRHR-1636
Strepsils MaxPro	8.75 mg Lozenge (honey and lemon flavor)	Over-the-Counter (OTC) Drug	DR-XY45838
Strepsils MaxPro	8.75 mg Lozenge (honey and lemon flavor)	Over-the-Counter (OTC) Drug	DR-XY45839

Brand name	Dosage strength and form	Classification	Reg. No.
Strepsils MaxPro Direct	8.75 mg/0.5 mL Oromucosal Spray (Honey and Lemon Flavor)	Over-the-Counter (OTC) Drug	DR-XY49658
Strepsils MaxPro Direct	8.75 mg/0.5 mL Oromucosal Spray (Mint and Cherry Flavor)	Over-the-Counter (OTC) Drug	DR-XY49229

Strepsils Max: Official leaflet (Package Insert / Patient Information Leaflet)

No PI/PIL is published on the FDA Verification Portal for registration DR-XY40212 as of 2026-10-22. Indication and dosage are not covered by this record.

Source

FDA Philippines Verification Portal, https://verification.fda.gov.ph (search: "Strepsils"), retrieved 2026-10-22. Fields above are reproduced from the official registration record without modification.

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Tuseran (Dextromethorphan hydrobromide + Phenylephrine hydrochloride + Paracetamol)

15 mg/10 mg/325 mg Capsule | Over-the-Counter (OTC) Drug | FDA Reg. No. DR-XY41123

Tuseran: Official FDA Philippines registration record

Brand name	Tuseran
Generic name	Dextromethorphan hydrobromide + Phenylephrine hydrochloride + Paracetamol
Dosage strength and form	15 mg/10 mg/325 mg Capsule
Classification	Over-the-Counter (OTC) Drug
Pharmacologic category	Antitussive/Nasal Decongestant/Analgesic/Antipyretic
Registration number	DR-XY41123
Manufacturer	PT Medifarma Laboratories

Tuseran: OTC forms registered under the Tuseran brand (capsule availability)

OTC capsule form registered under the Tuseran brand: YES. 3 OTC / Household Remedy product(s) registered for this brand on the FDA Verification Portal:

Brand name	Dosage strength and form	Classification	Reg. No.
Tuseran Forte	15 mg/10 mg/325 mg Capsule	Over-the-Counter (OTC) Drug	DR-XY29720
Tuseran	15 mg/10 mg/325 mg Capsule	Over-the-Counter (OTC) Drug	DR-XY41123
Tuseran Forte (Reformulated)	15 mg/25 mg/325 mg Capsule	Over-the-Counter (OTC) Drug	DR-XY16850

Tuseran: Official leaflet (Package Insert / Patient Information Leaflet)

No PI/PIL is published on the FDA Verification Portal for registration DR-XY41123 as of 2026-10-22. Indication and dosage are not covered by this record.

Source

FDA Philippines Verification Portal, https://verification.fda.gov.ph (search: "Tuseran"), retrieved 2026-10-22. Fields above are reproduced from the official registration record without modification.

Compiled for an academic RAG case study. Not medical advice. The portal states its information is for verification purposes. Verify current status on the portal before relying on it.

Tuseran Forte (Dextromethorphan hydrobromide + Phenylephrine hydrochloride + Paracetamol)

15 mg/10 mg/325 mg Capsule | Over-the-Counter (OTC) Drug | FDA Reg. No. DR-XY29720

Tuseran Forte: Official FDA Philippines registration record

Brand name	Tuseran Forte
Generic name	Dextromethorphan hydrobromide + Phenylephrine hydrochloride + Paracetamol
Dosage strength and form	15 mg/10 mg/325 mg Capsule
Classification	Over-the-Counter (OTC) Drug
Pharmacologic category	Antitussive / Nasal Decongestant / Analgesic-Antipyretic
Registration number	DR-XY29720
Manufacturer	UNILAB, Inc.

Tuseran Forte: OTC forms registered under the Tuseran brand (capsule availability)

OTC capsule form registered under the Tuseran brand: YES. 3 OTC / Household Remedy product(s) registered for this brand on the FDA Verification Portal:

Brand name	Dosage strength and form	Classification	Reg. No.
Tuseran Forte	15 mg/10 mg/325 mg Capsule	Over-the-Counter (OTC) Drug	DR-XY29720
Tuseran	15 mg/10 mg/325 mg Capsule	Over-the-Counter (OTC) Drug	DR-XY41123
Tuseran Forte (Reformulated)	15 mg/25 mg/325 mg Capsule	Over-the-Counter (OTC) Drug	DR-XY16850

Tuseran Forte: Official leaflet (Package Insert / Patient Information Leaflet)

No PI/PIL is published on the FDA Verification Portal for registration DR-XY29720 as of 2026-10-22. Indication and dosage are not covered by this record.

Source

FDA Philippines Verification Portal, https://verification.fda.gov.ph (search: "Tuseran"), retrieved 2026-10-22. Fields above are reproduced from the official registration record without modification.

Compiled for an academic RAG case study. Not medical advice. The portal states its information is for verification purposes. Verify current status on the portal before relying on it.

Vicks VapoDrops (Menthol + Glycyrrhiza glabra (Liquorice) Root + Zingiber officinale (Ginger) Rhizome + Phyllantus emblica (Indian Gooseberry) Fruit)

7 mg/ 15mg/ 10 mg/ 10 mg Lozenge (Orange Menthol Flavor) | Over-the-Counter (OTC) Drug | FDA Reg. No. DR-XY47938

Vicks VapoDrops: Official FDA Philippines registration record

Brand name	Vicks VapoDrops
Generic name	Menthol + Glycyrrhiza glabra (Liquorice) Root + Zingiber officinale (Ginger) Rhizome + Phyllantus emblica (Indian Gooseberry) Fruit
Dosage strength and form	7 mg/ 15mg/ 10 mg/ 10 mg Lozenge (Orange Menthol Flavor)
Classification	Over-the-Counter (OTC) Drug
Pharmacologic category	Throat Preparation
Registration number	DR-XY47938
Manufacturer	Unique Pharmaceutical Laboratories (Division of J.B. Chemicals & Pharmaceuticals Ltd.)

Vicks VapoDrops: OTC forms registered under the Vicks brand (capsule availability)

OTC capsule form registered under the Vicks brand: NO. 16 OTC / Household Remedy product(s) registered for this brand on the FDA Verification Portal:

Brand name	Dosage strength and form	Classification	Reg. No.
Vicks Inhaler	108.8 mg/108.8 mg per 0.5 mL Inhaler (Lavender Scent)	Household Remedy (HR)	DRHR-2545
Vicks Formula 44 Cough Control	15 mg/5 mL Syrup	Over-the-Counter (OTC) Drug	DR-XY9056
Vicks VapoDrops Xtra Strong	21 mg Lozenge (Honey Menthol Flavor)	Over-the-Counter (OTC) Drug	DR-XY49236
Vicks VapoDrops Xtra Strong	21 mg Lozenge (Lemon Menthol Flavor)	Over-the-Counter (OTC) Drug	DR-XY49234
Vicks VapoDrops Xtra Strong	21 mg Lozenge (Lemon Menthol Flavor)	Over-the-Counter (OTC) Drug	DR-XY49235
Vicks Vaporub Easy Applicator	5.21 g/2.75 g/1.5 g (5.21%/2.75%/1.5% w/w) per 100 g Ointment	Household Remedy (HR)	DRHR-2542
Vicks VapoRub	5.26 g/2.82 g/1.33 g (5.26 %/2.82 %/1.33 % w/w) per 100 g Ointment (Lavender)	Household Remedy (HR)	DRHR-2535
Vicks VapoRub Ointment	5.26 g/2.82 g/1.33 g per 100 g Ointment	Household Remedy (HR)	DRHR-2485
Vicks VapoRub Ointment	5.26 g/2.82 g/1.33 g per 100 g Ointment	Household Remedy (HR)	DRHR-279
Vicks VapoRub Xtra Strong	7 g/4.03 g/5.2 g (7%/4.03%/5.2% w/w) per 100 g Ointment	Household Remedy (HR)	DRHR-2214

Brand name	Dosage strength and form	Classification	Reg. No.
Vicks VapoDrops	7 mg/ 15mg/ 10 mg/ 10 mg Lozenge (Berry Menthol Flavor)	Over-the-Counter (OTC) Drug	DR-XY47940
Vicks VapoDrops	7 mg/ 15mg/ 10 mg/ 10 mg Lozenge (Honey Lemon Menthol Flavor)	Over-the-Counter (OTC) Drug	DR-XY47939
Vicks VapoDrops	7 mg/ 15mg/ 10 mg/ 10 mg Lozenge (Orange Menthol Flavor)	Over-the-Counter (OTC) Drug	DR-XY47938
Vicks VapoDrops	7 mg/15 mg/10 mg/10 mg Lozenges (Berry Menthol Flavor)	Over-the-Counter (OTC) Drug	DR-XY47698
Vicks VapoDrops	7mg/15mg/10mg/10mg Lozenge (Honey Lemon Menthol Flavor) Lozenges	Over-the-Counter (OTC) Drug	DR-XY47727
Vicks Vapo Drops	7mg/15mg/10mg/10mg Lozenge (Orange Menthol Flavor)	Over-the-Counter (OTC) Drug	DR-XY47726

Vicks VapoDrops: Official leaflet (Package Insert / Patient Information Leaflet)

An official leaflet for this registration is published on the FDA Verification Portal: PI_01 (https://drive.google.com/file/d/1BJqQZ-PuhihsjKHhHvEcPiE7vs1klLvr/view?usp=drivesdk). Its text (OCR) and the original leaflet pages follow.

Source

FDA Philippines Verification Portal, https://verification.fda.gov.ph (search: "Vicks"), retrieved 2026-10-22. Fields above are reproduced from the official registration record without modification.

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Vicks VapoDrops: Official leaflet text (OCR transcription)

Machine transcription of the official FDA leaflet for Reg. No. DR-XY47938. OCR may contain errors; the original leaflet pages are attached after this section.

Vicks VapoDrops leaflet PI_01

Product Description
Linagliptin + Metformin HCl (Trajenta Duo) film-coated tablets, 2.5 mg / 500 mg are light yellow, oval, biconvex film-coated tablets. One side is debossed with the code ‘D2 / 500’, the other side is debossed with the Boehringer Ingelheim company symbol.
Linagliptin + Metformin HCl (Trajenta Duo) film-coated tablets, 2.5 mg / 850 mg are light orange, oval, biconvex film-coated tablets. One side is debossed with the code ‘D2 / 850’, the other side is debossed with the Boehringer
Linagliptin + Metformin HCl (Trajenta Duo) film-coated tablets, 2.5 mg / 1 g are light pink, oval, biconvex film-coated tablets. One side is debossed with the code ‘D2 / 1000’, the other side is debossed with the Boehringer Ingelheim company symbol.
Formulation
Each film-coated tablet contains:
1H-Purine-2,6-dione, 8-[(3R)-3-amino-1-piperidinyl]-7-(2-butyn-1-yl)-3,7-dihydro-3-methyl-1-[(4- methyl-2-quinazolinyl)methyl]- (= linagliptin) 2.5 mg
and
N,N-dimethylimido-dicarbonimidic diamide hydrochloride
(= metformin hydrochloride) 500 mg, 850 mg or 1 g
Excipients: L-Arginine, Maize starch, Copovidone, Silica, colloidal anhydrous, Magnesium stearate (E470b), Hypromellose 2910, Propylene glycol, Titanium dioxide (E171), Talc, Iron oxide red/yellow (E172)
Indications
Linagliptin + Metformin HCl (Trajenta Duo) is indicated as an adjunct to diet and exercise to improve glycemic control in adults with type 2 diabetes mellitus when treatment with both linagliptin and metformin is appropriate, in patients inadequately controlled on metformin alone or those already being treated and well controlled with the free combination of linagliptin and metformin.
Linagliptin + Metformin HCl (Trajenta Duo) is indicated in combination with a sulphonylurea (i.e., triple combination therapy) as an adjunct to diet and exercise in adult patients inadequately controlled on their maximal tolerated dose of metformin and a sulphonylurea.
an SGLT2 inhibitor (i.e. triple combination therapy) as an adjunct to diet and exercise in patients inadequately controlled on their maximum tolerated dose of metformin and an SGLT2 inhibitor.
Linagliptin + Metformin HCl (Trajenta Duo) is indicated as add-on to insulin (i.e., triple combination therapy) as an adjunct to diet and exercise to improve glycaemic control in adult patients when insulin and metformin alone do not provide adequate glycaemic control.
Dosage and administration
Adults with normal renal function (GFR >= 90mL/min)
The recommended dose is 2.5/500 mg, 2.5/ 850 mg or 2.5/1 g twice daily. The dosage should be individualised on the basis of the patient’s current regimen, effectiveness, and tolerability. Maximum recommended daily dose metformin (see table 1 for additional dosing information).
Linagliptin + Metformin HCl (Trajenta Duo) should be given with meals to reduce the gastrointestinal undesirable effects associated with metformin.
For patients currently not treated with metformin
For patients currently not treated with metformin, the recommended starting dose is 2.5 mg linagliptin/500mg metformin hydrochloride twice daily.
For patients inadequately controlled on maximal tolerated dose of metformin monotherapy
For patients not adequately controlled on metformin alone, the usual starting dose of Linagliptin + Metformin HCl (Trajenta Duo) should provide linagliptin dosed as 2.5 mg twice daily (5 mg total daily dose) plus the dose of metformin already being taken.
For patients switching from co-administration of linagliptin and metformin to the fixed dose combination, Linagliptin + Metformin HCl (Trajenta Duo)

should be initiated at the dose of linagliptin and metformin already being taken.
For patients inadequately controlled on dual combination therapy with the maximal tolerated dose of metformin and a sulphonylurea
The dose of Linagliptin + Metformin HCl (Trajenta Duo) should provide linagliptin dosed as 2.5 mg twice daily (5 mg total daily dose) and a dose of metformin similar to the dose already being taken. When Linagliptin + Metformin HCl (Trajenta Duo) is used in combination with a sulphonylurea, a lower dose of the sulphonylurea may be required to reduce the risk of hypoglycaemia (see section on Special Warnings and Precautions).
For patients inadequately controlled on dual combination therapy with insulin and the maximal tolerated dose of metformin
The dose of Linagliptin + Metformin HCl (Trajenta Duo) should provide linagliptin dosed as 2.5 mg twice daily (5 mg total daily dose) and a dose of metformin similar to the dose already being taken. When Linagliptin + Metformin HCl (Trajenta Duo) is used in combination with insulin, a lower dose of insulin may be required to reduce the risk of hypoglycaemia (see section on Special Warnings and Precautions).
For the different doses on metformin, Linagliptin + Metformin HCl (Trajenta Duo) is available in strengths of 2.5 mg linagliptin plus 500 mg metformin hydrochloride, 850 mg metformin hydrochloride or 1 g metformin hydrochloride.
Renal impairment
A GFR should be assessed before initiation of treatment with metformin containing products and at least annually thereafter.
In patients at an increased risk of further progression of renal impairment and in the elderly, renal function should be assessed more frequently, e.g. every 3-6 months.
Factors that may increase the risk of lactic acidosis (see section Special Warnings and Precautions) should be reviewed before considering initiation of metformin in patients with GFR<60 mL/min.
Table 1 Posology for renally impaired patients*
eGFR mL/min Metformin Linagliptin
Dose reduction may be considered in relation to declining renal function.
No dose adjustment is required.
The starting dose is at most half of the maximum dose.
<30 Metformin is contraindicated.
*If no adequate strength of Linagliptin + Metformin HCl (Trajenta Duo) is available, individual monocomponents should be used instead of the fixed dose combination.
Hepatic impairment
Linagliptin + Metformin HCl (Trajenta Duo) is contraindicated in patients with hepatic impairment due to the metformin component (see section Contraindications).
Geriatric patients
As metformin is excreted via the kidney, and elderly patients have a tendency for decreased renal function, elderly patients taking Linagliptin + Metformin HCl (Trajenta Duo) should have their renal function monitored regularly (see section Special Warnings and Precautions).
Paediatric patients
Linagliptin + Metformin HCl (Trajenta Duo) is not recommended for use in children below 18 years due to lack of data on safety and efficacy.
Missed dose
If a dose is missed, it should be taken as soon as the patient remembers. However, a double dose should not be taken at the same time. In that case the missed dose should be skipped.
Contraindications
• Hypersensitivity to active ingredients linagliptin and/or metformin hydrochloride or to any of the excipients
• Any type of acute metabolic acidosis (such as lactic acidosis, diabetic ketoacidosis)
• Diabetic pre-coma

• Severe renal failure (CrCl < 30mL/min or eGFR < 30 mL/min/1.73m2)
• Acute conditions with the potential to alter renal function such as: dehydration, severe infection, shock, intravascular administration of iodinated contrast agents (see section Special Warnings and Precautions)
• Disease which may cause tissue hypoxia (especially acute disease, or worsening of chronic disease) such as: decompensated heart failure, respiratory failure, recent myocardial infarction, shock
• Hepatic impairment
• Acute alcohol intoxication
• Alcoholism

Special Warnings and Precautions

General

Linagliptin + Metformin HCl (Trajenta Duo) should not be used in patients with type 1 diabetes.

Pancreatitis

Acute pancreatitis has been observed in patients taking linagliptin.

If pancreatitis is suspected, Linagliptin + Metformin HCl (Trajenta Duo) should be discontinued.

Hypoglycaemia

Linagliptin alone showed a comparable incidence of hypoglycaemia to placebo. In clinical trials of linagliptin as part of combination therapy with agents not considered to cause hypoglycaemia (metformin, thiazolidinediones) rates of hypoglycaemia reported with linagliptin were similar to rates in patients taking placebo.

Sulphonylureas are known to cause hypoglycaemia. Therefore, caution is advised when Linagliptin + Metformin HCl (Trajenta Duo) is used in combination with a sulphonylurea. A dose reduction of the sulphonylurea may be considered.

Insulin is known to cause hypoglycaemia. Therefore, caution is advised when Linagliptin + Metformin HCl (Trajenta Duo) is used in combination with insulin. A dose reduction of insulin may be considered.

Metformin alone does not cause hypoglycaemia under usual circumstances of use, but hypoglycaemia could occur when caloric intake is deficient, when strenuous exercise is not compensated by caloric supplementation, or during concomitant use with other glcuose-lowering agents (such as sulphonylureas and insulin) or ethanol.

Lactic acidosis

Lactic acidosis, a very rare, but serious, metabolic complication, most often occurs at acute worsening of renal function or cardiorespiratory illness or sepsis. Metformin accumulation occurs at acute worsening of renal function and increases the risk of lactic acidosis.

In case of dehydration (severe diarrhoea or vomiting, fever or reduced fluid intake), metformin should be temporarily discontinued and contact with a health care professional is recommended.

Medicinal products that can acutely impair renal function (such as antihypertensives, diuretics and NSAIDs) should be initiated with caution in metformin-treated patients.

Other risk factors for lactic acidosis are excessive alcohol intake, hepatic insufficiency, inadequately controlled diabetes, ketosis, prolonged fasting and any conditions associated with hypoxia, as well as concomitant use of medicinal products that may cause lactic acidosis (see also section Contraindications and Interactions).

Patients and/or care-givers should be informed of the risk of lactic acidosis. Lactic acidosis is characterised by acidotic dyspnoea, abdominal pain, muscle cramps, asthenia and hypothermia followed by coma. In case of suspected symptoms, the patient should stop taking metformin and seek immediate medical attention. Diagnostic laboratory findings are decreased blood pH (<7.35), increased plasma lactate levels (>5 mmol/L), and an increased anion gap and lactate/pyruvate ratio.

Administration of iodinated contrast agent

Intravascular administration of iodinated contrast agents may lead to contrast induced nephropathy, resulting in metformin accumulation and an increased risk of lactic acidosis. Metformin should be discontinued prior to or at the time of the imaging procedure and not restarted until at least 48 hours after, provided that renal function has been re-evaluated and found to be stable, see sections Patients with renal impairment and Interactions.

Renal function

GFR should be assessed before treatment initiation and regularly thereafter, see

section Posology. Linagliptin + Metformin HCl (Trajenta Duo) is contraindicated in patients with GFR<30 ml/min and should be temporarily discontinued in the presence of conditions that alter renal function, see section Contraindications.

Cardiac function

Patients with heart failure are more at risk of hypoxia and renal insufficiency. In patients with stable chronic heart failure, Linagliptin + Metformin HCl (Trajenta Duo) may be used with a regular monitoring of cardiac and renal function. For patients with acute and unstable heart failure, Linagliptin + Metformin HCl (Trajenta Duo) is contraindicated due to the metformin component (see section Contraindications).

Surgery

Metformin must be discontinued at the time of surgery under general, spinal or epidural anaesthesia. Therapy may be restarted no earlier than 48 hours following surgery or resumption of oral nutrition and provided that renal function has been re-evaluated and found to be stable.

Bullous pemphigoid

Bullous pemphigoid has been observed in patients taking linagliptin. If bullous pemphigoid is suspected, Linagliptin + Metformin HCl (Trajenta Duo) should be discontinued.

Vitamin B12

The risk of low vitamin B12 levels increases with increasing metformin dose, treatment duration, and/or in patients with risk factors known to cause vitamin B12 deficiency. In case of suspicion of vitamin B12 deficiency (such as anaemia or neuropathy), vitamin B12 serum levels should be monitored. Periodic vitamin B12 monitoring could be necessary in patients with risk factors for vitamin B12 deficiency. Metformin therapy should be continued for as long as it is tolerated and not contra-indicated and appropriate corrective treatment for vitamin B12 deficiency provided in line with current clinical guidelines.

Use in Specific Populations

Pregnancy, Lactation and Fertility

Pregnancy

There are no adequate and well-controlled studies in pregnant women with Linagliptin + Metformin HCl (Trajenta Duo) or its individual components. Non-clinical reproduction studies in pregnant rats performed with the combined products in Linagliptin + Metformin HCl (Trajenta Duo) did not indicate a teratogenic effect attributed to the co-administration of linagliptin and metformin.

There are limited data from the use of linagliptin in pregnant women. Nonclinical studies do not indicate direct or indirect harmful effects with respect to reproductive toxicity.

There are limited data from the use of metformin in pregnant women. Metformin was not teratogenic in rats at a dose of 200 mg/kg/day associated with 4 times human exposure. At higher doses (500 and 1000 mg/kg/day, associated with 11 and 23 times human exposure), teratogenicity of metformin was observed in the rat.

As a precautionary measure, it is preferable to avoid the use of Linagliptin + Metformin HCl (Trajenta Duo) during pregnancy.

When the patient plans to become pregnant and during pregnancy, diabetes should not be treated with Linagliptin + Metformin HCl (Trajenta Duo) but insulin should be used to maintain blood glucose levels as close to normal as possible in order to lower the risk of fetal malformations associated with abnormal blood glucose levels.

Lactation

No studies on lactating animals have been performed with the combination of metformin and linagliptin. Non-clinical studies with the individual active substances have shown excretion of both metformin and linagliptin into milk in lactating rats. Metformin is excreted with milk in humans. It is not known whether linagliptin is excreted into human milk. Linagliptin + Metformin HCl (Trajenta Duo) should not be used during breast-feeding.

Fertility

No studies on the effect on human fertility have been conducted for Linagliptin + Metformin HCl (Trajenta Duo). No adverse effects of linagliptin on fertility were observed in non-clinical studies up to the highest tested dose of 240 mg/kg/day (> 900 times human exposure).

Driving and Using Machines

No studies on the effects on the ability to drive and use machines have been performed.

Interactions

Pharmacokinetic Interactions

General

Co-administration of multiple doses of linagliptin (10 mg once daily) and metformin (850 mg twice daily) did not meaningfully alter the pharmacokinetics of either linagliptin or metformin in healthy volunteers.

Pharmacokinetic drug interaction studies with Linagliptin + Metformin HCl (Trajenta Duo) have not been performed; however, such studies have been conducted with the individual active substances of Linagliptin + Metformin HCl (Trajenta Duo): linagliptin and metformin.

Linagliptin

In vitro assessment of drug interactions:

Linagliptin is a weak competitive and a weak to moderate mechanism-based inhibitor of CYP isozyme CYP3A4, but does not inhibit other CYP isozymes. It is not an inducer of CYP isozymes.

Linagliptin is a P-glycoprotein substrate, and inhibits P-glycoprotein mediated transport of digoxin with low potency. Based on these results and in vivo drug interaction studies, linagliptin is considered unlikely to cause interactions with other P-gp substrates.

In vivo assessment of drug interactions:

Clinical data described below suggest that the risk for clinically meaningful interactions by coadministered medicinal products is low. No clinically significant interactions requiring dose adjustment were observed.

Linagliptin had no clinically relevant effect on the pharmacokinetics of metformin, glibenclamide, simvastatin, pioglitazone, warfarin, digoxin or oral contraceptives providing in vivo evidence of a low propensity for causing drug interactions with substrates of CYP3A4, CYP2C9, CYP2C8, P-glycoprotein, and organic cationic transporter (OCT).

Metformin: Co-administration of multiple three-times-daily doses of 850 mg metformin with a supratherapeutic dose of 10 mg linagliptin once daily did not clinical meaningfully alter the pharmacokinetics of linagliptin or metformin in healthy volunteers. Therefore, linagliptin is not an inhibitor of OCT-mediated transport.

Sulphonylureas: The steady-state pharmacokinetics of 5 mg linagliptin was not changed by co-administration of a single 1.75 mg dose glibenclamide (glyburide) and multiple oral doses of 5 mg linagliptin. However there was a clinically not relevant reduction of 14% of both AUC and Cmax of glibenclamide. Because glibenclamide is primarily metabolized by CYP2C9, these data also support the conclusion that linagliptin is not a CYP2C9 inhibitor. Clinically meaningful interactions would not be expected with other sulphonylureas (e.g. glipizide, tolbutamide and glimepiride) which, like glibenclamide, are primarily eliminated by CYP2C9.

Thiazolidinediones: Co-administration of multiple daily doses of 10 mg linagliptin (supratherapeutic) with multiple daily doses of 45 mg pioglitazone, a CYP2C8 and CYP3A4 substrate, had no clinically relevant effect on the pharmacokinetics of either linagliptin or pioglitazone or the active metabolites of pioglitazone, indicating that linagliptin is not an inhibitor of CYP2C8-mediated metabolism in vivo and supporting the conclusion that the in vivo inhibition of CYP3A4 by linagliptin is negligible.

Ritonavir: A study was conducted to assess the effect of ritonavir, a potent inhibitor of P-glycoprotein and CYP3A4, on the pharmacokinetics of linagliptin. Co-administration of a single 5 mg oral dose of linagliptin and multiple 200 mg oral doses of ritonavir increased the AUC and Cmax of linagliptin approximately twofold and threefold, respectively. Simulations of steady-state plasma concentrations of linagliptin with and without ritonavir indicated that the increase in exposure will not be associated with an increased accumulation. These changes in linagliptin pharmacokinetics were not considered to be clinically relevant. Therefore, clinically relevant interactions would not be expected with other P-glycoprotein/CYP3A4 inhibitors and dose adjustment is not required.

Rifampicin: A study was conducted to assess the effect of rifampicin, a potent inductor of P-glycoprotein and CYP3A4, on the pharmacokinetics of 5 mg linagliptin. Multiple co-administration of linagliptin with rifampicin, resulted in a 39.6% and 43.8% decreased linagliptin steady-state AUC and Cmax and about 30% decreased DPP-4 inhibition at trough. Thus linagliptin in combination with strong P-gp inducers is expected to be clinically efficacious, although full efficacy might not be achieved.

Digoxin: Co-administration of multiple daily doses of 5 mg linagliptin with multiple doses of 0.25 mg digoxin had no effect on the pharmacokinetics

of digoxin in healthy volunteers. Therefore, linagliptin is not an inhibitor of P-glycoprotein-mediated transport in vivo.
Warfarin: Multiple daily doses of 5 mg linagliptin did not alter the pharmacokinetics of S(-) or R(+) warfarin, a CYP2C9 substrate, showing that linagliptin is not an inhibitor of CYP2C9.
Simvastatin: Multiple daily doses of 10 mg linagliptin (supratherapeutic) had a minimal effect on the steady state pharmacokinetics of simvastatin, a sensitive CYP3A4 substrate, in healthy volunteers. Following administration of 10 mg linagliptin concomitantly with 40 mg of simvastatin daily for 6 days, the plasma AUC of simvastatin was increased by 34%, and the plasma Cmax by 10%. Therefore, linagliptin is considered to be a weak inhibitor of CYP3A4-mediated metabolism, and dosage adjustment of concomitantly administered substances metabolised by CYP3A4 is considered unnecessary.
Oral Contraceptives: Co-administration with 5 mg linagliptin did not alter the steady-state pharmacokinetics of levonorgestrel or ethinylestradiol.
The absolute bioavailability of linagliptin is approximately 30%. Because coadministration of a high-fat meal with linagliptin had no clinically relevant
effect on the pharmacokinetics, linagliptin may be administered with or without food.
Metformin
Concomitant use not recommended
Alcohol
Alcohol intoxication is associated with an increased risk of lactic acidosis, particularly in cases of fasting, malnutrition or hepatic insufficiency.
Iodinated contrast agents
Linagliptin + Metformin HCl (Trajenta Duo) must be discontinued prior to, or at the time of the imaging procedure and not be restarted until at least 48 hours after, provided that renal function has been re-evaluated and found to be stable (see section Posology and Special Warnings and Precautions).
Combination requiring precautions for use
Some medicinal products can adversely affect renal function which may increase the risk of lactic acidosis, e.g. NSAIDs, including selective cyclo-oxygenase (COX) II inhibitors, ACE inhibitors, angiotensin II receptor antagonists and diuretics, especially loop diuretics. When starting or using such products in combination with metformin, close monitoring of renal function is necessary.
Organic cation transporters (OCT)
Metformin is a substrate of both transporters OCT1 and OCT2.
Co-administration of metformin with:
• Inhibitors of OCT1 (such as verapamil) may reduce efficacy of metformin.
• Inducers of OCT1 (such as rifampicin) may increase gastrointestinal absorption and efficacy of metformin.
• Inhibitors of OCT2 (such as cimetidine, dolutegravir, ranolazine, trimethoprime, vandetanib, isavuconazole) may decrease the renal elimination of metformin and thus lead to an increase in metformin plasma concentration.
• Inhibitors of both OCT1 and OCT2 (such as crizotinib, olaparib) may alter efficacy and renal elimination of metformin.
Caution is therefore advised, especially in patients with renal impairment, when these drugs are co-administered with metformin, as metformin plasma concentration may increase. If needed, dose adjustment of metformin may be considered as OCT inhibitors/inducers may alter the efficacy of metformin.
Adverse Reactions
Summary of the safety profile
The safety of linagliptin 2.5 mg twice daily (or its bioequivalent of 5 mg once daily) plus metformin has been evaluated in over 6800 patients with type 2 diabetes mellitus (T2DM).
In placebo-controlled studies, more than 1800 patients were treated with the therapeutic dose of either 2.5 mg linagliptin twice daily (or its bioequivalent of 5 mg linagliptin once daily) in combination with metformin for >=12/24 weeks.
In the pooled analysis of the placebo-controlled trials, the overall incidence of adverse events (AEs) in patients treated with placebo and metformin was comparable to linagliptin 2.5 mg and metformin (54.3% and 49.0%). Discontinuation of therapy due to AEs was comparable in patients who received placebo and metformin to patients treated with linagliptin and metformin (3.8% and 2.9%).
Due to the impact of the background therapy on AEs (e.g. on hypoglycaemias), adverse events were analysed and displayed based on the respective treatment regimens, add-on to metformin, and add-on to meformin plus sulphonylurea.

The placebo-controlled studies included 7 studies where linagliptin was given as add-on to metformin and 1 study where linagliptin was given as add-on to metformin + sulphonylurea.

Tabulated summary of adverse reactions

The following adverse reactions derived from the use of the linagliptin/ metformin combination or the use of the monocomponents (linagliptin or metformin) in clinical trials or from post-marketing experience are shown in the table below. Undesirable effects previously reported with one of the individual components may be potential undesirable effects with Linagliptin + Metformin HCl (Trajenta Duo) even if not observed in clinical trials with this product.

MedDRA System Organ Class terminology	Linagliptin and Metformin Adverse reactions
Infections and infestations	Nasopharyngitis 3
Immune system disorders	Hypersensitivity 1; 3 Angioedema 4 Urticaria 2;4
Metabolism and nutrition disorders	Lactic acidosis 2 Vitamin B12 decrease/deficiency2 Hypoglycaemia (when linagliptin and metformin were combined with a sulphonylurea)
Nervous system disorders	Taste disturbance 2
Respiratory, thoracic and mediastinal disorders	Cough 1; 3
MedDRA System Organ Class terminology	Linagliptin and Metformin Adverse reactions
Gastrointestinal disorders**	Decreased appetite 3; 5 Diarrhoea 3; 5 Constipation (when linagliptin and metformin were combined with insulin) Nausea 3; 5 Pancreatitis 3 Vomiting 3; 5 Abdominal pain 2 Mouth ulceration 4
Hepato-biliary disorders	Liver function test abnormal 2 Hepatitis 2
Skin and subcutaneous tissue disorders	Pruritus 3; 5 Erythema 2 Rash 4 Bullous pemphigoid 4; §
Investigations	Lipase increased 3;† Amylase increased

1 Adverse reactions reported also in patients treated with linagliptin monotherapy

2 Adverse reactions of metformin as monotherapy; refer to the Summary of Product Characteristics for metformin for additional information

3 Adverse reactions of FDC linagliptin + metformin (pooled analysis of placebo-controlled studies)

4 Adverse reactions identified from post-marketing experience with linagliptin

5 Adverse reactions reported in patients who received FDC linagliptin+metformin and also in patients who received metformin monotherapy

** Gastrointestinal disorders such as abdominal pain and nausea, vomiting, diarrhoea, decreased appetite occur most frequently during initiation of therapy with metformin hydrochloride and resolve spontaneously in most cases. To prevent them, it is recommended that metformin hydrochloride be taken in 2 daily doses during or after meals if administered as monotherapy.

§ See also Linagliptin cardiovascular and renal safety study (CARMELINA) below

† Based on lipase increase >3 times ULN.

In the CAROLINA study comparing linagliptin with active comparator glimepiride (see section Clinical Trials) laboratory analysis of amylase showed increases to >3xULN in 0.99% of patients treated with linagliptin and in 0.54% of patients treated with glimepiride.
In placebo-controlled studies the most frequently reported related adverse reaction for linagliptin+metformin was diarrhoea (1.6%) with comparable rate on metformin+placebo (2.4%).
Adverse reactions reported when linagliptin and metformin were combined with SU:
When linagliptin and metformin were administered in combination with a sulphonylurea, hypoglycaemia was the most commonly reported AE (linagliptin plus metformin plus sulphonylurea 23.9% vs 16.0% in the placebo group) and identified as an additional adverse reaction under these conditions. None of the hypoglycaemias episodes were classified as severe (requiring assistance).
Adverse reactions reported when linagliptin and metformin were combined with insulin:
When linagliptin and metformin were administered in combination with insulin, hypoglycaemia was the most commonly reported AE, but occurred at comparable rate when placebo and metformin were combined with insulin (linagliptin plus metformin plus insulin 29.5% vs 30.9% in the placebo plus metformin plus insulin group) with a low rate of severe (requiring assistance) episodes (1.5% vs. 0.9%).
Linagliptin cardiovascular outcome and renal safety study (CARMELINA)
The CARMELINA study evaluated the cardiovascular and renal safety of linagliptin versus placebo in patients with type 2 diabetes and with increased CV risk evidenced by a history of established macrovascular or renal disease (see section Clinical Trials). The study included 3494 patients treated with linagliptin (5 mg) and 3485 patients treated with placebo. Both treatments were added to standard of care targeting regional standards for HbA1c and CV risk factors; with 54% on metformin. The overall incidence of adverse events and serious adverse events in patients receiving linagliptin was similar to that in patients receiving placebo. Safety data from this study was in line with previous known safety profile of linagliptin.
In the treated population, severe hypoglycaemic events (requiring assistance) were reported in 3.0% of patients on linagliptin and in 3.1% on placebo. Among patients who were using sulfonylurea at baseline, the incidence of severe hypoglycaemia was 2.0% in linagliptin-treated patients and 1.7% in placebo treated patients. Among patients who were using insulin at baseline, the incidence of severe hypoglycaemia was 4.4% in linagliptin-treated patients and 4.9% in placebo treated patients.
In the overall study observation period adjudicated acute pancreatitis was reported in 0.3% of patients treated with linagliptin and in 0.1% of patients treated with placebo.
In the CARMELINA study, bullous pemphigoid was reported in 0.2% of patients treated with linagliptin and in no patient treated with placebo.
Overdose
Symptoms
During controlled clinical trials in healthy subjects, single doses of up to 600 mg linagliptin (equivalent to 120 times the recommended dose) were well tolerated. There is no experience with doses above 600 mg in humans.
Hypoglycaemia has not been seen with metformin hydrochloride doses of up to 85 g, although lactic acidosis has occurred in such circumstances. High overdose of metformin hydrochloride or concomitant risks may lead to lactic acidosis. Lactic acidosis is a medical emergency and must be treated in hospital.
Therapy
In the event of an overdose, it is reasonable to employ the usual supportive measures, e.g., remove unabsorbed material from the gastrointestinal tract, employ clinical monitoring and institute clinical measures as required. The most effective method to remove lactate and metformin hydrochloride is haemodialysis.
Pharmacological properties
Pharmacotherapeutic group: Combinations of oral blood glucose lowering drugs, ATC code: A10BD11
Mode of Action
Linagliptin is an inhibitor of the enzyme DPP-4 (Dipeptidyl peptidase 4, EC 3.4.14.5) an enzyme which is involved in the inactivation of the incretin

hormones GLP-1 and GIP (glucagon-like peptide-1, glucose-dependent insulinotropic polypeptide). These hormones are rapidly degraded by the enzyme DPP-4. Both incretin hormones are involved in the physiological regulation of glucose homeostasis. Incretins are secreted at a low basal level throughout the day and levels rise immediately after meal intake. GLP-1 and GIP increase insulin biosynthesis and secretion from pancreatic beta cells in the presence of normal and elevated blood glucose levels. Furthermore GLP-1 also reduces glucagon secretion from pancreatic alpha cells, resulting in a reduction in hepatic glucose output. Linagliptin binds very effectively to DPP-4 in a reversible manner and thus leads to a sustained increase and a prolongation of active incretin levels. Linagliptin glucose-dependently increases insulin secretion and lowers glucagon secretion thus resulting in an overall improvement in the glucose homoeostasis. Linagliptin binds selectively to DPP-4 and exhibits a >10000 fold selectivity versus DPP-8 or DPP-9 activity in vitro.
Metformin hydrochloride is a biguanide with antihyperglycaemic effects, lowering both basal and postprandial plasma glucose. It does not stimulate insulin secretion and therefore does not produce hypoglycaemia.
Metformin hydrochloride may act via 3 mechanisms:
(1)reduction of hepatic glucose production by inhibiting gluconeogenesis and glycogenolysis
(2)in muscle, by increasing insulin sensitivity, improving peripheral glucose uptake and utilisation
(3)and delay of intestinal glucose absorption.
Metformin hydrochloride stimulates intracellular glycogen synthesis by acting on glycogen synthase.
Metformin hydrochloride increases the transport capacity of all types of membrane glucose transporters (GLUTs) known to date.
In humans, independently of its action on glycaemia, metformin hydrochloride has favourable effects on lipid metabolism. This has been shown at therapeutic doses in controlled, medium-term or long-term clinical studies: metformin hydrochloride reduces total cholesterol, LDL cholesterol and triglyceride levels.
Clinical trials
Linagliptin as add-on to metformin therapy
The efficacy and safety of linagliptin in combination with metformin in patients with insufficient glycaemic control on metformin monotherapy was evaluated in a double blind placebo controlled study of 24 weeks duration. Linagliptin added to metformin provided significant improvements in HbA1c, (-0.64 % change compared to placebo), from a mean baseline HbA1c of 8 %. Linagliptin also showed significant improvements in fasting plasma glucose (FPG) by -21.1 mg/dL (-1.2 mmol/L) and 2-hour post-prandial glucose (PPG) by -67.1 mg/dL (-3.7 mmol/L) compared to placebo, as well as a greater portion of patients achieving a target HbA1c of < 7.0% (28.3% on linagliptin vs. 11.4% on placebo). The observed incidence of hypoglycaemia in patients treated with linagliptin was similar to placebo. Body weight did not differ significantly between the groups.
In a 24-week placebo-controlled factorial study of initial therapy, linagliptin 2.5 mg twice daily in combination with metformin (500 mg or 1 g twice daily) provided significant improvements in glycaemic parameters compared with either monotherapy as summarized in table 2 (mean baseline HbA1c 8.65%). [1]
Mean reductions from baseline in HbA1c were generally greater for patients with higher baseline HbA1c values. Effects on plasma lipids were generally neutral. The decrease in body weight with the combination of linagliptin and metformin was similar to that observed for metformin alone or placebo; there was no change from baseline for patients on linagliptin alone. The incidence of hypoglycaemia was similar across treatment groups (placebo 1.4%, linagliptin 5 mg 0%, metformin 2.1%, and linagliptin 2.5mg plus metformin twice daily 1.4%).
In addition, this study included patients (n=66) with more severe hyperglycaemia (HbA1c at baseline >= 11%) who were treated with twice daily open-label linagliptin 2.5 mg + metformin 1 g. In this group of patients, the mean baseline HbA1c value was 11.8% and mean FPG was 261.8 mg/dL (14.5 mmol/L). A mean decrease from baseline of -3.74% in HbA1c (n=48) and -81.2 mg/dL (-4.5 mmol/L) for FPG (n=41) was observed for patients completing the 24 week trial period without rescue therapy (n=48).

In the LOCF analysis including all patients with primary endpoint measurements (n=65) at last observation without rescue therapy changes from baseline were -3.19% for HbA1c and -73.6 mg/dL (- 4.1 mmol/L) for FPG.
The efficacy and safety of linagliptin 2.5 mg twice daily versus 5 mg once daily in combination with metformin in patients with insufficient glycaemic control on metformin monotherapy was evaluated in a double blind placebo controlled study of 12 weeks duration. Linagliptin (2.5 mg twice daily and 5 mg once daily) added to metformin provided significant improvements in glycaemic parameters compared to placebo. Linagliptin 5 mg once daily and 2.5 mg twice daily provided comparable (CI: -0.07; 0.19), significant HbA1c reductions of -0.80 % (from baseline 7.98%), and -0.74 (from baseline 7.96%) compared to placebo.
Linagliptin
Metformin HCl
Trajenta Duo®
Oral Blood Glucose Lowering Drug Combination abcd
P047772-004
Trajenta Duo® Oral Blood Glucose Lowering Drug Combination
abcd
Linagliptin
Metformin HCl
® Trajenta Duo
Oral Blood Glucose
File information
Issue date of TD:
PPM SKU:
PPM SKU version:
Issue date of artwork:
Print colors: PAN Black
08/MAR/2023
004
06/Jul/2022 Yes Yes
No Yes
No
Yes
TD Printfile
Mandatory in
Mat. No. Pack. Site:
Min. font size:
P047772-004
7pt BISansNEXT
Legend case version: V5.0 01/JAN/2021 (please do not change or remove it)
BI-Diecut-Legendcase Free area Gluepoints
Technical colours - not for printing
Additional Requirements of Packaging site
Control Code Type:
Laetus Code
Dimensions: 850 x 380 mm
In-process control code specification
Front/Reverse information should stay at BI Legendcase for further workflow. Please do not remove. FRONT
The observed incidence of hypoglycaemia in patients treated with linagliptin was similar to placebo (2.2% on linagliptin 2.5 mg twice daily, 0.9% on linagliptin 5 mg once daily, and 2.3% on placebo). Body weight did not differ significantly between the groups.
Linagliptin as add-on to a combination of metformin and sulphonylurea therapy
A placebo controlled study of 24 weeks in duration was conducted to evaluate the efficacy and safety of linagliptin 5 mg to placebo, in patients not sufficiently controlled with a combination with metformin and a sulphonylurea. Linagliptin provided significant improvements in HbA1c (-0.62 % change compared to placebo), from a mean baseline HbA1c of 8.14%.
Linagliptin also showed significant improvements in patients achieving a target HbA1c of < 7.0% (31.2% on linagliptin vs. 9.2% on placebo), and also for fasting plasma glucose (FPG) with -12.7 mg/dL (-0.7 mmol/L) reduction compared to placebo. Body weight did not differ significantly between the groups.
Linagliptin as add on to a combination of metformin and empagliflozin

In patients inadequately controlled with metformin and empagliflozin (10 mg (n=247) or 25 mg (n=217)), 24-weeks treatment with add-on therapy of linagliptin 5 mg provided adjusted mean HbA1c reductions from baseline by -0.53% (significant difference to add-on placebo -0.32% (95% CI -0.25, -0.13) and -0.58% (significant difference to add-on placebo -0.47% (95% CI -0.66; -0.28), respectively. A statistically significant greater proportion of patients with a baseline HbA1c >=7.0% and treated with linagliptin 5 mg achieved a target HbA1c of <7% compared to placebo.
In prespecified subgroups of patients with baseline HbA1c greater or equal than 8.5% (n=66 and n=42 patients on metformin plus empagliflozin 10 mg or 25 mg, respectively), the adjusted mean HbA1c reductions from baseline to 24 weeks on add-on therapy with linagliptin 5 mg were -0.97% (p=0.0875, for difference to add-on placebo) and -1.16% (p=0.0046 for difference to add-on placebo), respectively.
Linagliptin in combination with metformin and insulin
A 24-week placebo-controlled study was conducted to evaluate the efficacy and safety of linagliptin (5 mg once daily) added to insulin with or without metformin. 83 % of patients were taking metformin in combination with insulin in this trial. Linagliptin in combination with metformin plus insulin provided significant improvements in HbA1c in this subgroup with -0.68 (CI:-0.78; -0.57) adjusted mean change from baseline (mean baseline HbA1c 8.28%) compared to placebo in combination with metformin plus insulin. There was no meaningful change from baseline in body weight in either group.
Linagliptin 24 month data, as add- on to metformin in comparison with glimepiride
In a study comparing the efficacy and safety of the addition of linagliptin 5 mg or glimepiride (a sulphonylurea agent) in patients with inadequate glycaemic control on metformin monotherapy, linagliptin was similar to glimepiride in reducing HbA1c, with a mean treatment difference in HbA1c from baseline to 104 weeks for linagliptin compared to glimepiride of + 0.20%.
In this study, the proinsulin to insulin ratio, a marker of efficiency of insulin synthesis and release, showed a statistically significant improvement with linagliptin compared with glimepiride treatment. The incidence of hypoglycaemia in the linagliptin group (7.5%) was significantly lower than that in the glimepiride group (36.1%).
Patients treated with linagliptin exhibited a significant mean decrease from baseline in body weight compared to a significant weight gain in patients administered glimepiride (-1.39 vs + 1.29 kg).
Linagliptin as add-on therapy in elderly patients (age >= 70 years) with type 2 diabetes
The efficacy and safety of linagliptin in elderly (age >= 70 years) type 2 diabetes patients was evaluated in a double blind study of 24 weeks duration. Patients received metformin and/or sulphonylurea and/or insulin as background therapy. Doses of background antidiabetic medications were kept stable during the first 12 weeks, after which adjustments were permitted. Linagliptin provided significant improvements in HbA1c of -0.64 % (95% CI -0.81, -0.48; p<0.0001) compared to placebo after 24 weeks, from a mean baseline HbA1c of 7.8%.
Linagliptin also showed significant improvements in fasting plasma glucose (FPG) of -20.7mg/dL (95% CI -30.2, -11.2; p<0.0001) compared to placebo (-1.1 mmol/L). Body weight did not differ significantly between the groups.
Overall, the incidence of hypoglycaemia was comparable between linagliptin (2 of 45 patients, 4.4%) and placebo (none of 22 patients, 0%) on the background of metformin alone. Hypoglycaemia rates were also comparable on a background of insulin with or without metformin (13 of 35 patients, 37.1% treated with linagliptin and 6 of 15 patients, 40.0% treated with placebo). However, on a background of sulphonylurea with or without metformin, hypoglycaemia was reported in a higher proportion of patients treated with linagliptin (24 of 82 patients, 29.3%) compared to placebo (7 of 42 patients, 16.7%). There was no difference between linagliptin and placebo in severe hypoglycaemic events.
In a pooled analysis of elderly (age >=70 years) patients with type 2 diabetes (n=183) who were taking both metformin and basal insulin as background therapy, linagliptin in combination with metformin plus insulin provided significant improvements in HbA1c parameters with -0.81 (CI: - 1.01, -0.61) adjusted mean change from baseline (mean baseline HbA1c 8.13%) compared to placebo in combination with metformin plus insulin. There was no clinically meaningful difference in the incidence of hypoglycaemic events, in patient´s

>=70 years (37.2% on linagliptin in combination with metformin plus insulin vs. 39.8% on placebo in combination with metformin plus insulin).

Linagliptin and initial combination with Linagliptin and Metformin in recently diagnosed treatment naïve patients with marked hyperglycaemia:

The efficacy and safety of the initial combination of linagliptin 5 mg once daily and metformin twice daily (uptitrated in the first 6 weeks to 1500 mg or 2000 mg/d) compared to linagliptin 5 mg once has been studied in a 24 week trial in recently diagnosed treatment naive patients with T2DM and marked hyperglycaemia (baseline HbA1c 8.5–12.0%). After 24 weeks both linagliptin monotherapy as well as the initial combination of linagliptin and metformin significantly reduced HbA1c levels by –2.0% and –2.8% respectively, from a baseline HbA1c of 9.9% and 9.8% respectively. The treatment difference of –0.8% (95% CI –1.1 to –0.5) showed superiority for the initial combination over monotherapy (p<0.0001). Notably, 40% and 61% of patients in the monotherapy and combination arms achieved HbA1c <7.0%.

Linagliptin cardiovascular and renal safety study (CARMELINA)

CARMELINA was a randomized study in 6979 patients with type 2 diabetes with increased CV risk evidenced by a history of established macrovascular or renal disease who were treated with linagliptin 5 mg (3494) or placebo (3485) added to standard of care targeting regional standards for HbA1c , CV risk factors and renal disease. The study population included 1,211 (17.4%) patients >= 75 years of age and 4,348 (62.3%) patients with renal impairment. Approximately 19% of the population had eGFR >=45 to <60 mL/min/1.73 m2 , 28% of the population had eGFR >=30 to <45 mL/min/1.73 m2 and 15% had eGFR < 30 mL/min/1.73 m2.

The mean HbA1c at baseline was 8.0%.

The study was designed to demonstrate non-inferiority for the primary cardiovascular endpoint which was a composite of the first occurrence of cardiovascular death or a non-fatal myocardial infarction (MI) or a non-fatal stroke (3P-MACE). The renal composite endpoint was defined as renal death or sustained end stage renal disease or sustained decrease of 40% or more in eGFR. After a median follow up of 2.2 years, linagliptin, when added to standard of care, did not increase the risk of major adverse cardiovascular events or renal outcome events (Table 3 and Figure 1). There was no increased risk in hospitalization for heart failure which was an additional adjudicated endpoint observed compared to standard of care without linagliptin in patients with type 2 diabetes (Table 4).

Table 3 Major adverse cardiovascular events (MACE) and renal outcome events by treatment group in the CARMELINA study

	Linagliptin 5mg		Placebo		Hazard Ratio
	Number of Subjects (%)	Incidence Rate per 1000 PY*	Number of Subjects (%)	Incidence Rate per 1000 PY*	(95% CI)
Number of patients	3494		3485		
Primary CV composite (Cardiovascular death, non-fatal MI, non-fatal stroke)	434 (12.4)	57.7	420 (12.1)	56.3	1.02 (0.89, 1.17)**

Secondary renal composite (renal death, ESRD, 40% sustained decrease in eGFR) 327 (9.4) 48.9 306 (8.8) 46.6 1.04 (0.89, 1.17)

* PY=patient years

** Test on non-inferiority to demonstrate that the upper bound of the 95% CI for the hazard ratio is less than 1.3

Figure 1 Time to first occurrence of 3P-MACE in CARMELINA

Table 4 Hospitalization for heart failure and mortality by treatment group in the CARMELINA study

	Linagliptin 5mg		Placebo		Hazard Ratio
	Number of Subjects (%)	Incidence Rate per 1000 PY*	Number of Subjects (%)	Incidence Rate per 1000 PY*	(95% CI)
Number of patients	3494		3485		
All-cause mortality	367 (10.5)	46.9	373 (10.7)	48.0	0.98 (0.84, 1.13)
CV death	255 (7.3)	32.6	264 (7.6)	34	0.96 (0.81, 1.14)
Hospitalization for heart failure	209 (6.0)	27.7	226 (6.5)	30.4	0.90 (0.74, 1.08)

* PY=patient years

In analyses for albuminuria progression (change from normoalbuminuria to micro- or macroalbuminuria, or from microalbuminuria to macroalbuminuria) the estimated hazard ratio was 0.86 (95% CI 0.78, 0.95) for linagliptin versus placebo. The microvascular endpoint was defined as the composite of renal death, sustained ESRD, sustained decrease of >=50% in eGFR, albuminuria progression, use of retinal photocoagulation or intravitreal injections of an anti-VEGF therapy for diabetic retinopathy or vitreous haemorrhage or diabetes-related-blindness. The estimated hazard ratio for time to first occurrence for the composite microvascular endpoint was 0.86 (95% CI 0.78, 0.95) for linagliptin versus placebo, mainly driven by albuminuria progression.

Linagliptin cardiovascular safety study (CAROLINA)

CAROLINA was a randomized study in 6033 patients with early type 2 diabetes and increased CV risk or established complications who were treated with linagliptin 5 mg (3023) or glimepiride 1-4 mg (3010) added to standard of care (including background therapy with metformin in 83% of patients) targeting regional standards for HbA1c and CV risk factors. The mean age for study population was 64 years and included 2030 (34%) patients >= 70 years of age. The study population included 2089 (35%) patients with cardiovascular disease and 1130 (19%) patients with renal impairment with an eGFR < 60ml/min/1.73m2 at baseline. The mean HbA1c at baseline was 7.15%.

The study was designed to demonstrate non-inferiority for the primary cardiovascular endpoint which was a composite of the first occurrence of cardiovascular death or a non-fatal myocardial infarction (MI) or a non-fatal stroke (3P-MACE).
After a median follow up of 6.25 years, linagliptin, when added to standard of care, did not increase the risk of major adverse cardiovascular events (Table 5) as compared to glimepiride. Results were consistent for patients treated with or without metformin.
Table 5 Major adverse cardiovascular events (MACE) and mortality by treatment group in the CAROLINA study
Linagliptin 5mg Glimepiride (1-4mg) Hazard Ratio
Number of Subjects (%) Incidence Rate per 1000 PY* (95% CI) Number of patients
3023 3010
Primary CV composite (Cardiovascular death, non-fatal MI, non-fatal stroke)
356 (11.8) 20.7 362 (12.0) 21.2 0.98 (0.84, 1.14)**
* PY=patient years
** Test on non-inferiority to demonstrate that the upper bound of the 95% CI for the hazard ratio is less than 1.3
The composite of treatment sustainability, a key secondary endpoint, was defined as the proportion of patients on study treatment following initial titration period (16 weeks) that maintain glycaemic control (HbA1c <= 7.0%) at final visit without need for additional antidiabetic drug therapy (rescue medication) without any moderate (symptomatic with glucose value <= 70mg/dL) or severe (requiring assistance) hypoglycaemic episodes and without > 2% weight gain. A higher number of patients on linagliptin (481, 16.0%) achieved this key secondary endpoint compared to glimepiride (305, 10.2%).
For the entire treatment period (median time on treatment 5.9 years) the rate of patients with moderate or severe hypoglycaemia was 6.5% on linagliptin versus 30.9% on glimepiride, severe hypoglycaemia occurred in 0.3% of patients on linagliptin versus 2.2% on glimepiride.
Pharmacokinetics
Bioequivalence studies in healthy subjects demonstrated that the Linagliptin + Metformin HCl (Trajenta Duo) (linagliptin/metformin hydrochloride) combination tablets are bioequivalent to co-administration of linagliptin and metformin hydrochloride as individual tablets.
Administration of Linagliptin + Metformin HCl (Trajenta Duo) 2.5 mg/1 g with food resulted in no change in overall exposure of linagliptin. With metformin there was no change in AUC, however mean peak serum concentration of metformin was decreased by 18% when administered with food. A delayed time to peak serum concentrations by 2 hours was observed for metformin under fed conditions. These changes are not likely to be clinically significant.
The following statements reflect the pharmacokinetic properties of the individual active substances of Linagliptin + Metformin HCl (Trajenta Duo).
Linagliptin
The pharmacokinetics of linagliptin has been extensively characterized in healthy subjects and patients with type 2 diabetes. After oral administration of a 5 mg dose to healthy volunteers or patients, linagliptin was rapidly absorbed, with peak plasma concentrations (median Tmax) occurring 1.5 hours postdose. Plasma concentrations of linagliptin decline in at least a bi-phasic with a long terminal half- life (terminal half-life for linagliptin more than 100 hours), that is mostly related to the saturable, tight binding of linagliptin to DPP-4 and does not contribute to the accumulation of the drug. The effective half-life for accumulation

of linagliptin, as determined from oral administration of multiple doses of 5 mg linagliptin, is approximately 12 hours. After once-daily dosing, steady-state plasma concentrations of 5 mg linagliptin are reached by the third dose.
Plasma AUC of linagliptin increased approximately 33% following 5 mg doses at steady-state compared to the first dose. The intra-subject and inter-subject coefficients of variation for linagliptin AUC were small (12.6% and 28.5%, respectively).
Plasma AUC of linagliptin increased in a less than dose-proportional manner.
The pharmacokinetics of linagliptin was generally similar in healthy subjects and in patients with type 2 diabetes.
Absorption
The absolute bioavailability of linagliptin is approximately 30%. Because coadministration of a high-fat meal with linagliptin had no clinically relevant effect on the pharmacokinetics, linagliptin may be administered with or without food. In vitro studies indicated that linagliptin is a substrate of P-glycoprotein and of CYP3A4. Ritonavir, a potent inhibitor of P-glycoprotein and CYP3A4, led to a twofold increase in exposure (AUC) and multiple coadministration of linagliptin with rifampicin, a potent inducer of P-gp and
CYP3A, resulted in an about 40% decreased linagliptin steady-state AUC, presumably by increasing/decreasing the bioavailability of linagliptin by inhibition/induction of P-glycoprotein.
Distribution
As a result of tissue binding, the mean apparent volume of distribution at steady state following a single 5 mg intravenous dose of linagliptin to healthy subjects is approximately 1110 litres, indicating that linagliptin extensively distributes to the tissues. Plasma protein binding of linagliptin is concentration-dependent, decreasing from about 99% at 1 nmol/L to 75-89% at >=30 nmol/L, reflecting saturation of binding to DPP-4 with increasing concentration of linagliptin. At high concentrations, where DPP-4 is fully saturated, 70-80% of linagliptin was bound to other plasma proteins than DPP4, hence 30-20% were unbound in plasma.
Biotransformation
Following a [14C] linagliptin oral 10 mg dose, approximately 5% of the radioactivity was excreted in urine. Metabolism plays a subordinate role in the elimination of linagliptin. One main metabolite with a relative exposure of 13.3 % of linagliptin at steady state was detected which was found to be pharmacologically inactive and thus does not contribute to the plasma DPP-4 inhibitory activity of linagliptin.
Elimination
Following administration of an oral [14C] linagliptin dose to healthy subjects, approximately 85% of the administered radioactivity was eliminated in faeces (80%) or urine (5%) within 4 days of dosing. Renal clearance at steady state was approximately 70 mL/min.
Special populations
Renal Impairment
A multiple-dose, open-label study was conducted to evaluate the pharmacokinetics of linagliptin (5 mg dose) in patients with varying degrees of chronic renal impairment compared to normal healthy control subjects.
The study included patients with renal impairment classified on the basis of creatinine clearance as mild (50 to <80 mL/min), moderate (30 to <50 mL/min), and severe (<30 mL/min), as well as patients with ESRD on haemodialysis. In addition patients with T2DM and severe renal impairment (<30 mL/min) were compared to T2DM patients with normal renal function.
Creatinine clearance was measured by 24-hour urinary creatinine clearance measurements or estimated from serum creatinine based on the CockcroftGault formula:
CrCl = [140 - age (years)] x weight (kg) (x 0.85 for female patients) / [72 x serum creatinine (mg/dL)].
Under steady-state conditions, linagliptin exposure in patients with mild renal impairment was comparable to healthy subjects. In moderate renal impairment, a moderate increase in exposure of about 1.7 fold was observed compared with control.
Exposure in T2DM patients with severe RI was increased by about 1.4 fold compared to T2DM patients with normal renal function. Steady-state predictions for AUC of linagliptin in patients with ESRD indicated comparable exposure to that of patients with moderate or severe renal impairment.
In addition, linagliptin is not expected to be eliminated to a therapeutically significant degree by haemodialysis or peritoneal dialysis. Therefore, no dosage adjustment of linagliptin is necessary in patients with any degree of renal impairment.

In addition, mild renal impairment had no effect on linagliptin pharmacokinetics in patients with type 2 diabetes as assessed by population pharmacokinetic analyses.

Hepatic Impairment

In patients with mild moderate and severe hepatic impairment (according to the Child-Pugh classification), mean AUC and Cmax of linagliptin were similar to healthy matched controls following administration of multiple 5 mg doses of linagliptin. No dosage adjustment for linagliptin is necessary for patients with mild, moderate or severe hepatic impairment.

Body Mass Index (BMI)

No dosage adjustment is necessary based on BMI. Body mass index had no clinically relevant effect on the pharmacokinetics of linagliptin based on a population pharmacokinetic analysis of Phase I and Phase II data.

Gender

No dosage adjustment is necessary based on gender. Gender had no clinically relevant effect on the pharmacokinetics of linagliptin based on a population pharmacokinetic analysis of Phase I and Phase II data.

Geriatric

No dosage adjustment is required based on age, as age did not have a clinically relevant impact on the pharmacokinetics of linagliptin based on a population pharmacokinetic analysis of Phase I and Phase II data. Elderly subjects (65 to 80 years) had comparable plasma concentrations of linagliptin compared to younger subjects.

Paediatric

Studies characterizing the pharmacokinetics of linagliptin in paediatric patients have not been yet performed.

Race

No dosage adjustment is necessary based on race. Race had no obvious effect on the plasma concentrations of linagliptin based on a composite analysis of available pharmacokinetic data, including patients of Caucasian, Hispanic, African-American, and Asian origin. In addition the pharmacokinetic characteristics of linagliptin were found to be similar in dedicated phase I studies in Japanese, Chinese and Caucasian healthy volunteers and African American type 2 diabetes patients.

Metformin

Absorption

After an oral dose of metformin, Tmax is reached in 2.5 hours. Absolute bioavailability of a 500 mg or 850 mg metformin hydrochloride tablet is approximately 50-60% in healthy subjects. After an oral dose, the nonabsorbed fraction recovered in faeces was 20-30%.

After oral administration, metformin hydrochloride absorption is saturable and incomplete. It is assumed that the pharmacokinetics of metformin hydrochloride absorption is non-linear.

At the recommended metformin hydrochloride doses and dosing schedules, steady state plasma concentrations are reached within 24 to 48 hours and are generally less than 1 microgram/mL. In controlled clinical trials, maximum metformin hydrochloride plasma levels (Cmax) did not exceed 5 microgram/mL, even at maximum doses.

Food decreases the extent and slightly delays the absorption of metformin hydrochloride. Following administration of a dose of 850 mg, a 40% lower plasma peak concentration, a 25% decrease in AUC (area under the curve) and a 35 minute prolongation of the time to peak plasma concentration were observed. The clinical relevance of these decreases is unknown.

Distribution

Plasma protein binding is negligible. Metformin hydrochloride partitions into erythrocytes. The blood peak is lower than the plasma peak and appears at approximately the same time. The red blood cells most likely represent a secondary compartment of distribution. The mean volume of distribution (Vd) ranged between 63-276 L.

Biotransformation

Metformin hydrochloride is excreted unchanged in the urine. No metabolites have been identified in humans.

Elimination

Renal clearance of metformin hydrochloride is >400 mL/min, indicating that metformin hydrochloride is eliminated by glomerular filtration and tubular secretion. Following an oral dose, the apparent terminal elimination half-life is approximately 6.5 hours.

When renal function is impaired, renal clearance is decreased in proportion to that of creatinine and thus the elimination half-life is prolonged, leading to increased levels of metformin hydrochloride in plasma.

Special populations
Paediatric
Single dose study: After single doses of metformin 500 mg, paediatric patients have shown a similar pharmacokinetic profile to that observed in healthy adults.
Multiple dose study: Data are restricted to one study. After repeated doses of 500 mg twice daily for 7 days in paediatric patients the peak plasma concentration (Cmax) and systemic exposure (AUC0-t) were reduced by approximately 33% and 40%, respectively compared to diabetic adults who received repeated doses of 500 mg twice daily for 14 days. As the dose is individually titrated based on glycaemic control, this is of limited clinical relevance.
Renal impairment
The available data in subjects with moderate renal impairment are scarce and no reliable estimation of the systemic exposure to metformin in this subgroup as compared to subjects with normal renal function could be made. Therefore, the dose adaptation should be made upon clinical efficacy/tolerability considerations (see section Dosage and Administration).
Toxicology
General toxicity studies in rats up to 13 weeks were performed with the combined products in Linagliptin + Metformin HCl (Trajenta Duo). The only observed interaction between linagliptin and metformin was a reduction of body weight gain. No other additive toxicity caused by the combination of linagliptin and metformin was observed.
An animal reproduction study in pregnant rats did not indicate a teratogenic effect attributed to the co-administration of linagliptin and metformin.
The following data are findings in studies performed with linagliptin or metformin individually.
Linagliptin
Effects in non-clinical studies were observed only at exposures considered sufficiently in excess of the maximum human exposure indicating little relevance to clinical use.
Genotoxicity:
Linagliptin was not mutagenic or clastogenic with or without metabolic activation in the Ames bacterial mutagenicity assay, a chromosomal aberration test in human lymphocytes, and an in vivo micronucleus assay.
Carcinogenicity:
A two-year carcinogenicity study was conducted in male and female rats given oral doses of linagliptin of 6, 18, and 60 mg/kg/day. There was no increase in the incidence of tumours in any organ up to 60 mg/kg/day. This dose results in exposures approximately 418 times the human exposure at the maximum recommended daily adult human dose (MRHD) of 5 mg/day based on AUC comparisons. A two-year carcinogenicity study was conducted in male and female mice given oral doses of 8, 25 and 80 mg/kg/day. There was no evidence of a carcinogenic potential up to 80 mg/kg/day, approximately 242 times human exposure at the MRHD.
Reproduction toxicity:
In rat fertility studies with oral gavage doses of 10, 30 and 240 mg/kg/day, males were treated for 4 weeks prior to mating and during mating; females were treated 2 weeks prior to mating through gestation day 6. No adverse effect on early embryonic development, mating, fertility, and bearing live young were observed up to the highest dose of 240 mg/kg/day (approximately 943 times human exposure at the MRHD of 5 mg/day based on AUC comparisons). In the studies on embryo-foetal development in rats and rabbits, linagliptin was shown to be not teratogenic at dosages up to and including 240 mg/kg/day (943x MRHD) in the rat and 150 mg/kg/day (1943x MRHD) in the rabbit. A NOAEL of 30 mg/kg/day (49x MRHD) and 25 mg/kg (78x MRHD) was derived for embryo-foetal toxicity in the rat and the rabbit, respectively.
Metformin
Non-clinical data reveal no special hazard for humans based on conventional studies on safety pharmacology, genotoxicity, and carcinogenic potential. In a 13-week toxicity study in rats, metformin related toxicity was seen in heart, liver, kidneys, salivary glands, ovaries, thymus, gastrointestinal tract and adrenal glands at dosages associated with a systemic exposure of 7 times the MRHD or higher.
Metformin was not teratogenic in rats at a dose of 200 mg/kg/day associated with a systemic exposure of 4 times the MRHD (2 g metformin). At higher doses (500 and 1000 mg/kg/day, associated with 11 and 23 times the MRHD), teratogenicity of metformin was observed in the rat.

Availability
Tablets of 2.5/500 mg Bottles of 60’s
Tablets of 2.5/850 mg Bottles of 60’s
Tablets of 2.5/1 g Bottles of 60’s
Caution
Foods, Drugs, Devices and Cosmetics Act prohibits dispensing without prescription.
For any suspected adverse drug reaction, please seek medical attention immediately. Report this to the FDA: www.fda.gov.ph and email us at PV_local_Philippines@boehringer-ingelheim.com.
Store at temperatures not exceeding 30°C.
Manufactured by
Boehringer Ingelheim Promeco, S.A de C.V.
Calle del Maíz No. 49, Col. Barrio Xaltocan, C.P. 16090
Xochimilco, Mexico City, Mexico
Imported by:
Boehringer Ingelheim (Philippines), Inc.
23rd Floor BDO Towers Valero Bldg., 8741 Paseo de Roxas
Bel-Air, Makati City, Metro Manila
®
T.M. Reg. Phil. Pat. Off.
DR-XY48038 (2.5 mg/500 mg Film-coated Tablet)
Date of First Authorization: 25 May 2022
DR-XY47930 (2.5 mg/850 mg Film-coated Tablet)
DR-XY47929 (2.5 mg/1 g Film-coated Tablet)
Date of First Authorization: 12 April 2022
Date of Revision of Text: 03 August 2022
[1]
Table 2
Glycemic Parameters at Final Visit (24-Week Study) for Linagliptin and Metformin, Alone and in Combination in Patients with Type 2 Diabetes Mellitus
Inadequately Controlled on Diet and Exercise
Placebo Linagliptin 5 mg
Once Daily*
Metformin 500 mg
Twice Daily
Linagliptin
2.5 mg Twice Daily*
+ Metformin 500 mg
Metformin 1 g
HbA1c (%)
Baseline (mean) 8.7 8.7 8.7 8.7 8.5 8.7
Change from baseline (adjusted mean)
0.1 -0.5 -0.6 -1.2 -1.1 -1.6
Difference from placebo (adjusted mean) (95% CI)
-- -0.6
(-0.9, -0.3)
-0.8
(-1.0, -0.5)
-1.3
(-1.6, -1.1)
-1.2
(-1.5, -0.9)
-1.7
(-2.0, -1.4)
Patients (n, %) achieving HbA1c <7%
7 (10.8) 14 (10.4) 27 (19.1) 42 (30.7) 43 (31.2) 76 (54.3)
Patients (%) receiving rescue medication
29.2 11.1 13.5 7.3 8.0 4.3
FPG (mg/dL)
Baseline (mean) 203 195 191 199 191 196
Change from baseline (adjusted mean)

10 -9 -16 -33 -32 -49
Difference from placebo
(adjusted mean) (95% CI)
-- -19
(-31, -6)
-26
(-38, -14)
-43
(-56, -31)
-42
(-55, -30)
-60
(-72, -47)
* Total daily dose of linagliptin is equal to 5 mg P047772-004
REVERSE

Linagliptin Metformin HCl Trajanta Duo®

Oral Blood Glucose Lowering Drug Combination

Boehringer Ingelheim

Product Description
Linagliptin + Metformin HCl (Trajenta Duo) film-coated tablets, 2.5 mg / 500 mg are light yellow, oval, biconvex film-coated tablets. One side is debossed with the code 'D2 / 500', the other side is debossed with the Boehringer Ingelheim company symbol.

Linagliptin + Metformin HCl (Trajenta Duo) film-coated tablets, 2.5 mg / 850 mg are light orange, oval, biconvex film-coated tablets. One side is debossed with the code 'D2 / 850', the other side is debossed with the Boehringer Ingelheim company symbol.

Linagliptin + Metformin HCl (Trajenta Duo) film-coated tablets, 2.5 mg / 1 g are light pink, oval, biconvex film-coated tablets. One side is debossed with the code 'D2 / 1000', the other side is debossed with the Boehringer Ingelheim company symbol.

Formulation
Each film-coated tablet contains:
1H-Purine-2,6-dione, 8-[(3R)-3-amino-1-piperidinyl]-7-(2-butyn-1-yl)-3,7-dihydro-3-methyl-1-[(4-methyl-2-quinazolinyl)methyl]- (= linagliptin) 2.5 mg and
N,N-dimethylimidio-dicarbonimidic diamide hydrochloride
(= metformin hydrochloride) 500 mg, 850 mg or 1 g

Excipients: L-Arginine, Maize starch, Copovidone, Silica, colloidal anhydrous, Magnesium stearate (E470b), Hypromellose 2910, Propylene glycol, Titanium dioxide (E171), Talc, Iron oxide red/yellow (E172)

Indications
Linagliptin + Metformin HCl (Trajenta Duo) is indicated as an adjunct to diet and exercise to improve glycemic control in adults with type 2 diabetes mellitus when treatment with both linagliptin and metformin is appropriate, in patients inadequately controlled on metformin alone or those already being treated and well controlled with the free combination of linagliptin and metformin.

Linagliptin + Metformin HCl (Trajenta Duo) is indicated in combination with a sulphonylurea (i.e., triple combination therapy) as an adjunct to diet and exercise in adult patients inadequately controlled on their maximal tolerated dose of metformin and a sulphonylurea.

Linagliptin + Metformin HCl (Trajenta Duo) is indicated in combination with an SGLT2 inhibitor (i.e. triple combination therapy) as an adjunct to diet and exercise in patients inadequately controlled on their maximum tolerated dose of metformin and an SGLT2 inhibitor.

Linagliptin + Metformin HCl (Trajenta Duo) is indicated as add-on to insulin (i.e., triple combination therapy) as an adjunct to diet and exercise to improve glycemic control in adult patients when insulin and metformin alone do not provide adequate glycaemic control.

Dosage and administration

Adults with normal renal function (GFR ≥ 90mL/min)
The recommended dose is 2.5/500 mg, 2.5/ 850 mg or 2.5/1 g twice daily.

The dosage should be individualised on the basis of the patient's current regimen, effectiveness, and tolerability. Maximum recommended daily dose of Linagliptin + Metformin HCl (Trajenta Duo) is 5 mg of linagliptin and 2 g of metformin (see table 1 for additional dosing information).

Linagliptin + Metformin HCl (Trajenta Duo) should be given with meals to reduce the gastrointestinal undesirable effects associated with metformin.

For patients currently not treated with metformin
For patients currently not treated with metformin, the recommended starting dose is 2.5 mg linagliptin/500mg metformin hydrochloride twice daily.

For patients inadequately controlled on maximal tolerated dose of metformin monotherapy
For patients not adequately controlled on metformin alone, the usual starting dose of Linagliptin + Metformin HCl (Trajenta Duo) should provide linagliptin dosed as 2.5 mg twice daily (5 mg total daily dose) plus the dose of metformin already being taken.

For patients switching from co-administration of linagliptin and metformin
For patients switching from co-administration of linagliptin and metformin to the fixed dose combination, Linagliptin + Metformin HCl (Trajenta Duo) should be initiated at the dose of linagliptin and metformin already being taken.

For patients inadequately controlled on dual combination therapy with the maximal tolerated dose of metformin and a sulphonylurea
The dose of Linagliptin + Metformin HCl (Trajenta Duo) should provide linagliptin dosed as 2.5 mg twice daily (5 mg total daily dose) and a dose of metformin similar to the dose already being taken. When Linagliptin + Metformin HCl (Trajenta Duo) is used in combination with a sulphonylurea, a lower dose of the sulphonylurea may be required to reduce the risk of hypoglycaemia (see section Special Warnings and Precautions).

For patients inadequately controlled on dual combination therapy with insulin and the maximal tolerated dose of metformin
The dose of Linagliptin + Metformin HCl (Trajenta Duo) should provide linagliptin dosed as 2.5 mg twice daily (5 mg total daily dose) and a dose of metformin similar to the dose already being taken. When Linagliptin + Metformin HCl (Trajenta Duo) is used in combination with insulin, a lower dose of insulin may be required to reduce the risk of hypoglycaemia (see section on Special Warnings and Precautions).
Insulin is known to cause hypoglycaemia. Therefore, caution is advised when Linagliptin + Metformin HCl (Trajenta Duo) is used in combination with a sulphonylurea. A dose reduction of the sulphonylurea may be considered.

For the different doses on metformin, Linagliptin + Metformin HCl (Trajenta Duo) is available in strengths of 2.5 mg linagliptin plus 500 mg metformin hydrochloride, 850 mg metformin hydrochloride or 1 g metformin hydrochloride.

Renal impairment
A GFR should be assessed before initiation of treatment with metformin containing products and at least annually thereafter.

In patients at an increased risk of further progression of renal impairment and in the elderly, renal function should be assessed more frequently, e.g. every 3-6 months.

Factors that may increase the risk of lactic acidosis (see section Special Warnings and Precautions) should be reviewed before considering initiation of metformin in patients with GFR<60 mL/min.

Table 1 Posology for renally impaired patients*

eGFR mL/min	Metformin	Linagliptin
60 - 89	**Maximum daily dose is 3 g*. Dose reduction may be considered in relation to declining renal function.**	No dose adjustment is required.
45 - 59	**Maximum daily dose is 2 g. The starting dose is at most half of the maximum dose.**	No dose adjustment is required.
30 - 44	**Maximum daily dose is 1 g*. The starting dose is at most half of the maximum dose.**	No dose adjustment is required.
<30	**Metformin is contraindicated.**	No dose adjustment is required.

*If no adequate strength of Linagliptin + Metformin HCl (Trajenta Duo) is available, individual monocomponents should be used instead of the fixed dose combination.

Hepatic impairment
Linagliptin + Metformin HCl (Trajenta Duo) is contraindicated in patients with hepatic impairment due to the metformin component (see section Contraindications).

Geriatric patients
As metformin is excreted via the kidney, and elderly patients have a tendency for decreased renal function, elderly patients taking Linagliptin + Metformin HCl (Trajenta Duo) should have their renal function monitored regularly (see section Special Warnings and Precautions).

Paediatric patients
Linagliptin + Metformin HCl (Trajenta Duo) is not recommended for use in children below 18 years due to lack of data on safety and efficacy.

Missed dose
If a dose is missed, it should be taken as soon as the patient remembers. However, a double dose should not be taken at the same time. In that case the missed dose should be skipped.

Contraindications
- Hypersensitivity to active ingredients linagliptin and/or metformin hydrochloride or to any of the excipients
- Any type of acute metabolic acidosis (such as lactic acidosis, diabetic ketoacidosis)
- Diabetic pre-coma
- Severe renal failure (CrCl < 30mL/min or eGFR < 30 mL/min/1.73m²)
- Acute conditions with the potential to alter renal function such as: dehydration, severe infection, shock, intravascular administration of iodinated contrast agents (see section Special Warnings and Precautions)
- Disease which may cause tissue hypoxia (especially acute disease, or worsening of chronic disease) such as: decompensated heart failure, respiratory failure, recent myocardial infarction, shock
- Hepatic impairment
- Acute alcohol intoxication
- Alcoholism

Special Warnings and Precautions
General
Linagliptin + Metformin HCl (Trajenta Duo) should not be used in patients with type 1 diabetes.

Pancreatitis
Acute pancreatitis has been observed in patients taking linagliptin.
If pancreatitis is suspected, Linagliptin + Metformin HCl (Trajenta Duo) should be discontinued.

Hypoglycaemia
Linagliptin alone showed a comparable incidence of hypoglycaemia to placebo. In clinical trials of linagliptin as part of combination therapy with agents not considered to cause hypoglycaemia (metformin, thiazolidinediones) rates of hypoglycaemia reported with linagliptin were similar to rates in patients taking placebo.

Sulphonylureas are known to cause hypoglycaemia. Therefore, caution is advised when Linagliptin + Metformin HCl (Trajenta Duo) is used in combination with a sulphonylurea. A dose reduction of the sulphonylurea may be considered.
Insulin is known to cause hypoglycaemia. Therefore, caution is advised when Linagliptin + Metformin HCl (Trajenta Duo) is used in combination with insulin. A dose reduction of insulin may be considered.
Metformin alone does not cause hypoglycaemia under usual circumstances of use, but hypoglycaemia could occur when caloric intake is deficient, when strenuous exercise is not compensated by caloric supplementation, or during concomitant use with other glucose-lowering agents (such as sulphonylureas and insulin) or ethanol.

Lactic acidosis
Lactic acidosis, a very rare, but serious, metabolic complication, most often occurs at acute worsening of renal function or cardiorespiratory illness or sepsis. Metformin accumulation occurs at acute worsening of renal function and increases the risk of lactic acidosis.

In case of dehydration (severe diarrhoea or vomiting, fever or reduced fluid intake), metformin should be temporarily discontinued and contact with a health care professional is recommended.

Medicinal products that can acutely impair renal function (such as antihypertensives, diuretics and NSAIDs) should be initiated with caution in metformin-treated patients.

Other risk factors for lactic acidosis are excessive alcohol intake, hepatic insufficiency, inadequately controlled diabetes, ketosis, prolonged fasting and any conditions associated with hypoxia, as well as concomitant use of medicinal products that may cause lactic acidosis (see also section Contraindications and Interactions).

Patients and/or care-givers should be informed of the risk of lactic acidosis. Lactic acidosis is characterised by acidotic dyspnoea, abdominal pain, muscle cramps, asthenia and hypothermia followed by coma. In case of suspected symptoms, the patient should stop taking metformin and seek immediate medical attention.

Diagnostic laboratory findings are decreased blood pH (<7.35), increased plasma lactate levels (>5 mmol/L), and an increased anion gap and lactate/pyruvate ratio.

Administration of iodinated contrast agent
Intravascular administration of iodinated contrast agents may lead to contrast induced nephropathy, resulting in metformin accumulation and an increased risk of lactic acidosis. Metformin should be discontinued prior to or at the time of the imaging procedure and not restarted until at least 48 hours after, provided that renal function has been re-evaluated and found to be stable, see sections Patients with renal impairment and Interactions.

Renal function
GFR should be assessed before treatment initiation and regularly thereafter, see section Posology. Linagliptin + Metformin HCl (Trajenta Duo) is contraindicated in patients with GFR<30 ml/min and should be temporarily discontinued in the presence of conditions that alter renal function, see section Contraindications.

Cardiac function
Patients with heart failure are more at risk of hypoxia and renal insufficiency. In patients with stable chronic heart failure, Linagliptin + Metformin HCl (Trajenta Duo) may be used with a regular monitoring of cardiac and renal function. For patients with acute and unstable heart failure, Linagliptin + Metformin HCl (Trajenta Duo) is contraindicated due to the metformin component (see section Contraindications).

Surgery
Metformin must be discontinued at the time of surgery under general, spinal or epidural anaesthesia. Therapy may be restarted no earlier than 48 hours following surgery or resumption of oral nutrition and provided that renal function has been re-evaluated and found to be stable.

Bullous pemphigoid
Bullous pemphigoid has been observed in patients taking linagliptin.
If bullous pemphigoid is suspected, Linagliptin + Metformin HCl (Trajenta Duo) should be discontinued.

Vitamin B12
The risk of low vitamin B12 levels increases with increasing metformin dose, treatment duration, and/or in patients with risk factors known to cause vitamin B12 deficiency. In case of suspicion of vitamin B12 deficiency (such as anaemia or neuropathy), vitamin B12 serum levels should be monitored. Periodic vitamin B12 monitoring could be necessary in patients with risk factors for vitamin B12 deficiency. Metformin therapy should be continued for as long as it is tolerated and not contra-indicated and appropriate corrective treatment for vitamin B12 deficiency provided in line with current clinical guidelines.

Use in Specific Populations

Pregnancy, Lactation and Fertility

Pregnancy
There are no adequate and well-controlled studies in pregnant women with Linagliptin + Metformin HCl (Trajenta Duo) or its individual components. Non-clinical reproduction studies in pregnant rats performed with the combined products in Linagliptin + Metformin HCl (Trajenta Duo) did not indicate a teratogenic effect attributed to the co-administration of linagliptin and metformin.

There are limited data from the use of linagliptin in pregnant women. Non-clinical studies do not indicate direct or indirect harmful effects with respect to reproductive toxicity.

There are limited data from the use of metformin in pregnant women. Metformin was not teratogenic in rats at a dose of 200 mg/kg/day associated with 4-times human exposure. At higher doses (500 and 1000 mg/kg/day, associated with 11 and 23 times human exposure), teratogenicity of metformin was observed in the rat.

As a precautionary measure, it is preferable to avoid the use of Linagliptin + Metformin HCl (Trajenta Duo) during pregnancy.

When the patient plans to become pregnant and during pregnancy, diabetes should not be treated with Linagliptin + Metformin HCl (Trajenta Duo) but insulin should be used to maintain blood glucose levels as close to normal as possible in order to lower the risk of fetal malformations associated with abnormal blood glucose levels.

Lactation
No studies on lactating animals have been performed with the combination of metformin and linagliptin. Non-clinical studies with the individual active substances have shown excretion of both metformin and linagliptin into milk in lactating rats. Metformin is excreted with milk in humans. It is not known whether linagliptin is excreted into human milk. Linagliptin + Metformin HCl (Trajenta Duo) should not be used during breast-feeding.

Fertility
No studies on the effect on human fertility have been conducted for Linagliptin + Metformin HCl (Trajenta Duo). No adverse effects of linagliptin on fertility were observed in non-clinical studies up to the highest tested dose of 240 mg/kg/day (> 900 times human exposure).

Driving and Using Machines
No studies on the effects on the ability to drive and use machines have been performed.

Interactions
Pharmacokinetic Interactions

General
Co-administration of multiple doses of linagliptin (10 mg once daily) and metformin (850 mg twice daily) did not meaningfully alter the pharmacokinetics of either linagliptin or metformin in healthy volunteers. Pharmacokinetic drug interaction studies with Linagliptin + Metformin HCl (Trajenta Duo) have not been performed; however, such studies have been conducted with the individual active substances of Linagliptin + Metformin HCl (Trajenta Duo): linagliptin and metformin.

Linagliptin
In vitro assessment of drug interactions:
Linagliptin is a weak competitive and a weak to moderate mechanism-based inhibitor of CYP isozyme CYP3A4, but does not inhibit other CYP isozymes. It is not an inducer of CYP isozymes.
Linagliptin is a P-glycoprotein substrate, and inhibits P-glycoprotein mediated transport of digoxin with low potency. Based on these results and *in vivo* drug interaction studies, linagliptin is considered unlikely to cause interactions with other P-gp substrates.

In vivo assessment of drug interactions:
Clinical data described below suggest that the risk for clinically meaningful interactions by coadministered medicinal products is low. No clinically significant interactions requiring dose adjustment were observed. Linagliptin had no clinically relevant effect on the pharmacokinetics of metformin, glibenclamide, simvastatin, pioglitazone, warfarin, digoxin or oral contraceptives providing in vivo evidence of a low propensity for causing drug interactions with substrates of CYP3A4, CYP2C9, CYP2C8, P-glycoprotein, and organic cationic transporter (OCT).

Metformin: Co-administration of multiple three-times-daily doses of 850 mg metformin with a supratherapeutic dose of 10 mg linagliptin once daily did not clinical meaningfully alter the pharmacokinetics of linagliptin or metformin in healthy volunteers. Therefore, linagliptin is not an inhibitor of OCT-mediated transport.

Sulphonylureas: The steady-state pharmacokinetics of 5 mg linagliptin was not changed by co-administration of a single 1.75 mg dose glibenclamide (glyburide) and multiple oral doses of 5 mg linagliptin. However there was a clinically not relevant reduction of 14% of both AUC and C_{max} of glibenclamide. Because glibenclamide is primarily metabolized by CYP2C9, these data also support the conclusion that linagliptin is not a CYP2C9 inhibitor. Clinically meaningful interactions would not be expected with other sulphonylureas (e.g. glipizide, tolbutamide and glimepiride) which, like glibenclamide, are primarily eliminated by CYP2C9.

*Thiazolidinediones: Co-administration of multiple daily doses of 10 mg linagliptin (supratherapeutic) with multiple daily doses of 45 mg pioglitazone, a CYP2C8 and CYP3A4 substrate, had no clinically relevant effect on the pharmacokinetics of either linagliptin or pioglitazone or the active metabolites of pioglitazone, indicating that linagliptin is not an inhibitor of CYP2C8-mediated metabolism *in vivo* and supporting the conclusion that the *in vivo* inhibition of CYP3A4 by linagliptin is negligible.*

Ritonavir: A study was conducted to assess the effect of ritonavir, a potent inhibitor of P-glycoprotein and CYP3A4, on the pharmacokinetics of linagliptin. Co-administration of a single 5 mg oral dose of linagliptin and multiple 200 mg oral doses of ritonavir increased the AUC and C_{max} of linagliptin approximately twofold and threefold, respectively. Simulations of steady-state plasma concentrations of linagliptin with and without ritonavir indicated that the increase in exposure will not be associated with an increased accumulation. These changes in linagliptin pharmacokinetics were not considered to be clinically relevant. Therefore, clinically relevant interactions would not be expected with other P-glycoprotein/CYP3A4 inhibitors and dose adjustment is not required.

Rifampicin: A study was conducted to assess the effect of rifampicin, a potent inducer of P-glycoprotein and CYP3A4, on the pharmacokinetics of 5 mg linagliptin. Multiple co-administration of linagliptin with rifampicin, resulted in a 39.6% and 43.8% decreased linagliptin steady-state AUC and C_{max} and about 30% decreased DPP-4 inhibition at trough. Thus linagliptin in combination with strong P-gp inducers is expected to be clinically efficacious, although full efficacy might not be achieved.

*Digoxin: Co-administration of multiple daily doses of 5 mg linagliptin with multiple doses of 0.25 mg digoxin had no effect on the pharmacokinetics of digoxin in healthy volunteers. Therefore, linagliptin is not an inhibitor of P-glycoprotein-mediated transport *in vivo*.*

Warfarin: Multiple daily doses of 5 mg linagliptin did not alter the pharmacokinetics of S(-) or R(+) warfarin, a CYP2C9 substrate, showing that linagliptin is not an inhibitor of CYP2C9.

Simvastatin: Multiple daily doses of 10 mg linagliptin (supratherapeutic) had a minimal effect on the steady state pharmacokinetics of simvastatin, a sensitive CYP3A4 substrate, in healthy volunteers. Following administration of 10 mg linagliptin concomitantly with 40 mg of simvastatin daily for 6 days, the plasma AUC of simvastatin was increased by 34%, and the plasma C_{max} by 10%. Therefore, linagliptin is considered to be a weak inhibitor of CYP3A4-mediated metabolism, and dosage adjustment of concomitantly administered substances metabolised by CYP3A4 is considered unnecessary.

Oral Contraceptives: Co-administration with 5 mg linagliptin did not alter the steady-state pharmacokinetics of levonorgestrel or ethinylestradiol.

The absolute bioavailability of linagliptin is approximately 30%. Because co-administration of a high-fat meal with linagliptin had no clinically relevant effect on the pharmacokinetics, linagliptin may be administered with or without food.

Metformin
Concomitant use not recommended

Alcohol
Alcohol intoxication is associated with an increased risk of lactic acidosis, particularly in cases of fasting, malnutrition or hepatic insufficiency.

Iodinated contrast agents
Linagliptin + Metformin HCl (Trajenta Duo) must be discontinued prior to, or at the time of the imaging procedure and not be restarted until at least 48 hours after, provided that renal function has been re-evaluated and found to be stable (see section Posology and Special Warnings and Precautions).

Combination requiring precautions for use
Some medicinal products can adversely affect renal function which may increase the risk of lactic acidosis, e.g. NSAIDs, including selective cyclo-oxygenase (COX) II inhibitors, ACE inhibitors, angiotensin II receptor antagonists and diuretics, especially loop diuretics. When starting or using such products in combination with metformin, close monitoring of renal function is necessary.

Organic cation transporters (OCT)
Metformin is a substrate of both transporters OCT1 and OCT2.
Co-administration of metformin with:
- Inhibitors of OCT1 (such as verapamil) may reduce efficacy of metformin.
- Inducers of OCT1 (such as rifampicin) may increase gastrointestinal absorption and efficacy of metformin.
- Inhibitors of OCT2 (such as cimetidine, dolutegravir, ranolazine, trimethoprim, vandetanib, isavuconazole) may decrease the renal elimination of metformin and thus lead to an increase in metformin plasma concentration.
- Inhibitors of both OCT1 and OCT2 (such as crizotinib, olaparib) may alter efficacy and renal elimination of metformin.

Caution is therefore advised, especially in patients with renal impairment, when these drugs are co-administered with metformin, as metformin plasma concentration may increase. If needed, dose adjustment of metformin may be considered as OCT inhibitors/inducers may alter the efficacy of metformin.

Adverse Reactions
Summary of the safety profile
The safety of linagliptin 2.5 mg twice daily (or its bioequivalent of 5 mg once daily) plus metformin has been evaluated in over 6800 patients with type 2 diabetes mellitus (T2DM).

In placebo-controlled studies, more than 1800 patients were treated with the therapeutic dose of either 2.5 mg linagliptin twice daily (or its bioequivalent of 5 mg linagliptin once daily) in combination with metformin for ≥12/24 weeks.

In the pooled analysis of the placebo-controlled trials, the overall incidence of adverse events (AEs) in patients treated with placebo and metformin was comparable to linagliptin 2.5 mg and metformin (54.3% and 49.0%). Discontinuation of therapy due to AEs was comparable in patients who received placebo and metformin to patients treated with linagliptin and metformin (3.8% and 2.9%).

Due to the impact of the background therapy on AEs (e.g. on hypoglycaemias), adverse events were analysed and displayed based on the respective treatment regimens, add-on to metformin, and add-on to metformin plus sulphonylurea. The placebo-controlled studies included 7 studies where linagliptin was given as add-on to metformin and 1 study where linagliptin was given as add-on to metformin + sulphonylurea.

Tabulated summary of adverse reactions
The following adverse reactions derived from the use of the linagliptin/metformin combination or the use of the monocomponents (linagliptin or metformin) in clinical trials or from post-marketing experience are shown in the table below. Undesirable effects previously reported with one of the individual components may be potential undesirable effects with Linagliptin + Metformin HCl (Trajenta Duo) even if not observed in clinical trials with this product.

MedDRA System Organ Class terminology	*Linagliptin and Metformin Adverse reactions*
Infections and infestations	Nasopharyngitis [1]
Immune system disorders	Hypersensitivity [1, 3] Angioedema [4] Urticaria [2,4]
Metabolism and nutrition disorders	Lactic acidosis [2] Vitamin B12 decrease/deficiency[2] Hypoglycaemia (when linagliptin and metformin were combined with a sulphonylurea)
Nervous system disorders	Taste disturbance [2]
Respiratory, thoracic and mediastinal disorders	Cough [1, 3]
Gastrointestinal disorders**	Decreased appetite [2, 5] Diarrhoea [1, 5] Constipation (when linagliptin and metformin were combined with insulin) Nausea [1, 5] Pancreatitis [3] Vomiting [1, 5] Abdominal pain [2] Mouth ulceration [4]
Hepato-biliary disorders	Liver function test abnormal [2] Hepatitis [2]
Skin and subcutaneous tissue disorders	Pruritus [3, 5] Erythema [2] Rash [4] Bullous pemphigoid [4, §]
Investigations	Lipase increased [2, ¶] Amylase increased [¶]

MedDRA System Organ Class terminology | *Linagliptin and Metformin Adverse reactions*

1 Adverse reactions reported also in patients treated with linagliptin monotherapy
2 Adverse reactions of metformin as monotherapy; refer to the Summary of Product Characteristics for metformin for additional information
3 Adverse reactions of FDC linagliptin + metformin (pooled analysis of placebo-controlled studies)
4 Adverse reactions identified from post-marketing experience with linagliptin
5 Adverse reactions reported in patients who received FDC linagliptin+metformin and also in patients who received metformin monotherapy
** Gastrointestinal disorders such as abdominal pain and nausea, vomiting, diarrhoea, decreased appetite occur most frequently during initiation of therapy with metformin hydrochloride and resolve spontaneously in most cases. To prevent them, it is recommended that metformin hydrochloride be taken in 2 daily doses during or after meals if administered as monotherapy.
§ See also Linagliptin cardiovascular and renal safety study (CARMELINA) below
¶ Based on lipase increase >3 times ULN.
ψ In the CAROLINA study comparing linagliptin with active comparator glimepiride (see section Clinical Trials) laboratory analysis of amylase showed increases to >3xULN in 0.99% of patients treated with linagliptin and in 0.54% of patients treated with glimepiride.

In placebo-controlled studies the most frequently reported related adverse reaction for linagliptin+metformin was diarrhoea (1.6%) with comparable rate on metformin+placebo (2.4%).

Adverse reactions reported when linagliptin and metformin were combined with SU:
When linagliptin and metformin were administered in combination with a sulphonylurea, hypoglycaemia was the most commonly reported AE (linagliptin plus metformin plus sulphonylurea 23.9% vs 16.0% in the placebo group) and identified as an additional adverse reaction under these conditions. None of the hypoglycaemias episodes were classified as severe (requiring assistance).

Adverse reactions reported when linagliptin and metformin were combined with insulin:
When linagliptin and metformin were administered in combination with insulin, hypoglycaemia was the most commonly reported AE, but occurred at comparable rate when placebo and metformin were combined with insulin (-67.1 mg/dL (-3.7 mmol/L) compared to placebo, as well as a greater portion of patients achieving a target HbA_{1c} of < 7.0% (28.3% on linagliptin vs. 11.4% on placebo). The observed incidence of hypoglycaemia in patients treated with linagliptin was similar to placebo. Body-weight did not differ significantly between the groups.

In a 24-week placebo-controlled factorial study of initial therapy, linagliptin 2.5 mg twice daily in combination with metformin (500 mg or 1 g twice daily) provided significant improvements in glycaemic parameters compared with either monotherapy as summarized in table 2 (mean baseline HbA_{1c} 8.65%).

[1]

Mean reductions from baseline in HbA_{1c} were generally greater for patients with higher baseline HbA_{1c} values. Effects on plasma lipids were generally neutral. The decrease in body-weight with the combination of linagliptin and metformin was similar to that observed for metformin alone or placebo; there was no change from baseline for patients on linagliptin alone. The incidence of hypoglycaemia was similar across treatment groups (placebo 1.4%, linagliptin 5 mg 0%, metformin 2.1%, and linagliptin 2.5mg plus metformin twice daily 1.4%).

In addition, this study included patients (n=66) with more severe hyperglycaemia (HbA_{1c} at baseline ≥ 11%) who were treated with twice daily open-label linagliptin 2.5 mg + metformin 1 g. In this group of patients, the mean baseline HbA_{1c} value was 11.8% and mean FPG was 261.8 mg/dL (14.5 mmol/L). A mean decrease from baseline of -3.74% in HbA_{1c} (n=48) and -81.2 mg/dL (-4.5 mmol/L) for FPG (n=41) was observed for patients completing the 24 week trial period without rescue therapy (n=48).

In the LOCF analysis including all patients with primary endpoint measurements (n=65) at last observation without rescue therapy changes from baseline were -3.19% for HbA_{1c} and -73.6 mg/dL (- 4.1 mmol/L) for FPG.

The efficacy and safety of linagliptin 2.5 mg twice daily versus 5 mg once daily in combination with metformin in patients with insufficient glycaemic control on metformin monotherapy was evaluated in a double blind placebo controlled study of 12 weeks duration. Linagliptin (2.5 mg twice daily and 5 mg once daily) added to metformin provided significant improvements in glycaemic parameters compared to placebo. Linagliptin 5 mg once daily and 2.5 mg twice daily provided comparable (CI: -0.07; 0.19), significant HbA_{1c} reductions of -0.80 % (from baseline 7.98%), and -0.74 (from baseline 7.96%) compared to placebo.

employ clinical monitoring and institute clinical measures as required. The most effective method to remove lactate and metformin hydrochloride is haemodialysis.

Pharmacological properties
Pharmacotherapeutic group: Combinations of oral blood glucose lowering drugs, ATC code: A10BD11

Mode of Action
Linagliptin is an inhibitor of the enzyme DPP-4 (Dipeptidyl peptidase 4, EC 3.4.14.5) an enzyme which is involved in the inactivation of the incretin hormones GLP-1 and GIP (glucagon-like peptide-1, glucose-dependent insulinotropic polypeptide). These hormones are rapidly degraded by the enzyme DPP-4. Both incretin hormones are involved in the physiological regulation of glucose homeostasis. Incretins are secreted at a low basal level throughout the day and levels rise immediately after meal intake. GLP-1 and GIP increase insulin biosynthesis and secretion from pancreatic beta cells in the presence of normal and elevated blood glucose levels. Furthermore GLP-1 also reduces glucagon secretion from pancreatic alpha cells, resulting in a reduction in hepatic glucose output. Linagliptin binds very effectively to DPP-4 in a reversible manner and thus leads to a sustained increase and a prolongation of active incretin levels. Linagliptin glucose-dependently increases insulin secretion and lowers glucagon secretion thus resulting in an overall improvement in the glucose homoeostasis. Linagliptin binds selectively to DPP-4 and exhibits a >10000 fold selectivity versus DPP-8 or DPP-9 activity in vitro.
Metformin hydrochloride is a biguanide with antihyperglycaemic effects, lowering both basal and postprandial plasma glucose. It does not stimulate insulin secretion and therefore does not produce hypoglycaemia.

Metformin hydrochloride may act via 3 mechanisms:
(1) reduction of hepatic glucose production by inhibiting gluconeogenesis and glycogenolysis
(2) in muscle, by increasing insulin sensitivity, improving peripheral glucose uptake and utilisation
(3) and delay of intestinal glucose absorption.

Metformin hydrochloride stimulates intracellular glycogen synthesis by acting on glycogen synthase.

Metformin hydrochloride increases the transport capacity of all types of membrane glucose transporters (GLUTs) known to date.

In humans, independently of its action on glycaemia, metformin hydrochloride has favourable effects on lipid metabolism. This has been shown at therapeutic doses in controlled, medium-term or long-term clinical studies: metformin hydrochloride reduces total cholesterol, LDL cholesterol and triglyceride levels.

Clinical trials
Linagliptin as add-on to metformin therapy
The efficacy and safety of linagliptin in combination with metformin in patients with insufficient glycaemic control on metformin monotherapy was evaluated in a double blind placebo controlled study of 24 weeks duration.

Linagliptin added to metformin provided significant improvements in HbA_{1c}, (-0.64 % compared to placebo), from a mean baseline HbA_{1c} of 8 %. Linagliptin also showed significant improvements in fasting plasma glucose (FPG) by -21.1 mg/dL (-1.2 mmol/L) and 2-hour post-prandial glucose (PPG) by -67.1 mg/dL (-3.7 mmol/L) compared to placebo, as well as a greater portion of patients achieving a target HbA_{1c} of < 7.0% (28.3% on linagliptin vs. 11.4% on placebo). The observed incidence of hypoglycaemia in patients treated with linagliptin was similar to placebo. Body-weight did not differ significantly between the groups.

Linagliptin cardiovascular outcome and renal safety study (CARMELINA)
The CARMELINA study evaluated the cardiovascular and renal safety of linagliptin versus placebo in patients with type 2 diabetes and with increased CV risk evidenced by a history of established macrovascular or renal disease (see section Clinical Trials). The study included 3494 patients treated with linagliptin (5 mg) and 3485 patients treated with placebo. Both treatments were added to standard of care targeting regional standards for HbA_{1c} and CV risk factors; with 54% on metformin. The overall incidence of adverse events and serious adverse events in patients receiving linagliptin was similar to that in patients receiving placebo. Safety data from this study was in line with previous known safety profile of linagliptin.

In the treated population, severe hypoglycaemic events (requiring assistance) were reported in 3.0% of patients on linagliptin and in 3.1% on placebo. Among patients who were using sulphonylureas at baseline, the incidence of severe hypoglycaemia was 2.0% in linagliptin-treated patients and 1.7% in placebo treated patients. Among patients who were using insulin at baseline, the incidence of severe hypoglycaemia was 4.4% in linagliptin-treated patients and 4.9% in placebo treated patients.

In the overall study observation period adjudicated acute pancreatitis was reported in 0.3% of patients treated with linagliptin and in 0.1% of patients treated with placebo.

In the CARMELINA study, bullous pemphigoid was reported in 0.2% of patients treated with linagliptin and in no patient treated with placebo.

Overdose
Symptoms
During controlled clinical trials in healthy subjects, single doses of up to 600 mg linagliptin (equivalent to 120 times the recommended dose) were well tolerated. There is no experience with doses above 600 mg in humans.

Hypoglycaemia has not been seen with metformin hydrochloride doses of up to 85 g, although lactic acidosis has occurred in such circumstances. High overdose of metformin hydrochloride or concomitant risks may lead to lactic acidosis. Lactic acidosis is a medical emergency and must be treated in hospital.

Therapy
In the event of an overdose, it is reasonable to employ the usual supportive measures, e.g., remove unabsorbed material from the gastrointestinal tract,



FRONT

File information

	Mandatory in	
	TD	Proofing
Issue date of TD: 08/Jul/2022	Yes	No
PPM SKU: PSN1772	No	Yes
PPM SKU version: 004	Yes	Yes
Issue date of artwork: 08/AMR/2023	Yes	No
Print colors: PAN Black	Yes	No
Mat. No. Pack. Site: PSN1772-004	Yes	No
Min. font size: 7pt (ISO 21857)		
Legend case version: V5.0 01 JAN/2021 (please do not change or remove it)		

In-process control code specification

Control Code Type: Lantus Code



Technical colours - not for printing

80 Diecut Legend case | Free area | Glueprints

Additional Requirements of Packaging site

Dimensions: 850 x 180 mm

Front/Reverse Information should stay at 80 Legend case for further workflow. Please do not remove.

The observed incidence of hypoglycaemia in patients treated with linagliptin was similar to placebo (2.2% on linagliptin 2.5 mg twice daily, 0.9% on linagliptin 5 mg once daily, and 2.3% on placebo). Body weight did not differ significantly between the groups.

Linagliptin as add-on to a combination of metformin and sulphonylurea therapy

A placebo controlled study of 24 weeks in duration was conducted to evaluate the efficacy and safety of linagliptin 5 mg to placebo, in patients not sufficiently controlled with a combination with metformin and a sulphonylurea. Linagliptin provided significant improvements in HbA_{1c} (-0.62 % change compared to placebo), from a mean baseline HbA_{1c} of 8.14%.

Linagliptin also showed significant improvements in patients achieving a target HbA_{1c} of < 7.0% (31.2% on linagliptin vs. 9.2% on placebo), and also for fasting plasma glucose (FPG) with -12.7 mg/dL (-0.7 mmol/L) reduction compared to placebo. Body weight did not differ significantly between the groups.

Linagliptin as add on to a combination of metformin and empagliflozin

In patients inadequately controlled with metformin and empagliflozin (10 mg (n=247) or 25 mg (n=217)), 24-weeks treatment with add-on therapy of linagliptin 5 mg provided adjusted mean HbA_{1c} reductions from baseline by -0.53% (significant difference to add-on placebo -0.32% (95% CI -0.25, -0.13) and -0.58% (significant difference to add-on placebo -0.47% (95% CI -0.66; -0.28), respectively. A statistically significant greater proportion of patients with a baseline HbA_{1c} ≥ 7.0% and treated with linagliptin 5 mg achieved a target HbA_{1c} of <7% compared to placebo.

In prespecified subgroups of patients with baseline HbA_{1c} greater or equal than 8.5% (n=66 and n=42 patients on metformin plus empagliflozin 10 mg or 25 mg, respectively), the adjusted mean HbA_{1c} reductions from baseline to 24-weeks on add-on therapy with linagliptin 5 mg were -0.97% (p<0.0875, for difference to add-on placebo) and -1.16% (p<0.0044 for difference to add-on placebo), respectively.

Linagliptin in combination with metformin and insulin

A 24-week placebo-controlled study was conducted to evaluate the efficacy and safety of linagliptin (5 mg once daily) added to insulin with or without metformin. 83 % of patients were taking metformin in combination with insulin in this trial. Linagliptin in combination with metformin plus insulin provided significant improvements in HbA_{1c} in this subgroup with -0.68 (CI -0.78; -0.57) additional change from baseline (mean baseline HbA_{1c} 8.28%) compared to placebo in combination with metformin plus insulin. There was no meaningful change from baseline in body weight in either group.

Linagliptin 24 month data, as add- on to metformin in comparison with glimepiride

In a study comparing the efficacy and safety of the addition of linagliptin 5 mg or glimepiride (a sulphonylurea agent) in patients with inadequate glycaemic control on metformin monotherapy, linagliptin was similar to glimepiride in reducing HbA_{1c}, with a mean treatment difference in HbA_{1c} from baseline to 104 weeks for linagliptin compared to glimepiride of + 0.20%.

In this study, the proinsulin to insulin ratio, a marker of efficiency of insulin synthesis and release, showed a statistically significant improvement with linagliptin compared with glimepiride treatment. The incidence of hypoglycaemia in the linagliptin group (7.5%) was significantly lower than that in the glimepiride group (36.1%).

Patients treated with linagliptin exhibited a significant mean decrease from baseline in body weight compared to a significant weight gain in patients administered glimepiride (-1.39 vs + 1.29 kg).

Linagliptin as add-on therapy in elderly patients (age ≥ 70 years) with type 2 diabetes

The efficacy and safety of linagliptin in elderly (age ≥ 70 years) type 2 diabetes patients was evaluated in a double blind study of 24 weeks duration. Patients received metformin and/or sulphonylurea and/or insulin as background therapy. Doses of background antidiabetic medications were kept stable during the first 12 weeks, after which adjustments were permitted. Linagliptin provided significant improvements in HbA_{1c} of -0.64 % (95% CI -0.81, -0.48; p<0.0001) compared to placebo after 24 weeks, from a mean baseline HbA_{1c} of 7.8%.

Linagliptin also showed significant improvements in fasting plasma glucose (FPG) of -20.7mg/dL (95% CI -30.2, -11.2; p<0.0001) compared to placebo (-1.1 mmol/L). Body weight did not differ significantly between the groups. Overall, the incidence of hypoglycaemia was comparable between linagliptin (2 of 45 patients, 4.4%) and placebo (none of 22 patients, 0%) on the background of metformin alone. Hypoglycaemia rates were also comparable on a background of insulin with or without metformin (13 of 35 patients, 37.1% treated with linagliptin and 6 of 15 patients, 40.0% treated with placebo). However, on a background of sulphonylurea with or without metformin, hypoglycaemia was reported in a higher proportion of patients treated with linagliptin (24 of 82 patients, 29.3%) compared to placebo (7 of 42 patients, 16.7%). There was no difference between linagliptin and placebo in severe hypoglycaemic events.

In a pooled analysis of elderly (age ≥70 years) patients with type 2 diabetes (n=183) who were taking both metformin and basal insulin as background therapy, linagliptin in combination with metformin plus insulin provided significant improvements in HbA_{1c} parameters with -0.81 (CI: - 1.01, -0.61) adjusted mean change from baseline (mean baseline HbA_{1c} 8.13%) compared to placebo in combination with metformin plus insulin. There was no clinically meaningful difference in the incidence of hypoglycaemic events, in patient´s ≥70 years (37.2% on linagliptin in combination with metformin plus insulin vs. 39.8% on placebo in combination with metformin plus insulin).

Linagliptin and initial combination with Linagliptin and Metformin in recently diagnosed treatment naïve patients with marked hyperglycaemia:

The efficacy and safety of the initial combination of linagliptin 5 mg once daily and metformin twice daily (uptitrated in the first 6 weeks to 1500 mg or 2000 mg/d) compared to linagliptin 5 mg once has been studied in a 24 week trial in recently diagnosed treatment naïve patients with T2DM and marked hyperglycaemia (baseline HbA_{1c} 8.5–12.0%). After 24 weeks both linagliptin monotherapy as well as the initial combination of linagliptin and metformin significantly reduced HbA_{1c} levels by -2.0% and -2.8% respectively, from a baseline HbA_{1c} of 9.9% and 9.8% respectively. The treatment difference of −0.8% (95% CI −1.1 to −0.5) showed superiority for the initial combination over monotherapy (p<0.0001). Notably, 40% and 61% of patients in the monotherapy and combination arms achieved HbA_{1c} <7.0%.

Linagliptin cardiovascular and renal safety study (CARMELINA)

CARMELINA was a randomized study in 6979 patients with type 2 diabetes with increased CV risk evidenced by a history of established macrovascular or renal disease who were treated with linagliptin 5 mg (3494) or placebo (3485) added to standard of care targeting regional standards for HbA_{1c}. CV risk factors and renal disease. The study population included 1,211 (17.4%) patients ≥ 75 years of age and 4,348 (62.3%) patients with renal impairment. Approximately 19% of the population had eGFR ≥45 to <60 mL/min/1.73 m², 28% of the population had eGFR ≥30 to <45 mL/min/1.73 m² and 15% had eGFR < 30 mL/ min/1.73 m².

The mean HbA_{1c} at baseline was 8.0%.

The study was designed to demonstrate non-inferiority for the primary cardiovascular endpoint which was a composite of the first occurrence of cardiovascular death or a non-fatal myocardial infarction (MI) or a non-fatal stroke (3P-MACE). The renal composite endpoint was defined as renal death or sustained end stage renal disease or sustained decrease of 40% or more in eGFR.

After a median follow up of 2.2 years, linagliptin, when added to standard of care, did not increase the risk of major adverse cardiovascular events or renal outcome events (Table 3 and Figure 1). There was no increased risk in hospitalization for heart failure which was an additional adjudicated endpoint observed compared to standard of care without linagliptin in patients with type 2 diabetes (Table 4).

[1]

Table 2

Glycemic Parameters at Final Visit (24-Week Study) for Linagliptin and Metformin, Alone and in Combination in Patients with Type 2 Diabetes Mellitus Inadequately Controlled on Diet and Exercise

	Placebo	Linagliptin 5 mg Once Daily*	Metformin 500 mg Twice Daily	Linagliptin 2.5 mg Twice Daily* + Metformin 500 mg Twice Daily	Metformin 1 g Twice Daily	Linagliptin 2.5 mg Twice Daily* + Metformin 1 g Twice Daily
HbA_{1c} (%)						
Number of patients	n = 65	n = 135	n = 141	n = 137	n = 138	n = 140
Baseline (mean)	8.7	8.7	8.7	8.7	8.5	8.7
Change from baseline (adjusted mean)	0.1	-0.5	-0.6	-1.2	-1.1	-1.6
Difference from placebo (adjusted mean) (95% CI)	--	-0.6 (-0.9, -0.3)	-0.8 (-1.0, -0.5)	-1.3 (-1.6, -1.1)	-1.2 (-1.5, -0.9)	-1.7 (-2.0, -1.4)
Patients (%) achieving HbA_{1c} <7%	7 (10.8)	14 (10.4)	27 (19.1)	42 (30.7)	43 (31.2)	76 (54.3)
Patients (%) receiving rescue medication	29.2	11.1	13.5	7.3	8.0	4.3
FPG (mg/dL)						
Number of patients	n = 61	n = 134	n = 136	n = 135	n = 132	n = 136
Baseline (mean)	203	195	191	199	191	196
Change from baseline (adjusted mean)	10	-9	-16	-33	-32	-49
Difference from placebo (adjusted mean) (95% CI)	--	-19 (-31, -6)	-26 (-38, -14)	-43 (-56, -31)	-42 (-55, -30)	-60 (-72, -47)

* Total daily dose of linagliptin is equal to 5 mg

Table 3 Major adverse cardiovascular events (MACE) and renal outcome events by treatment group in the CARMELINA study

	Linagliptin 5mg		Placebo		Hazard Ratio
	Number of Subjects (%)	Incidence Rate per 1000 PY*	Number of Subjects (%)	Incidence Rate per 1000 PY*	(95% CI)
Number of patients	3494		3485		
Primary CV composite (Cardiovascular death, non-fatal MI, non-fatal stroke)	434 (12.4)	57.7	420 (12.1)	56.3	1.02 (0.89, 1.17)**
Secondary renal composite (renal death, ESRD, 40% sustained decrease in eGFR)	327 (9.4)	48.9	306 (8.8)	46.6	1.04 (0.89, 1.17)

* PY=patient years

** Test on non-inferiority to demonstrate that the upper bound of the 95% CI for the hazard ratio is less than 1.3

Figure 1 Time to first occurrence of 3P-MACE in CARMELINA

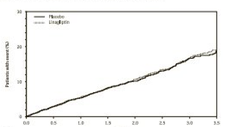


Table 4 Hospitalization for heart failure and mortality by treatment group in the CARMELINA study

	Linagliptin 5mg		Placebo		Hazard Ratio
	Number of Subjects (%)	Incidence Rate per 1000 PY*	Number of Subjects (%)	Incidence Rate per 1000 PY*	(95% CI)
Number of patients	3494		3485		
All-cause mortality	367 (10.5)	46.9	373 (10.7)	48.0	0.98 (0.84, 1.13)
CV death	255 (7.3)	32.6	264 (7.6)	34	0.96 (0.81, 1.14)
Hospitalization for heart failure	209 (6.0)	27.7	226 (6.5)	30.4	0.90 (0.74, 1.08)

* PY=patient years

In analyses for albuminuria progression (change from normoalbuminuria to micro- or macroalbuminuria, or from microalbuminuria to macroalbuminuria) the estimated hazard ratio was 0.86 (95% CI 0.78, 0.95) for linagliptin versus placebo. The microvascular endpoint was defined as the composite of renal death, sustained ESRD, sustained decrease of ≥50% in eGFR, albuminuria progression, use of retinal photocoagulation or intravitreal injections of an anti-VEGF therapy for diabetic retinopathy or vitreous haemorrhage or diabetes-related-blindness. The estimated hazard ratio for time to first occurrence for the composite microvascular endpoint was 0.86 (95% CI 0.78, 0.95) for linagliptin versus placebo, mainly driven by albuminuria progression.

Linagliptin cardiovascular safety study (CAROLINA)

CAROLINA was a randomized study in 6033 patients with early type 2 diabetes and increased CV risk or established complications who were treated with linagliptin 5 mg (3023) or glimepiride 1–4 mg (3010) added to standard of care (including background therapy with metformin in 83% of patients) targeting regional standards for HbA_{1c} and CV risk factors. The mean age for study population was 64 years and included 2030 (34%) patients ≥ 70 years of age. The study population included 2089 (35%) patients with cardiovascular disease and 1130 (19%) patients with renal impairment with an eGFR < 60ml/ min/1.73m² at baseline. The mean HbA_{1c} at baseline was 7.15%.

The study was designed to demonstrate non-inferiority for the primary cardiovascular endpoint which was a composite of the first occurrence of cardiovascular death or a non-fatal myocardial infarction (MI) or a non-fatal stroke (3P-MACE).

After a median follow up of 6.25 years, linagliptin, when added to standard of care, did not increase the risk of major adverse cardiovascular events (Table 5) as compared to glimepiride. Results were consistent for patients treated with or without metformin.

Table 5 Major adverse cardiovascular events (MACE) and mortality by treatment group in the CAROLINA study

	Linagliptin 5mg		Glimepiride (1-4mg)		Hazard Ratio
	Number of Subjects (%)	Incidence Rate per 1000 PY*	Number of Subjects (%)	Incidence Rate per 1000 PY*	(95% CI)
Number of patients	3023		3010		
Primary CV composite (Cardiovascular death, non-fatal MI, non-fatal stroke)	356 (11.8)	20.7	362 (12.0)	21.2	0.98 (0.84, 1.14)**

* PY=patient years

** Test on non-inferiority to demonstrate that the upper bound of the 95% CI for the hazard ratio is less than 1.3

The composite of treatment sustainability, a key secondary endpoint, was defined as the proportion of patients on study treatment following initial titration period (16 weeks) that maintain glycaemic control (HbA_{1c} ≤ 7.0%) at final visit without need for additional antidiabetic drug therapy (rescue medication) without any moderate (symptomatic with glucose value ≤ 70mg/ dL) or severe (requiring assistance) hypoglycaemic episodes and without > 2% weight gain. A higher number of patients on linagliptin (481, 16.0%) achieved this key secondary endpoint compared to glimepiride (305, 10.2%).

For the entire treatment period (median time on treatment 5.9 years) the rate of patients with moderate or severe hypoglycaemia was 6.5% on linagliptin versus 30.9% on glimepiride, severe hypoglycaemia occurred in 0.3% of patients on linagliptin versus 2.2% on glimepiride.

Pharmacokinetics

Bioequivalence studies in healthy subjects demonstrated that the Linagliptin + Metformin HCl (Trajenta Duo) (linagliptin/metformin hydrochloride) combination tablets are bioequivalent to co-administration of linagliptin and metformin hydrochloride as individual tablets.

Administration of Linagliptin + Metformin HCl (Trajenta Duo) 2.5 mg/1 g with food resulted in no change in overall exposure of linagliptin. With metformin there was no change in AUC, however mean peak serum concentration of metformin was decreased by 18% when administered with food. A delayed time to peak serum concentrations by 2 hours was observed for metformin under fed conditions. These changes are not likely to be clinically significant.

The following statements reflect the pharmacokinetic properties of the individual active substances of Linagliptin + Metformin HCl (Trajenta Duo).

Linagliptin

The pharmacokinetics of linagliptin has been extensively characterized in healthy subjects and patients with type 2 diabetes. After oral administration of a 5 mg dose to healthy volunteers or patients, linagliptin was rapidly absorbed, with peak plasma concentrations (median T_{max}) occurring 1.5 hours postdose. Plasma concentrations of linagliptin decline in at least a bi-phasic with a long terminal half-life (terminal half-life for linagliptin more than 100 hours), that is mostly related to the saturable, tight binding of linagliptin to DPP-4 and does not contribute to the accumulation of the drug. The effective half-life for accumulation of linagliptin, as determined from oral administration of multiple doses of 5 mg linagliptin, is approximately 12 hours. After once-daily dosing, steady-state plasma concentrations of 5 mg linagliptin are reached by the third dose.

Plasma AUC of linagliptin increased approximately 33% following 5 mg doses at steady-state compared to the first dose. The intra-subject and inter-subject coefficients of variation for linagliptin AUC were small (12.6% and 28.5%, respectively).

Plasma AUC of linagliptin increased in a less than dose-proportional manner. The pharmacokinetics of linagliptin was generally similar in healthy subjects and in patients with type 2 diabetes.

Absorption

The absolute bioavailability of linagliptin is approximately 30%. Because coadministration of a high-fat meal with linagliptin had no clinically relevant effect on the pharmacokinetics, linagliptin may be administered with or without food. *In vitro* studies indicated that linagliptin is a substrate of P-glycoprotein and of CYP3A4. Ritonavir, a potent inhibitor of P-glycoprotein and CYP3A4, led to a twofold increase in exposure (AUC) and multiple co-administration of linagliptin with rifampicin, a potent inducer of P-gp and CYP3A, resulted in an about 40% decreased linagliptin steady-state AUC, presumably by increasing/decreasing the bioavailability of linagliptin by inhibition/induction of P-glycoprotein.

Distribution

As a result of tissue binding, the mean apparent volume of distribution at steady state following a single 5 mg intravenous dose of linagliptin to healthy subjects is approximately 1110 litres, indicating that linagliptin extensively distributes to the tissues. Plasma protein binding of linagliptin is concentration-dependent, decreasing from about 99% at 1 nmol/L to 75-89% at ≥30 nmol/L, reflecting saturation of binding to DPP-4 with increasing concentration of linagliptin. At high concentrations, where DPP-4 is fully saturated, 70-80% of linagliptin was bound to other plasma proteins than DPP-4, hence 30-20% were unbound in plasma.

Biotransformation

Following a [^{14}C] linagliptin oral 10 mg dose, approximately 5% of the radioactivity was excreted in urine. Metabolism plays a subordinate role in the elimination of linagliptin. One main metabolite with a relative exposure of 13.3 % of linagliptin at steady state was detected which was found to be pharmacologically inactive and thus does not contribute to the plasma DPP-4 inhibitory activity of linagliptin.

PA-7772-004

Elimination

Following administration of an oral [^{14}C] linagliptin dose to healthy subjects, approximately 85% of the administered radioactivity was eliminated in faeces (80%) or urine (5%) within 4 days of dosing. Renal clearance at steady state was approximately 70 mL/min.

Special populations

Renal Impairment

A multiple-dose, open-label study was conducted to evaluate the pharmacokinetics of linagliptin (5 mg dose) in patients with varying degrees of chronic renal impairment compared to normal healthy control subjects. The study included patients with renal impairment classified on the basis of creatinine clearance as mild (50 to <80 mL/min), moderate (30 to <50 mL/min), and severe (<30 mL/min), as well as patients with ESRD on haemodialysis. In addition patients with T2DM and severe renal impairment (<30 mL/min) were compared to T2DM patients with normal renal function.

Creatinine clearance was measured by 24-hour urinary creatinine clearance measurements or estimated from serum creatinine based on the Cockcroft-Gault formula:

$CrCl = [140 - age\ (years)] \times weight\ (kg) \times (0.85\ for\ female\ patients) / [72 \times serum\ creatinine\ (mg/dL)]$.

Under steady-state conditions, linagliptin exposure in patients with mild renal impairment was comparable to healthy subjects. In moderate renal impairment, a moderate increase in exposure of about 1.7 fold was observed compared with control.

Exposure in T2DM patients with severe RI was increased by about 1.4 fold compared to T2DM patients with normal renal function. Steady-state predictions for AUC of linagliptin in patients with ESRD indicated comparable exposure to that of patients with moderate or severe renal impairment. In addition, linagliptin is not expected to be eliminated to a therapeutically significant degree by haemodialysis or peritoneal dialysis. Therefore, no dosage adjustment of linagliptin is necessary in patients with any degree of renal impairment.

In addition, mild renal impairment had no effect on linagliptin pharmacokinetics in patients with type 2 diabetes as assessed by population pharmacokinetic analyses.

Hepatic Impairment

In patients with mild moderate and severe hepatic impairment (according to the Child-Pugh classification), mean AUC and C_{max} of linagliptin were similar to healthy matched controls following administration of multiple 5 mg doses of linagliptin. No dosage adjustment for linagliptin is necessary for patients with mild, moderate or severe hepatic impairment.

Body Mass Index (BMI)

No dosage adjustment is necessary based on BMI. Body mass index had no clinically relevant effect on the pharmacokinetics of linagliptin based on a population pharmacokinetic analysis of Phase I and Phase II data.

Gender

No dosage adjustment is necessary based on gender. Gender had no clinically relevant effect on the pharmacokinetics of linagliptin based on a population pharmacokinetic analysis of Phase I and Phase II data.

Geriatric

No dosage adjustment is required based on age, as age did not have a clinically relevant impact on the pharmacokinetics of linagliptin based on a population pharmacokinetic analysis of Phase I and Phase II data. Elderly subjects (65 to 80 years) had comparable plasma concentrations of linagliptin compared to younger subjects.

Paediatric

Studies characterizing the pharmacokinetics of linagliptin in paediatric patients have not been yet performed.

Race

No dosage adjustment is necessary based on race. Race had no obvious effect on the plasma concentrations of linagliptin based on a composite analysis of available pharmacokinetic data, including patients of Caucasian, Hispanic, African-American, and Asian origin. In additional the pharmacokinetic characteristics of linagliptin were found to be similar in dedicated phase I studies in Japanese, Chinese and Caucasian healthy volunteers and African American type 2 diabetes patients.

Metformin

Absorption

After an oral dose of metformin, T_{max} is reached in 2.5 hours. Absolute bioavailability of a 500 mg or 850 mg metformin hydrochloride tablet is approximately 50-60% in healthy subjects. After an oral dose, the non-absorbed fraction recovered in faeces was 20-30%.

After oral administration, metformin hydrochloride absorption is saturable and incomplete. It is assumed that the pharmacokinetics of metformin hydrochloride absorption is non-linear.

At the recommended metformin hydrochloride doses and dosing schedules, steady state plasma concentrations are reached within 24 to 48 hours and are generally less than 1 microgram/mL. In controlled clinical trials, maximum metformin hydrochloride plasma levels (C_{max}) did not exceed 5 microgram/mL, even at maximum doses.

Food decreases the extent and slightly delays the absorption of metformin hydrochloride. Following administration of a dose of 850 mg, a 40% lower plasma peak concentration, a 25% decrease in AUC (area under the curve) and a 35 minute prolongation of the time to peak plasma concentration were observed. The clinical relevance of these decreases is unknown.

Distribution

Plasma protein binding is negligible. Metformin hydrochloride partitions into erythrocytes. The blood peak is lower than the plasma peak and appears at approximately the same time. The red blood cells most likely represent a secondary compartment of distribution. The mean volume of distribution (Vd) ranged between 63-276 L.

Biotransformation

Metformin hydrochloride is excreted unchanged in the urine. No metabolites have been identified in humans.

Elimination

Renal clearance of metformin hydrochloride is >400 mL/min, indicating that metformin hydrochloride is eliminated by glomerular filtration and tubular secretion. Following an oral dose, the apparent terminal elimination half-life is approximately 6.5 hours.

When renal function is impaired, renal clearance is decreased in proportion to that of creatinine and thus the elimination half-life is prolonged, leading to increased levels of metformin hydrochloride in plasma.

Special populations

Paediatric

Single dose study: After single doses of metformin 500 mg, paediatric patients have shown a similar pharmacokinetic profile to that observed in healthy adults. Multiple dose study: Data are restricted to one study. After repeated doses of 500 mg twice daily for 7 days in paediatric patients the peak plasma concentration (C_{max}) and systemic exposure (AUC_{0-t}) were reduced by approximately 33% and 40%, respectively compared to diabetic adults who received repeated doses of 500 mg twice daily for 14 days. As the dose is individually titrated based on glycaemic control, this is of limited clinical relevance.

Renal Impairment

The available data in subjects with moderate renal impairment are scarce and no reliable estimation of the systemic exposure to metformin in this subgroup as compared to subjects with normal renal function could be made. Therefore, the dose adaptation should be made upon clinical efficacy/tolerability considerations (see section Dosage and Administration).

Toxicology

General toxicity studies in rats up to 13 weeks were performed with the combined products in Linagliptin + Metformin HCl (Trajenta Duo). The only observed interaction between linagliptin and metformin was a reduction of body weight gain. No other additive toxicity caused by the combination of linagliptin and metformin was observed.

An animal reproduction study in pregnant rats did not indicate a teratogenic effect attributed to the co-administration of linagliptin and metformin.

The following data are findings in studies performed with linagliptin or metformin individually.

Linagliptin

Effects in non-clinical studies were observed only at exposures considered sufficiently in excess of the maximum human exposure indicating little relevance to clinical use.

Genotoxicity:

Linagliptin was not mutagenic or clastogenic with or without metabolic activation in the Ames bacterial mutagenicity assay, a chromosomal aberration test in human lymphocytes, and an in vivo micronucleus assay.

Carcinogenicity:

A two-year carcinogenicity study was conducted in male and female rats given oral doses of linagliptin of 6, 18, and 60 mg/kg/day. There was no increase in the incidence of tumours in any organ up to 60 mg/kg/day. This dose results in exposures approximately 418 times the human exposure at the maximum recommended daily adult human dose (MRHD) of 5 mg/day based on AUC comparisons. A two-year carcinogenicity study was conducted in male and female mice given oral doses of 8, 25 and 80 mg/kg/day. There was no evidence of a carcinogenic potential up to 80 mg/kg/day, approximately 242 times human exposure at the MRHD.

Reproduction toxicity:

In rat fertility studies with oral gavage doses of 10, 30 and 240 mg/kg/day, males were treated for 4 weeks prior to mating and during mating; females were treated 2 weeks prior to mating through gestation day 6. No adverse effect on early embryonic development, mating, fertility, and bearing live young were observed up to the highest dose of 240 mg/kg/day (approximately 943 times human exposure at the MRHD of 5 mg/day based on AUC comparisons). In the studies on embryo-foetal development in rats and rabbits, linagliptin was shown to be not teratogenic at dosages up to and including 240 mg/kg/day (943x MRHD) in the rat and 150 mg/kg/day (194x MRHD) in the rabbit. A NOAEL of 30 mg/kg/day (49x MRHD) and 25 mg/kg (78x MRHD) was derived for embryo-foetal toxicity in the rat and the rabbit, respectively.

Metformin

Non-clinical data reveal no special hazard for humans based on conventional studies on safety pharmacology, genotoxicity, and cancinogenic potential. In a 13-week toxicity study in rats, metformin related toxicity was seen in heart, liver, kidneys, salivary glands, ovaries, thymus, gastrointestinal tract and adrenal glands at dosages associated with a systemic exposure of 7 times the MRHD or higher.

Metformin was not teratogenic in rats at a dose of 200 mg/kg/day associated with a systemic exposure of 4 times the MRHD (2 g metformin). At higher doses (500 and 1000 mg/kg/day, associated with 11 and 23 times the MRHD), teratogenicity of metformin was observed in the rat.

Availability

Tablets of 2.5/500 mg — Bottles of 60's
Tablets of 2.5/850 mg — Bottles of 60's
Tablets of 2.5/1 g — Bottles of 60's

Caution

Foods, Drugs, Devices and Cosmetics Act prohibits dispensing without prescription.

For any suspected adverse drug reaction, please seek medical attention immediately. Report this to the FDA: www.fda.gov.ph and email us at *PV_local_Philippines@boehringer-ingelheim.com*.

Store at temperatures not exceeding 30°C.

Manufactured by
Boehringer Ingelheim Promeco, S.A de C.V.
Calle del Maiz No. 49, Col. Barrio Xaltocan, C.P. 16090
Xochimilco, Mexico City, Mexico

Imported by:
Boehringer Ingelheim (Philippines), Inc.
23rd Floor BDO Towers Valero Bldg., 8741 Paseo de Roxas
Bel-Air, Makati City, Metro Manila

* T.M. Reg. Phil. Pat. Off.

DR-XY48038 (2.5 mg/500 mg Film-coated Tablet)
Date of First Authorization: 25 May 2022

DR-XY47930 (2.5 mg/850 mg Film-coated Tablet)
DR-XY47929 (2.5 mg/1 g Film-coated Tablet)
Date of First Authorization: 12 April 2022

Date of Revision of Text: 03 August 2022

REVERSE

Vicks Inhaler (Menthol + Eucalyptus Oil)

108.8 mg/108.8 mg per 0.5 mL Inhaler (Lavender Scent) | Household Remedy (HR) | FDA Reg. No. DRHR-2545

Vicks Inhaler: Official FDA Philippines registration record

Brand name	Vicks Inhaler
Generic name	Menthol + Eucalyptus Oil
Dosage strength and form	108.8 mg/108.8 mg per 0.5 mL Inhaler (Lavender Scent)
Classification	Household Remedy (HR)
Pharmacologic category	Counterirritant
Registration number	DRHR-2545
Manufacturer	Makson Health Care Private Limited

Vicks Inhaler: OTC forms registered under the Vicks brand (capsule availability)

OTC capsule form registered under the Vicks brand: NO. 16 OTC / Household Remedy product(s) registered for this brand on the FDA Verification Portal:

Brand name	Dosage strength and form	Classification	Reg. No.
Vicks Inhaler	108.8 mg/108.8 mg per 0.5 mL Inhaler (Lavender Scent)	Household Remedy (HR)	DRHR-2545
Vicks Formula 44 Cough Control	15 mg/5 mL Syrup	Over-the-Counter (OTC) Drug	DR-XY9056
Vicks VapoDrops Xtra Strong	21 mg Lozenge (Honey Menthol Flavor)	Over-the-Counter (OTC) Drug	DR-XY49236
Vicks VapoDrops Xtra Strong	21 mg Lozenge (Lemon Menthol Flavor)	Over-the-Counter (OTC) Drug	DR-XY49234
Vicks VapoDrops Xtra Strong	21 mg Lozenge (Lemon Menthol Flavor)	Over-the-Counter (OTC) Drug	DR-XY49235
Vicks Vaporub Easy Applicator	5.21 g/2.75 g/1.5 g (5.21%/2.75%/1.5% w/w) per 100 g Ointment	Household Remedy (HR)	DRHR-2542
Vicks VapoRub	5.26 g/2.82 g/1.33 g (5.26 %/2.82 %/1.33 % w/w) per 100 g Ointment (Lavender)	Household Remedy (HR)	DRHR-2535
Vicks VapoRub Ointment	5.26 g/2.82 g/1.33 g per 100 g Ointment	Household Remedy (HR)	DRHR-2485
Vicks VapoRub Ointment	5.26 g/2.82 g/1.33 g per 100 g Ointment	Household Remedy (HR)	DRHR-279
Vicks VapoRub Xtra Strong	7 g/4.03 g/5.2 g (7%/4.03%/5.2% w/w) per 100 g Ointment	Household Remedy (HR)	DRHR-2214
Vicks VapoDrops	7 mg/ 15mg/ 10 mg/ 10 mg Lozenge (Berry Menthol Flavor)	Over-the-Counter (OTC) Drug	DR-XY47940
Vicks VapoDrops	7 mg/ 15mg/ 10 mg/ 10 mg Lozenge (Honey Lemon Menthol Flavor)	Over-the-Counter (OTC) Drug	DR-XY47939
Vicks VapoDrops	7 mg/ 15mg/ 10 mg/ 10 mg Lozenge (Orange Menthol Flavor)	Over-the-Counter (OTC) Drug	DR-XY47938

Brand name	Dosage strength and form	Classification	Reg. No.
Vicks VapoDrops	7 mg/15 mg/10 mg/10 mg Lozenges (Berry Menthol Flavor)	Over-the-Counter (OTC) Drug	DR-XY47698
Vicks VapoDrops	7mg/15mg/10mg/10mg Lozenge (Honey Lemon Menthol Flavor) Lozenges	Over-the-Counter (OTC) Drug	DR-XY47727
Vicks Vapo Drops	7mg/15mg/10mg/10mg Lozenge (Orange Menthol Flavor)	Over-the-Counter (OTC) Drug	DR-XY47726

Vicks Inhaler: Official leaflet (Package Insert / Patient Information Leaflet)

No PI/PIL is published on the FDA Verification Portal for registration DRHR-2545 as of 2026-10-22. Indication and dosage are not covered by this record.

Source

FDA Philippines Verification Portal, https://verification.fda.gov.ph (search: "Vicks"), retrieved 2026-10-22. Fields above are reproduced from the official registration record without modification.

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Vicks VapoRub (Camphor + Menthol + Eucalyptus Oil)

5.26 g/2.82 g/1.33 g (5.26 %/2.82 %/1.33 % w/w) per 100 g Ointment (Lavender) | Household Remedy (HR) | FDA Reg. No. DRHR-2535

Vicks VapoRub: Official FDA Philippines registration record

Brand name	Vicks VapoRub
Generic name	Camphor + Menthol + Eucalyptus Oil
Dosage strength and form	5.26 g/2.82 g/1.33 g (5.26 %/2.82 %/1.33 % w/w) per 100 g Ointment (Lavender)
Classification	Household Remedy (HR)
Pharmacologic category	Counterirritant
Registration number	DRHR-2535
Manufacturer	Makson Health Care Private Limited

Vicks VapoRub: OTC forms registered under the Vicks brand (capsule availability)

OTC capsule form registered under the Vicks brand: NO. 16 OTC / Household Remedy product(s) registered for this brand on the FDA Verification Portal:

Brand name	Dosage strength and form	Classification	Reg. No.
Vicks Inhaler	108.8 mg/108.8 mg per 0.5 mL Inhaler (Lavender Scent)	Household Remedy (HR)	DRHR-2545
Vicks Formula 44 Cough Control	15 mg/5 mL Syrup	Over-the-Counter (OTC) Drug	DR-XY9056
Vicks VapoDrops Xtra Strong	21 mg Lozenge (Honey Menthol Flavor)	Over-the-Counter (OTC) Drug	DR-XY49236
Vicks VapoDrops Xtra Strong	21 mg Lozenge (Lemon Menthol Flavor)	Over-the-Counter (OTC) Drug	DR-XY49234
Vicks VapoDrops Xtra Strong	21 mg Lozenge (Lemon Menthol Flavor)	Over-the-Counter (OTC) Drug	DR-XY49235
Vicks Vaporub Easy Applicator	5.21 g/2.75 g/1.5 g (5.21%/2.75%/1.5% w/w) per 100 g Ointment	Household Remedy (HR)	DRHR-2542
Vicks VapoRub	5.26 g/2.82 g/1.33 g (5.26 %/2.82 %/1.33 % w/w) per 100 g Ointment (Lavender)	Household Remedy (HR)	DRHR-2535
Vicks VapoRub Ointment	5.26 g/2.82 g/1.33 g per 100 g Ointment	Household Remedy (HR)	DRHR-2485
Vicks VapoRub Ointment	5.26 g/2.82 g/1.33 g per 100 g Ointment	Household Remedy (HR)	DRHR-279
Vicks VapoRub Xtra Strong	7 g/4.03 g/5.2 g (7%/4.03%/5.2% w/w) per 100 g Ointment	Household Remedy (HR)	DRHR-2214
Vicks VapoDrops	7 mg/ 15mg/ 10 mg/ 10 mg Lozenge (Berry Menthol Flavor)	Over-the-Counter (OTC) Drug	DR-XY47940
Vicks VapoDrops	7 mg/ 15mg/ 10 mg/ 10 mg Lozenge (Honey Lemon Menthol Flavor)	Over-the-Counter (OTC) Drug	DR-XY47939
Vicks VapoDrops	7 mg/ 15mg/ 10 mg/ 10 mg Lozenge (Orange Menthol Flavor)	Over-the-Counter (OTC) Drug	DR-XY47938

Brand name	Dosage strength and form	Classification	Reg. No.
Vicks VapoDrops	7 mg/15 mg/10 mg/10 mg Lozenges (Berry Menthol Flavor)	Over-the-Counter (OTC) Drug	DR-XY47698
Vicks VapoDrops	7mg/15mg/10mg/10mg Lozenge (Honey Lemon Menthol Flavor) Lozenges	Over-the-Counter (OTC) Drug	DR-XY47727
Vicks Vapo Drops	7mg/15mg/10mg/10mg Lozenge (Orange Menthol Flavor)	Over-the-Counter (OTC) Drug	DR-XY47726

Vicks VapoRub: Official leaflet (Package Insert / Patient Information Leaflet)

No PI/PIL is published on the FDA Verification Portal for registration DRHR-2535 as of 2026-10-22. Indication and dosage are not covered by this record.

Source

FDA Philippines Verification Portal, https://verification.fda.gov.ph (search: "Vicks"), retrieved 2026-10-22. Fields above are reproduced from the official registration record without modification.

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Virlix (Cetirizine Dihydrochloride)

10 mg Tablet | Over-the-Counter (OTC) Drug | FDA Reg. No. DR-XY13333

Virlix: Official FDA Philippines registration record

Brand name	Virlix
Generic name	Cetirizine Dihydrochloride
Dosage strength and form	10 mg Tablet
Classification	Over-the-Counter (OTC) Drug
Pharmacologic category	Antihistamine
Registration number	DR-XY13333
Manufacturer	UCB Farchim SA

Virlix: OTC forms registered under the Virlix brand (capsule availability)

OTC capsule form registered under the Virlix brand: NO. 2 OTC / Household Remedy product(s) registered for this brand on the FDA Verification Portal:

Brand name	Dosage strength and form	Classification	Reg. No.
Virlix	10 mg Tablet	Over-the-Counter (OTC) Drug	DR-XY13333
Virlix	10 mg/mL Solution (Oral Drops)	Over-the-Counter (OTC) Drug	DR-XY16259

Virlix: Official leaflet (Package Insert / Patient Information Leaflet)

No PI/PIL is published on the FDA Verification Portal for registration DR-XY13333 as of 2026-10-22. Indication and dosage are not covered by this record.

Source

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Virlix (Cetirizine Dihydrochloride)

10 mg/mL Solution (Oral Drops) | Over-the-Counter (OTC) Drug | FDA Reg. No. DR-XY16259

Virlix: Official FDA Philippines registration record

Brand name	Virlix
Generic name	Cetirizine Dihydrochloride
Dosage strength and form	10 mg/mL Solution (Oral Drops)
Classification	Over-the-Counter (OTC) Drug
Pharmacologic category	Antihistamine
Registration number	DR-XY16259
Manufacturer	Aesica Pharmaceuticals S.R.L.

Virlix: OTC forms registered under the Virlix brand (capsule availability)

OTC capsule form registered under the Virlix brand: NO. 2 OTC / Household Remedy product(s) registered for this brand on the FDA Verification Portal:

Brand name	Dosage strength and form	Classification	Reg. No.
Virlix	10 mg Tablet	Over-the-Counter (OTC) Drug	DR-XY13333
Virlix	10 mg/mL Solution (Oral Drops)	Over-the-Counter (OTC) Drug	DR-XY16259

Virlix: Official leaflet (Package Insert / Patient Information Leaflet)

No PI/PIL is published on the FDA Verification Portal for registration DR-XY16259 as of 2026-10-22. Indication and dosage are not covered by this record.

Source

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